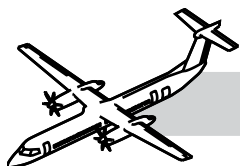


# **CHAPTER 28**

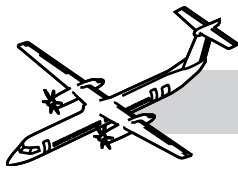
## **AIRFRAME FUEL SYSTEM**

### **CONTENTS**

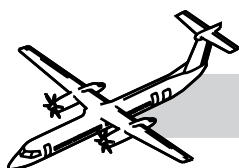
|                                                                           | <b>Page</b>  |
|---------------------------------------------------------------------------|--------------|
| <b>28-00-00 INTRODUCTION</b> .....                                        | <b>28-1</b>  |
| System Description .....                                                  | <b>28-3</b>  |
| <b>28-11-00 FUEL TANKS</b> .....                                          | <b>28-5</b>  |
| Introduction .....                                                        | <b>28-5</b>  |
| General .....                                                             | <b>28-5</b>  |
| System Description .....                                                  | <b>28-5</b>  |
| Inspection and Classification of Fuel Leaks .....                         | <b>28-5</b>  |
| Component Description.....                                                | <b>28-9</b>  |
| Drain the Water from the Wing Tanks and the Dry Bay .....                 | <b>28-18</b> |
| Water Contamination Fuel Check .....                                      | <b>28-18</b> |
| Functional Check of the Fuel Tank Components for Electrical Bonding ..... | <b>28-19</b> |
| <b>28-12-00 VENTS</b> .....                                               | <b>28-21</b> |
| General .....                                                             | <b>28-21</b> |
| System Description .....                                                  | <b>28-21</b> |
| Component Description.....                                                | <b>28-21</b> |
| Operation .....                                                           | <b>28-25</b> |
| <b>28-21-00 ENGINE FUEL FEED</b> .....                                    | <b>28-26</b> |
| General .....                                                             | <b>28-26</b> |
| System Description .....                                                  | <b>28-26</b> |
| Component Description.....                                                | <b>28-27</b> |
| Operation .....                                                           | <b>28-35</b> |



|                                                            | <b>Page</b>  |
|------------------------------------------------------------|--------------|
| Operational Test of the Engine Feed Shut-off Valve .....   | <b>28-35</b> |
| <b>28-22-00 APU FUEL FEED .....</b>                        | <b>28-36</b> |
| General .....                                              | <b>28-36</b> |
| System Description .....                                   | <b>28-36</b> |
| Component Description.....                                 | <b>28-37</b> |
| Operation .....                                            | <b>28-37</b> |
| Operational Check of the APU Fuel Feed Shut-off Valve..... | <b>28-38</b> |
| <b>28-23-00 FUEL TRANSFER.....</b>                         | <b>28-40</b> |
| General .....                                              | <b>28-40</b> |
| System Description .....                                   | <b>28-40</b> |
| Component Description.....                                 | <b>28-41</b> |
| Operation .....                                            | <b>28-45</b> |
| Operational Check of the Fuel Transfer System.....         | <b>28-49</b> |
| <b>28-24-00 REFUEL/DEFUEL.....</b>                         | <b>28-51</b> |
| General .....                                              | <b>28-51</b> |
| System Description .....                                   | <b>28-51</b> |
| Component Description.....                                 | <b>28-53</b> |
| Controls and Indications.....                              | <b>28-63</b> |
| Operation .....                                            | <b>28-65</b> |
| <b>28-40-00 INDICATING.....</b>                            | <b>28-77</b> |
| General .....                                              | <b>28-77</b> |
| System Description .....                                   | <b>28-77</b> |
| Component Description.....                                 | <b>28-78</b> |
| Operation .....                                            | <b>28-89</b> |
| <b>28-00-00 APPENDIX .....</b>                             | <b>28-96</b> |



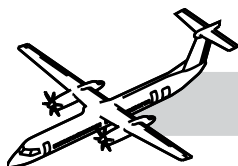
|                                                          | <b>Page</b>  |
|----------------------------------------------------------|--------------|
| Maintenance Consideration .....                          | <b>28-96</b> |
| <b>28-00-00 SPECIAL TOOLS &amp; TEST EQUIPMENT .....</b> | <b>28-97</b> |
| <b>28-00-00 MAINTENANCE PRACTICES.....</b>               | <b>28-97</b> |



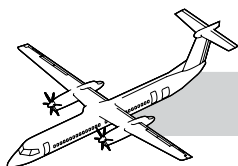
## ILLUSTRATIONS

| <b>Figure</b> | <b>Title</b>                                 | <b>Page</b>  |
|---------------|----------------------------------------------|--------------|
| <b>28-1</b>   | Fuel System Block Diagram .....              | <b>28-2</b>  |
| <b>28-2</b>   | Fuel Tanks .....                             | <b>28-4</b>  |
| <b>28-3</b>   | Fuel Tank - Sections .....                   | <b>28-6</b>  |
| <b>28-4</b>   | Flapper Check Valves .....                   | <b>28-8</b>  |
| <b>28-5</b>   | Motive Flow and Scavenge System .....        | <b>28-10</b> |
| <b>28-6</b>   | Motive Flow Check Valve .....                | <b>28-12</b> |
| <b>28-7</b>   | Scavenge Ejector .....                       | <b>28-14</b> |
| <b>28-8</b>   | Ejector Pump .....                           | <b>28-15</b> |
| <b>28-9</b>   | Gravity Refill Cap .....                     | <b>28-16</b> |
| <b>28-10</b>  | Water Drain Valve.....                       | <b>28-17</b> |
| <b>28-11</b>  | Vent System.....                             | <b>28-20</b> |
| <b>28-12</b>  | Vent Float Valve and Standpipe.....          | <b>28-22</b> |
| <b>28-13</b>  | Vent Float Valve.....                        | <b>28-23</b> |
| <b>28-14</b>  | Surge Bay NACA Vents .....                   | <b>28-24</b> |
| <b>28-15</b>  | Surge Bay NACA Vents (Interior) .....        | <b>28-24</b> |
| <b>28-16</b>  | Surge Bay NACA Vents (Exterior) .....        | <b>28-24</b> |
| <b>28-17</b>  | Engine Fuel Feed System.....                 | <b>28-26</b> |
| <b>28-18</b>  | Primary Ejector Pump and Inlet Strainer..... | <b>28-27</b> |
| <b>28-19</b>  | Engine Fuel Feed System.....                 | <b>28-28</b> |
| <b>28-20</b>  | Engine Feed Shut-Off Valve.....              | <b>28-29</b> |
| <b>28-21</b>  | Auxiliary Pump .....                         | <b>28-30</b> |
| <b>28-22</b>  | Fuel Control Transfer Panel .....            | <b>28-31</b> |
| <b>28-23</b>  | Auxiliary Pump Pressure Switch.....          | <b>28-31</b> |

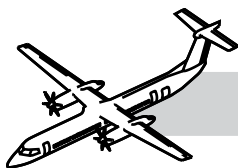




| <b>Figure</b> | <b>Title</b>                                   | <b>Page</b>  |
|---------------|------------------------------------------------|--------------|
| <b>28-24</b>  | MFD Fuel Page.....                             | <b>28-32</b> |
| <b>28-25</b>  | Engine Fuel Feed System Operation .....        | <b>28-34</b> |
| <b>28-26</b>  | APU Fuel Feed Locator .....                    | <b>28-36</b> |
| <b>28-27</b>  | APU Shut-Off Valve .....                       | <b>28-37</b> |
| <b>28-28</b>  | Fuel Transfer System .....                     | <b>28-40</b> |
| <b>28-29</b>  | Refuel/Defuel and Transfer Shut-Off Valve..... | <b>28-41</b> |
| <b>28-30</b>  | Level Control Solenoid .....                   | <b>28-42</b> |
| <b>28-31</b>  | Fuel Transfer Schematic .....                  | <b>28-43</b> |
| <b>28-32</b>  | Fuel Transfer Operation (Sheet 1 of 3).....    | <b>28-44</b> |
| <b>28-33</b>  | Fuel Transfer Operation (Sheet 2 of 3).....    | <b>28-46</b> |
| <b>28-34</b>  | Level Control Valve Operation.....             | <b>28-47</b> |
| <b>28-35</b>  | Fuel Transfer Operation (Sheet 3 of 3).....    | <b>28-48</b> |
| <b>28-36</b>  | Refuel/Defuel System .....                     | <b>28-50</b> |
| <b>28-37</b>  | Refuel/Defuel Control Panel .....              | <b>28-52</b> |
| <b>28-38</b>  | Refuel/Defuel Shut-Off Valve .....             | <b>28-54</b> |
| <b>28-39</b>  | Refuel/Defuel Adapter .....                    | <b>28-56</b> |
| <b>28-40</b>  | No-Flow Pressure Switch.....                   | <b>28-57</b> |
| <b>28-41</b>  | Refuel Vent Valves .....                       | <b>28-58</b> |
| <b>28-42</b>  | High Level Sensor .....                        | <b>28-59</b> |
| <b>28-43</b>  | Fuel Quantity Probes .....                     | <b>28-60</b> |
| <b>28-44</b>  | Fuel System Fault Codes.....                   | <b>28-62</b> |
| <b>28-45</b>  | Preselect Refueling (1 of 2) .....             | <b>28-64</b> |
| <b>28-46</b>  | Preselect Refueling (2 of 2) .....             | <b>28-66</b> |
| <b>28-47</b>  | Refuel/Defuel Operation (Sheet 1 of 2).....    | <b>28-68</b> |
| <b>28-48</b>  | Refuel/Defuel Operation (Sheet 2 of 2).....    | <b>28-70</b> |

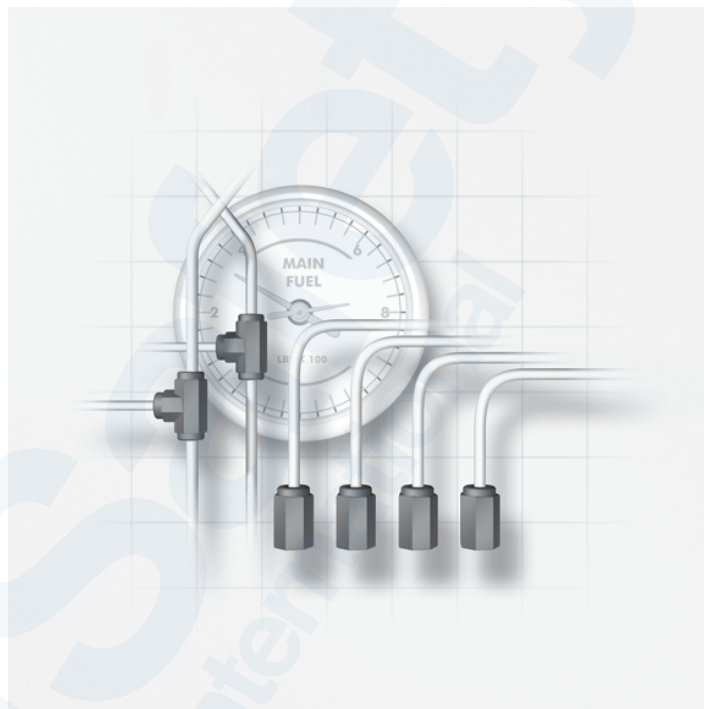


| <b>Figure</b> | <b>Title</b>                               | <b>Page</b>  |
|---------------|--------------------------------------------|--------------|
| <b>28-49</b>  | Refuel/Defuel System Schematic .....       | <b>28-72</b> |
| <b>28-50</b>  | Fuel Indicating System .....               | <b>28-76</b> |
| <b>28-51</b>  | Fuel Quantity Computer (FQC) .....         | <b>28-78</b> |
| <b>28-52</b>  | High Level Control Unit .....              | <b>28-79</b> |
| <b>28-53</b>  | Float Switch .....                         | <b>28-80</b> |
| <b>28-54</b>  | Temperature Sensor .....                   | <b>28-82</b> |
| <b>28-55</b>  | Magnastick .....                           | <b>28-84</b> |
| <b>28-56</b>  | Fuel Quantity Probe.....                   | <b>28-86</b> |
| <b>28-57</b>  | Fuel Indication.....                       | <b>28-88</b> |
| <b>28-58</b>  | Fuel Indicating System (Sheet 1 of 2)..... | <b>28-90</b> |
| <b>28-59</b>  | Fuel Indicating System (Sheet 2 of 2)..... | <b>28-92</b> |
| <b>28-60</b>  | Fuel System Synoptic .....                 | <b>28-94</b> |



# CHAPTER 28

## AIRFRAME FUEL SYSTEM



### 28-00-00 INTRODUCTION

Fuel is contained in two integral main wing tanks designated No. 1 (Left) and No. 2 (Right). The fuel system provides for indicating, storing, venting, fuel feeding and scavenging, refueling/defueling, and transfer. Only tank to tank transfer is available; there is no engine cross-feed capability. The tanks may be pressure or gravity refueled.

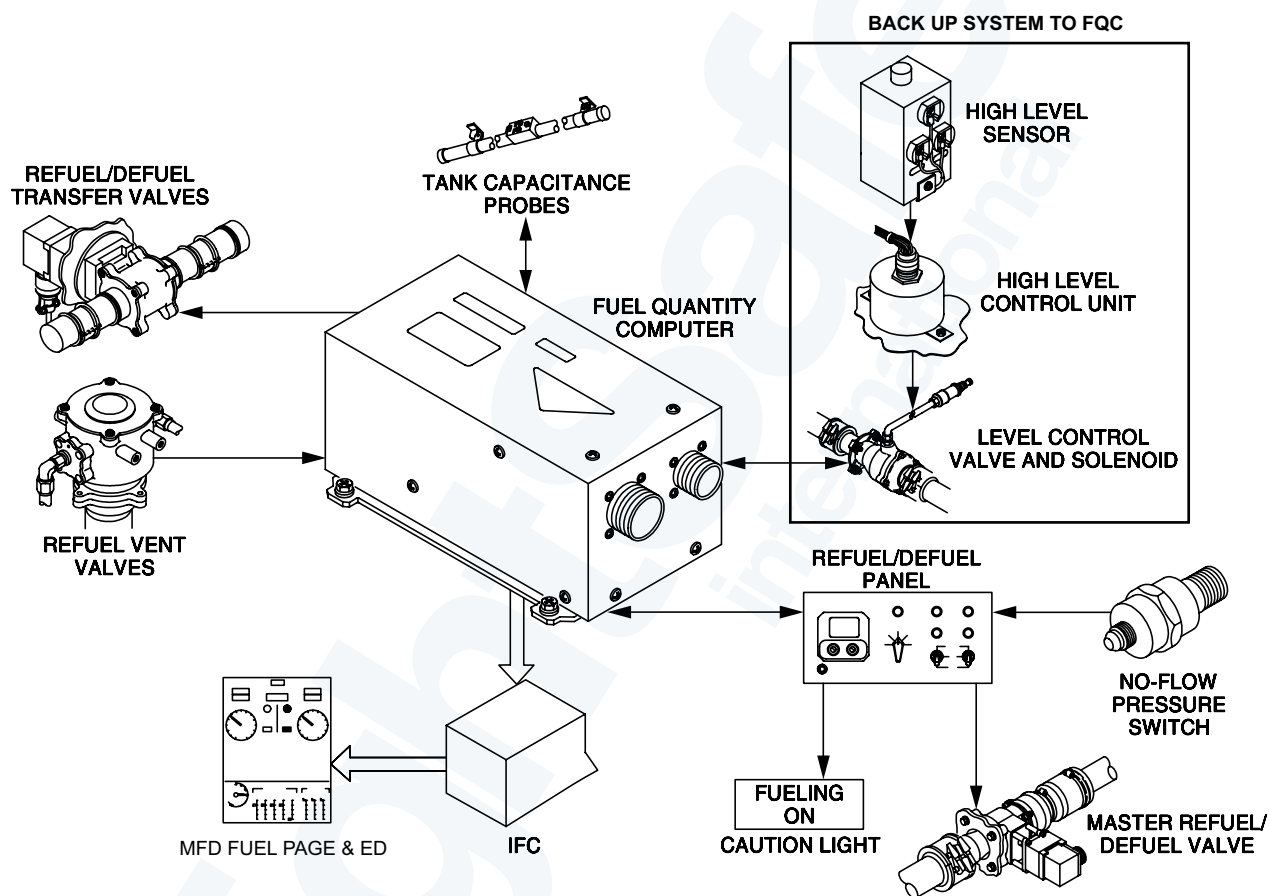
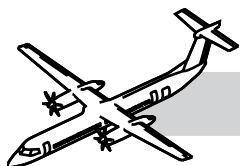
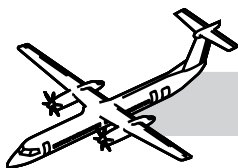


Figure 28-1. Fuel System Block Diagram



## SYSTEM DESCRIPTION

Refer to Figure 28-1. Fuel System Block Diagram.

Fuel is contained in two integral main wing tanks designated No.1 (Left) and No.2 (Right).

Each wing tank includes a surge bay and a collector bay. The left tank supplies fuel to the left engine and APU. The right tank supplies fuel to the right engine.

There are three drain valves at the lowest points of each tank to permit checks for water accumulation and to drain residual fuel from the tanks when required.

Normal engine feed is by means of an ejector pump in each collector bay. The engine Fuel Metering Unit (FMU) supplies the motive flow.

No electrical power is required for normal engine feed operation. In the event of an ejector pump failure, the auxiliary AC electrical pump in the collector bay is selected ON to supply engine fuel feed.

Four scavenge ejector pumps transfer fuel inboard to the collector bay. This maintains a constant level in the collector bay to ensure engine fuel feed regardless of aircraft attitude and minimizes the quantity of unusable fuel.

A low fuel level sensor is in each collector bay to provide a caution indication to the crew.

Pressure refueling and defueling is accomplished through a single point refuel/defuel adaptor in the right engine nacelle and is controlled through an adjacent refuel/defuel control panel. Both automatic and manual refueling modes are possible.

Selected and actual fuel tank quantities are displayed on the control panel. The Fuel Quantity Computer (FQC), under the center aisle floorboards, controls the logic during the automatic refuel/defuel process. Refuel operations can be accomplished using aircraft battery power.

A high level sensor shutoff system ensures the maximum allowable tank capacity is not exceeded.

As an alternative to pressure refueling and defueling, each tank can be filled through an overwing filler port.

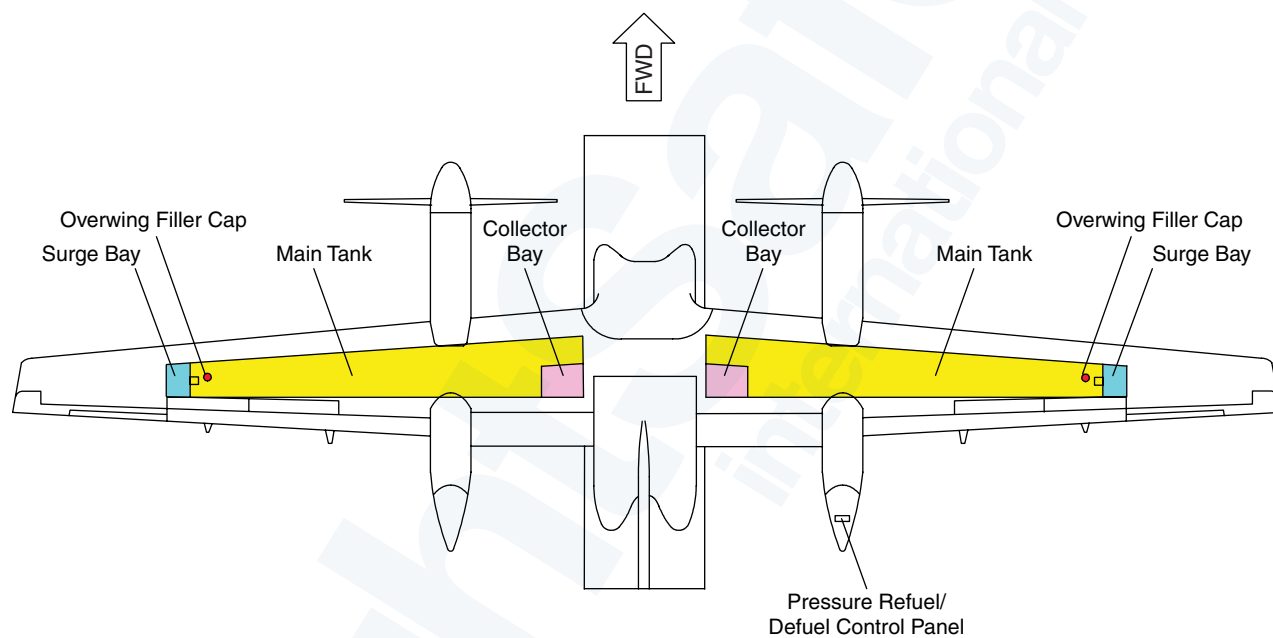
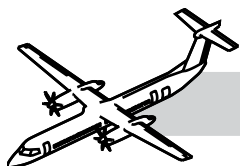
Each fuel tank is vented through a surge bay in the outboard section of the main tank. Fixed vent lines and float vent valves between the fuel tanks and the surge bays provide adequate capacity for in flight venting. During pressure refueling, tank venting is supplemented by a high capacity vent valve which is opened by refuel pressure.

The Fuel Quantity Gauging System (FQGS) accurately measures the quantity of fuel in the main tanks. It supplies the fuel quantity data to the flight compartment and the refuel/defuel panel for display.

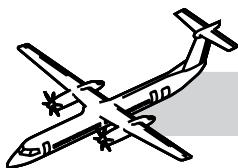
Fuel quantity may also be checked on the ground by use of the magnetic dipsticks on the lower surface of the wing. The maximum usable fuel capacity of the aircraft is approximately 5,338 kg (11,724 lb).

A wing-to-wing fuel transfer system is provided to correct lateral fuel imbalance. The auxiliary pump in the donor tank transfers the fuel to the receiving tank.

The FUEL CONTROL TRANSFER panel in the flight compartment controls the operation of the fuel transfer system. The maximum allowable tank to tank imbalance is 272 kg (600 lb).



**Figure 28-2. Fuel Tanks**



## 28-11-00 FUEL TANKS

### INTRODUCTION

The fuel tanks contain the fuel to be used by the aircraft engines and optional Auxiliary Power Unit (APU).

### GENERAL

The fuel tanks have the necessary components to move the fuel from the main tanks to the collector bay for use by the fuel distribution system.

Each wing tank has a surge bay and a collector bay. There are vent lines in the fuel storage system to prevent over-pressurization of the fuel tanks.

### SYSTEM DESCRIPTION

Refer to Figure 28-2. Fuel Tanks.

Each fuel tank is formed by the structure of the wing itself. This is known as an integral wet wing tank.

The fuel tanks include the following components:

- Flapper check valves
- Motive flow lines
- Motive flow check valves
- Scavenge flow lines
- Forward scavenge ejector
- Aft scavenge ejector
- Mid wing scavenge ejector
- Outboard scavenge ejector
- Gravity refill cap
- Water drain valve.

The fuel tanks are formed by the front and rear wing spars and the upper and lower wing skins. The inboard end of the main fuel tank is the wing rib at WS 42 and the outboard end is the wing rib at WS 407.

The main tanks and the collector bays are the storage areas for the fuel. The total volume

of each wing tank is approximately 3391 L (896 US gallons). The high level shut off is set at 3300 L (872 US gallons) to ensure that a minimum of 2% of the total volume is maintained for expansion of the fuel.

### INSPECTION AND CLASSIFICATION OF FUEL LEAKS

*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

Fuel leaks are divided into four groups for flight safety analysis. The four groups of fuel leaks are stain, seep, heavy seep and running leak. Each group is determined by visual examination of the wet areas around the leak source.

Areas must be wiped clean after each examination for accurate identification and analysis of each leak. The size patterns of the leaks are based on examinations 15 minutes after the leak area is wiped clean.

#### NOTE

Fuel tanks must be filled before you do the fuel leak analysis.

The four groups of fuel leaks are identified and analyzed as follows:

#### 1. Stain:

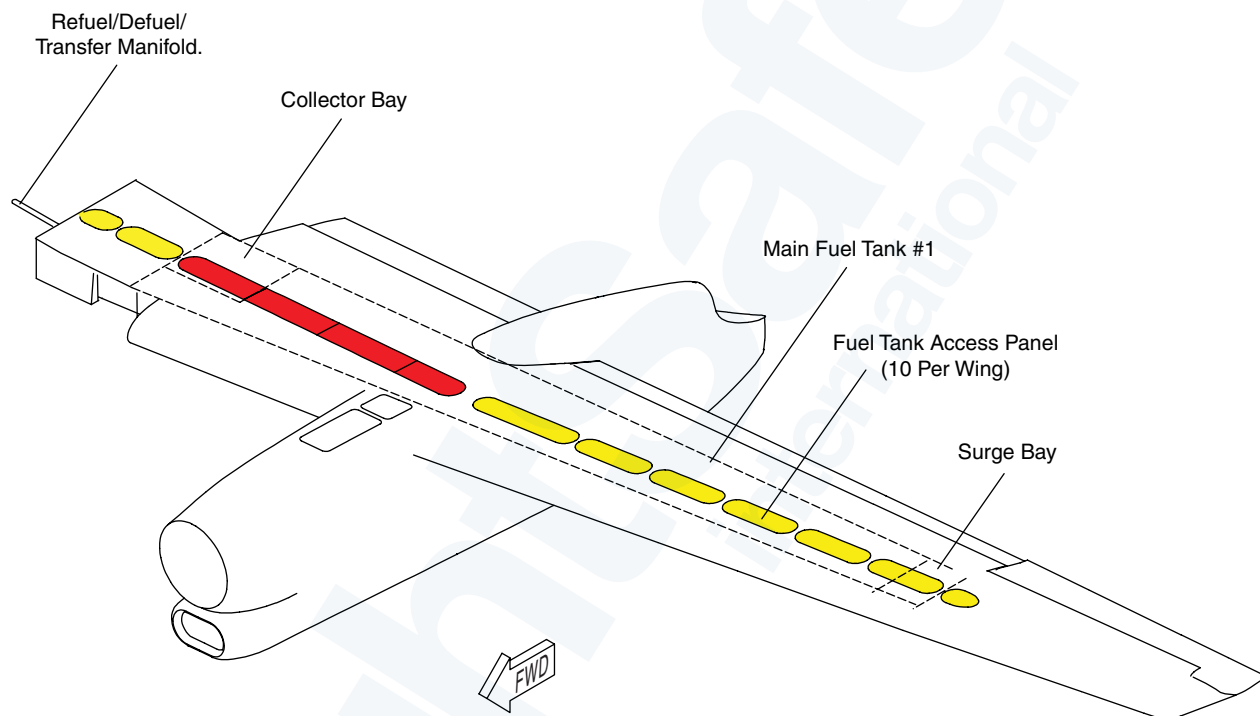
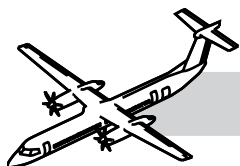
A stain is a leak where the wet area has a dimension that is no larger than 1.5 inches (3.8 cm) after the time interval (15 minutes).

#### NOTE

Small stains can be repaired when necessary.

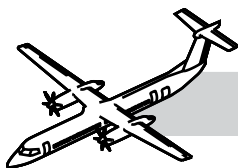
#### 2. Seep:

A seep is a leak where the wet area has a dimension that is no larger than 4 inches (10.1 cm) after the time interval (15 minutes).



**Figure 28-3. Fuel Tank - Sections**





### NOTE

Small seeps can be repaired when necessary.

#### 3. Heavy Seep:

A heavy seep is a leak where the wet area has a dimension that is no larger than 6 inches (15.2 cm) after the time interval (15 minutes).

### NOTE:

Heavy seep conditions must be monitored continuously and repaired as soon as possible.

#### 4. Running Leak:

A running leak is any leak in excess of heavy seep. Fuel will appear immediately after the area has been wiped dry and can run or drip from surface.

### NOTE

Running leaks must be repaired before the next flight.

Refer to Figure 28-3. Fuel Tank - Sections.

Each tank is divided into three sections:

- Main tank
- Collector bay
- Surge bay.

The inboard sections of the main tank form the collector bay. The collector bay holds a constant supply of fuel for the engine fuel feed system. The collector bay is located laterally between WS 42 and WS 79 and longitudinally between the rear wing spar and the wing box structure aft of fuselage station FS 396. The capacity of the collector bay is approximately 227 L (60 US gallons).

The surge bay is outboard of the main tank between WS 407 and WS 425. Each surge bay is connected through integral standpipes to two separate NACA vents on the bottom of the wings. The standpipes prevent fuel spillage overboard. Vent lines, routed from each surge bay to its main tank, control the pressure inside the tanks.

The wing box section between the WS 42 left and WS 42 right is called the dry bay. It separates the wing tanks from the fuselage. There are two openings in the bottom wing skin inboard of each WS 42. These openings provide overboard draining of minor fuel leaks, area ventilation and pressure relief resulting from altitude changes.

Removable rectangular panels along the top of the wing provide access to the inside of the main tank, surge bay and dry bay. The openings at the top of the wing rib structures inside the tanks minimize the amount of air pockets during refueling and aircraft maneuvers.

Rib baffles slow the flow of fuel to the outboard end of the wing tank during uncoordinated manoeuvres or accelerated rolls. One-way flapper valves installed near the bottom of the ribs allow the fuel to flow inboard only. This allows the scavenge ejectors to maximize the amount of fuel they can scavenge at low fuel and wing down conditions.

The wet wing tanks are prevented from leaking by sealing all the ribs, stringer joints, spars and fasteners with sealing compound. Adequate drainage paths are provided in the ribs and stringers to allow for the drainage of water by gravity to the sump areas of the tank.

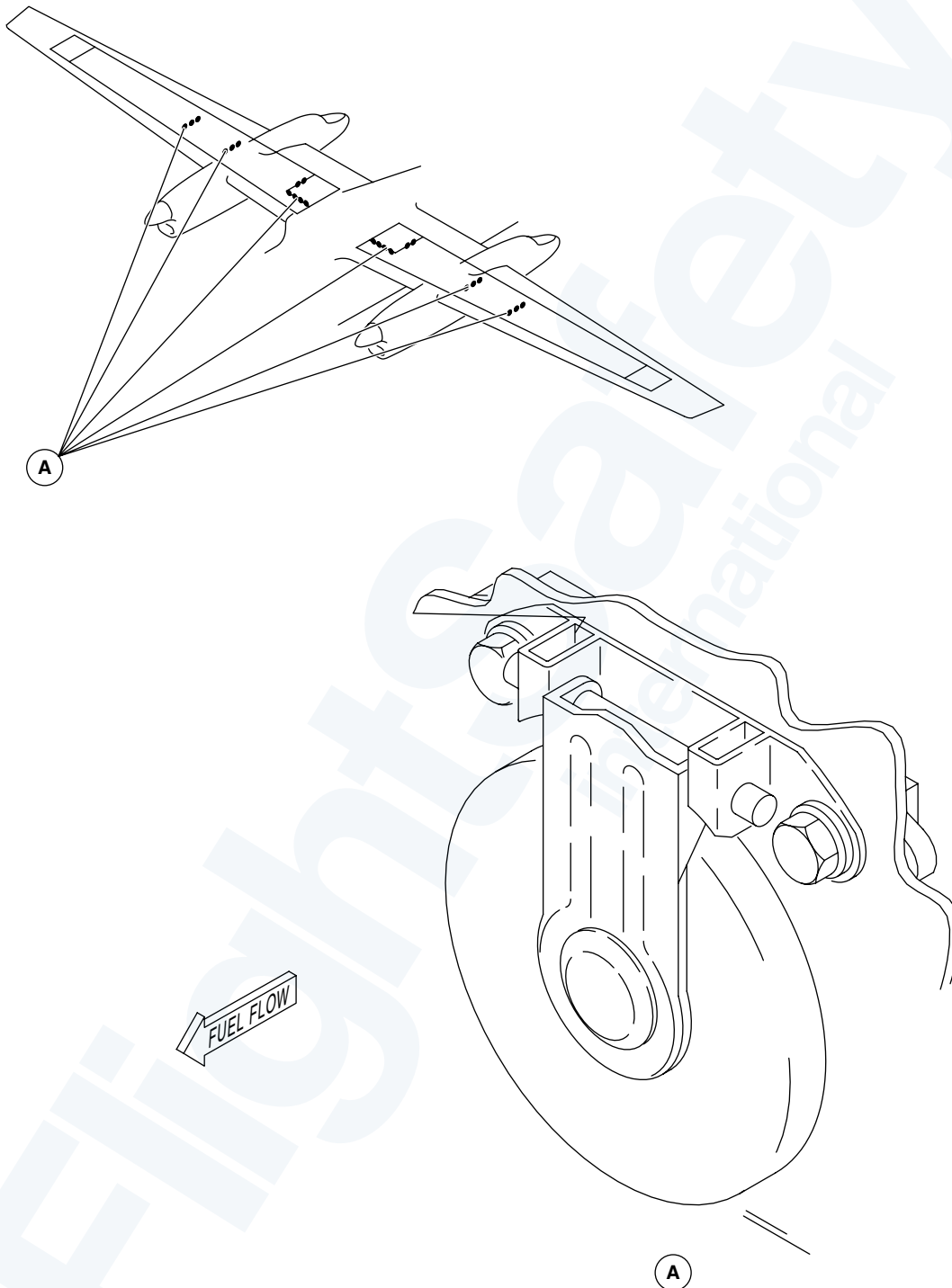
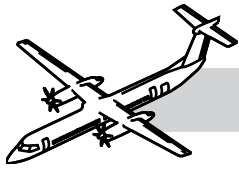
The sumps are in the following areas:

- Surge bay
- Main tank, immediately outboard of rib at WS 191
- Collector bay, close to the main ejector pump.

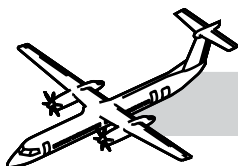
A drain valve is in each sump area to allow for the drainage of water when the aircraft is on the ground.

### CAUTION

Ensure all wing inspection panels are replaced before moving the aircraft.



**Figure 28-4. Flapper Check Valves**



## COMPONENT DESCRIPTION

## NOTES

### Flapper Check Valves

Refer to Figure 28-4. Flapper Check Valves.

The function of the flapper check valves is to allow the fuel to flow inboard only.

The flapper check valves are simple mechanical valves at the bottom of the wing ribs and along the forward and side walls of the collector bay. There are 18 flapper check valves in each wing tank.

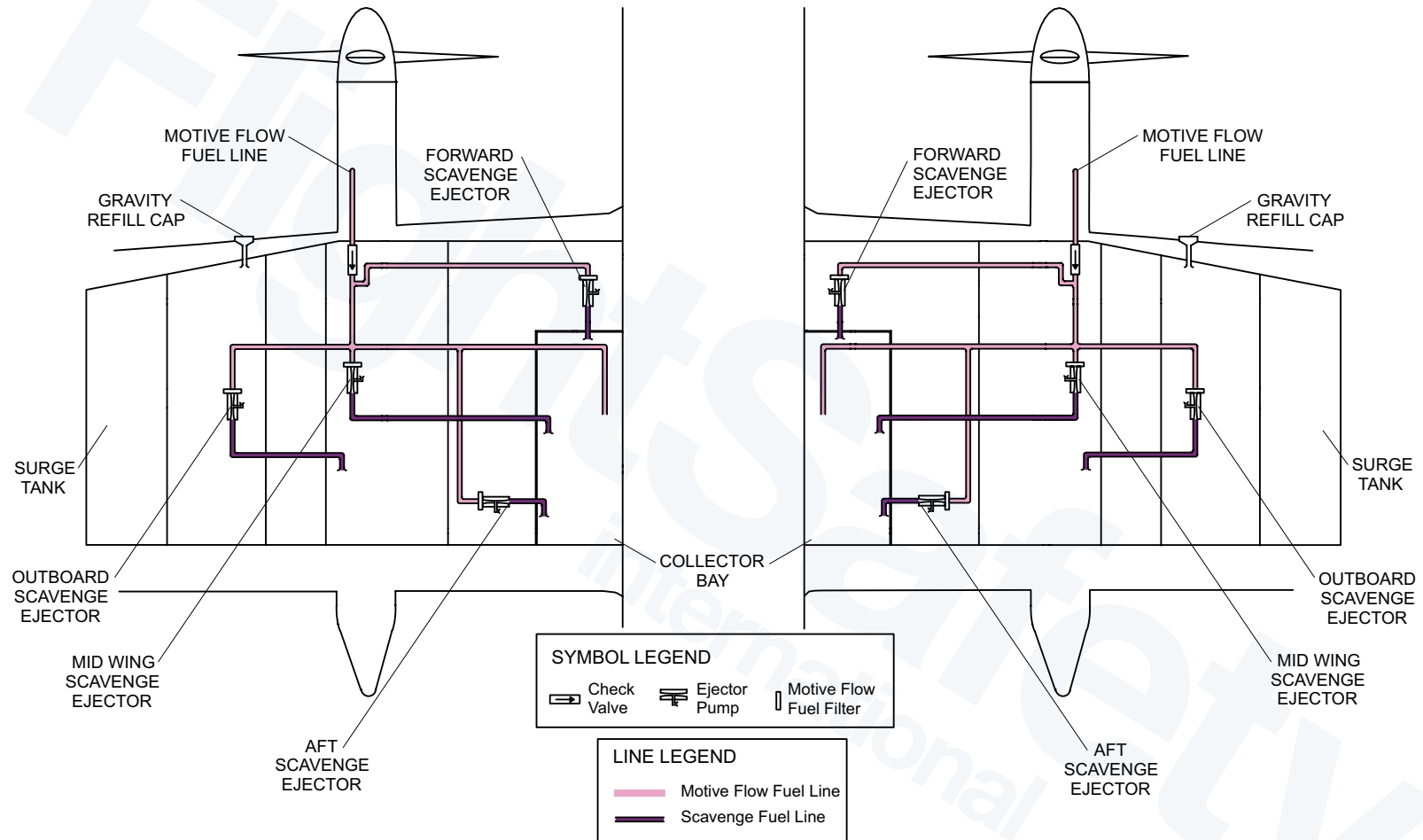
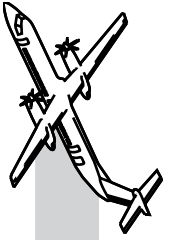
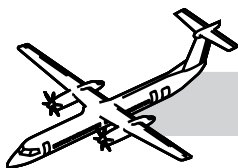


Figure 28-5. Motive Flow and Scavenge System



## Motive Flow Lines

## NOTES

Refer to Figure 28-5. Motive Flow and Scavenge System.

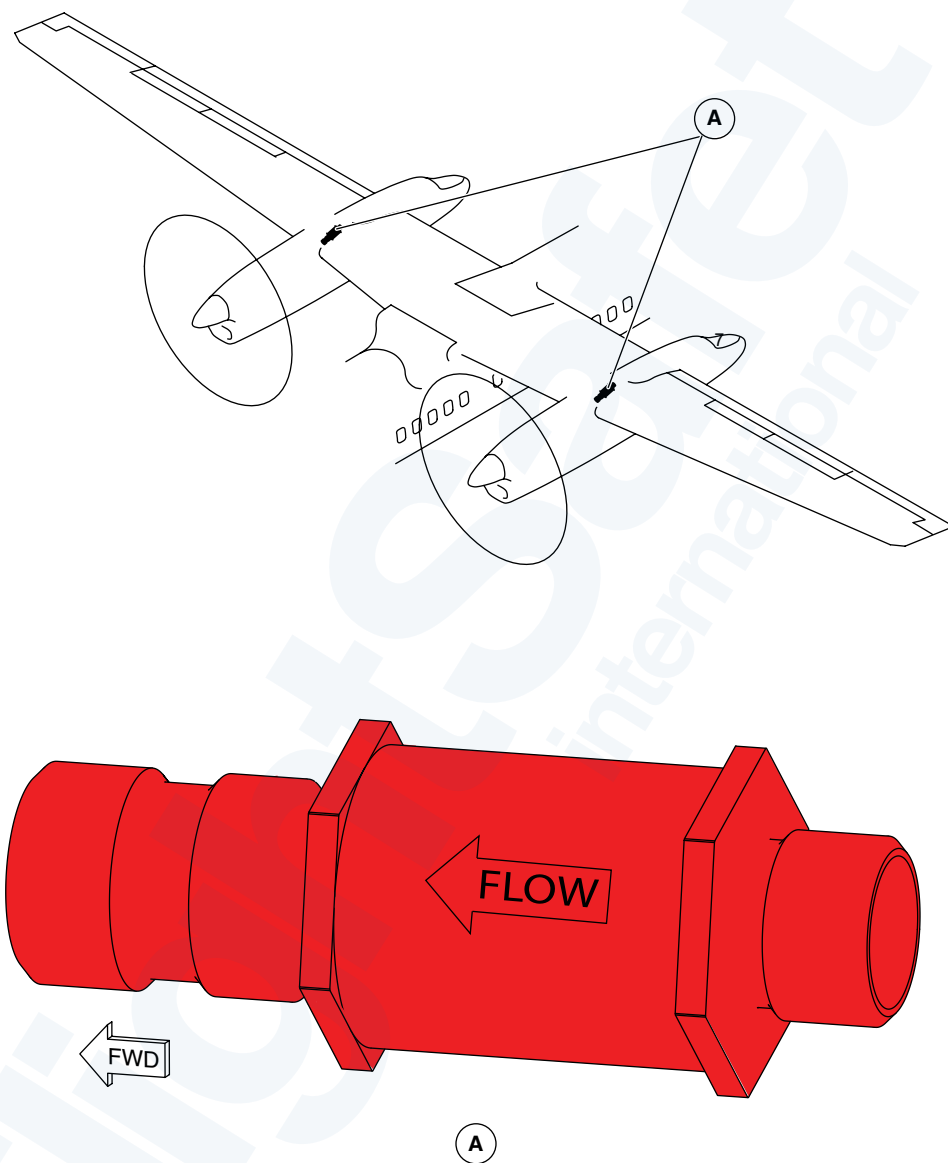
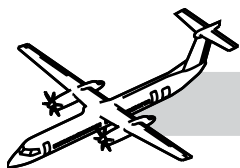
The Motive flow lines are in the main wing tanks.

Scavenge ejectors and engine fuel feed system primary ejectors receive motive flow through the motive flow lines.

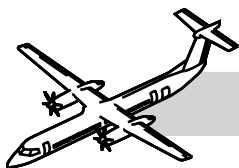
## Scavenge Flow Lines

Refer to Figure 28-5. Motive Flow and Scavenge System.

The scavenge flow lines are in the main wing tanks. The scavenge flow lines transfer fuel from various areas of the main tanks to the inboard part of the main tanks and to the collector bays.



**Figure 28-6. Motive Flow Check Valve**



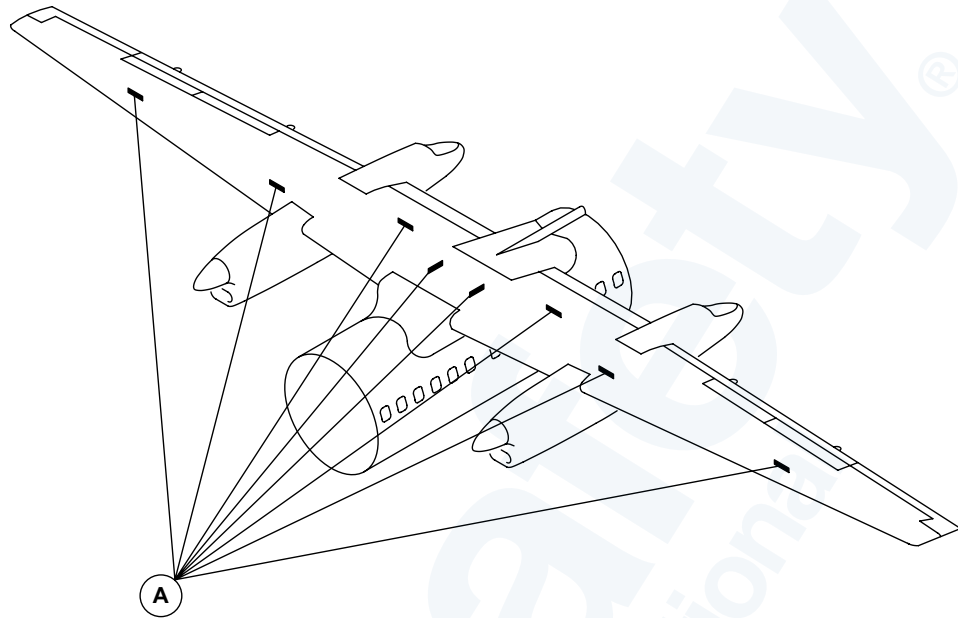
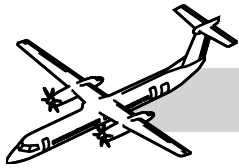
## Motive Flow Check Valve

## NOTES

Refer to Figure 28-6. Motive Flow Check Valve.

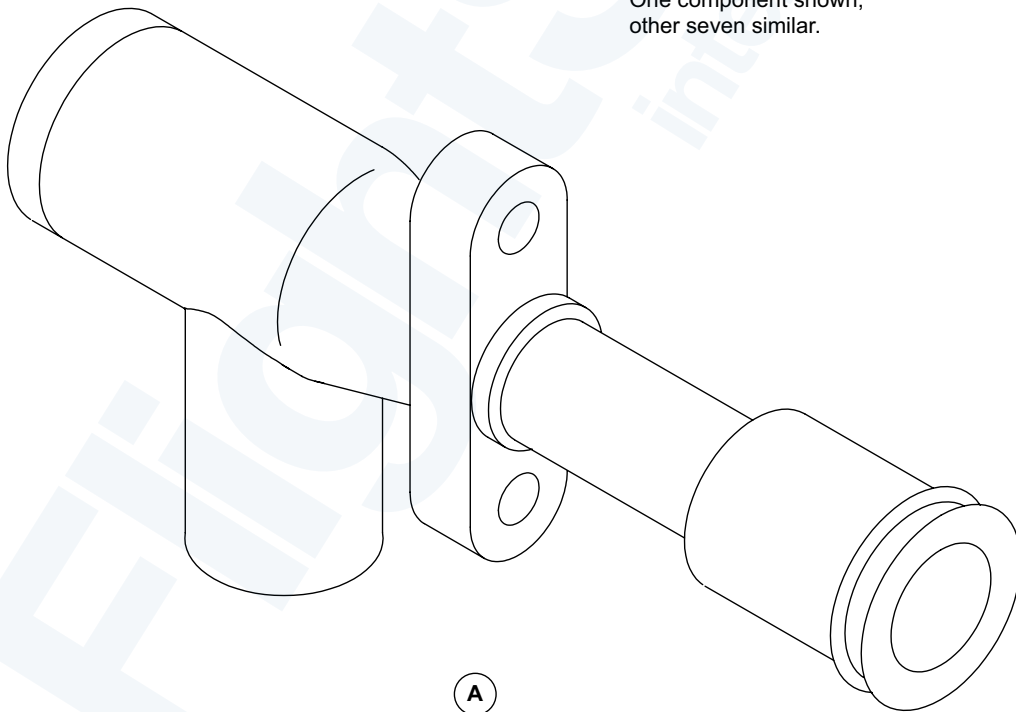
The purpose of the motive flow check valve is to prevent any reverse flow from the fuel tank to the engine.

There is one motive flow check valve in each tank. It is in the motive flow supply line between the engine and the ejector pumps.



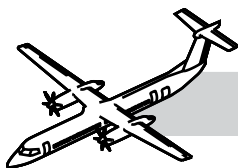
**NOTE**

One component shown,  
other seven similar.



**Figure 28-7. Scavenge Ejector**





## Forward Scavenge Ejector

Refer to Figure 28-7. Scavenge Ejector.

The purpose of the forward scavenge ejector is to keep the collector bay full of fuel.

There is one forward scavenge ejector in each tank. Each ejector is in the forward area of the wing just outboard of the closure rib at WS 42 and immediately aft of the front spar.

## Aft Scavenge Ejector

Refer to Figure 28-7. Scavenge Ejector.

The purpose of the aft scavenge ejector is to keep the collector bay full of fuel.

There is one aft scavenge ejector in each tank. The ejector is in the aft area of the wing near the rear wing spar inboard of the baffle rib at WS 117.

## Mid Wing Scavenge Ejector

Refer to:

- Figure 28-7. Scavenge Ejector.
- Figure 28-8. Ejector Pump.

The purpose of the mid-wing scavenge ejector is to transfer fuel inboard to the collector bay.

There is one mid wing scavenge ejector in each tank. The ejector is just inboard of WS 191 at the low point of the wing. Motive flow is pumped to the ejector through the motive flow lines.

## Outboard Scavenge Ejector

Refer to:

- Figure 28-7. Scavenge Ejector.
- Figure 28-8. Ejector Pump.

The function of the single outboard scavenge ejector is to supply fuel to the area of the wing inboard of WS 191.

There is one outboard scavenge ejector in each tank. The ejector is in the outboard area of the wing at the tank closure rib WS 407. Motive flow is pumped to the scavenge ejectors through the motive flow lines.

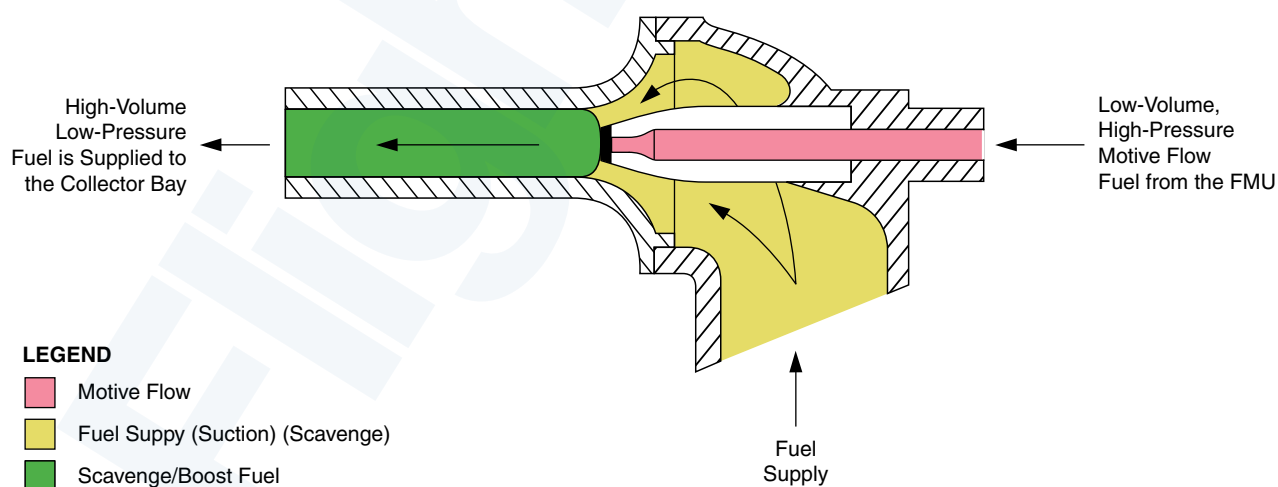
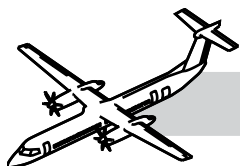


Figure 28-8. Ejector Pump

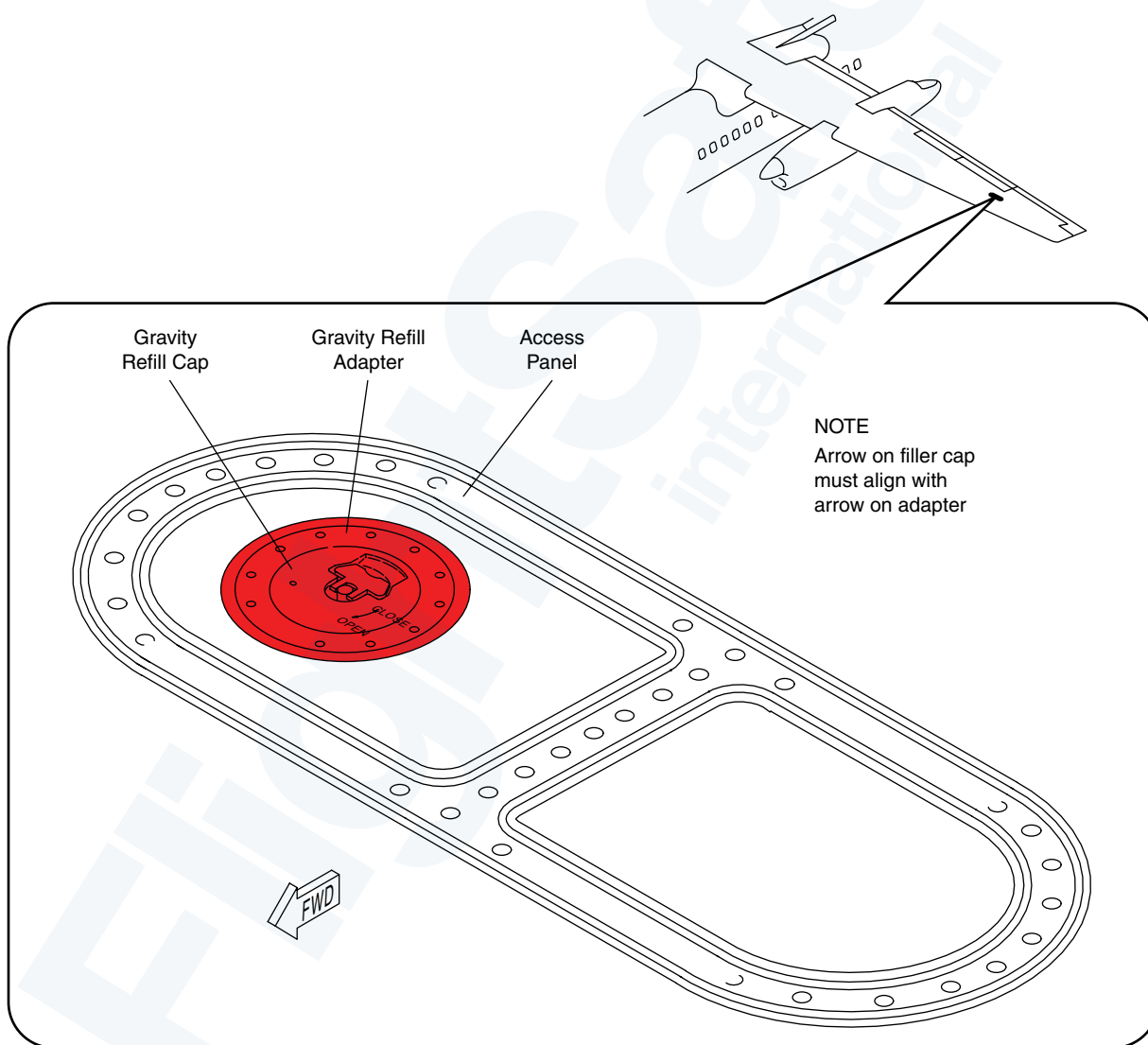


## Gravity Refill Cap

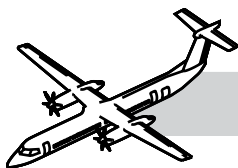
Refer to Figure 28-9. Gravity Refill Cap.

The gravity refill cap allows for gravity refueling of the aircraft.

There is one gravity refill cap installed at the outboard section of each wing near WS 407. The lightning proof refill cap mates with the refuel/defuel adaptor and is flush with the skin of the wing.



**Figure 28-9. Gravity Refill Cap**



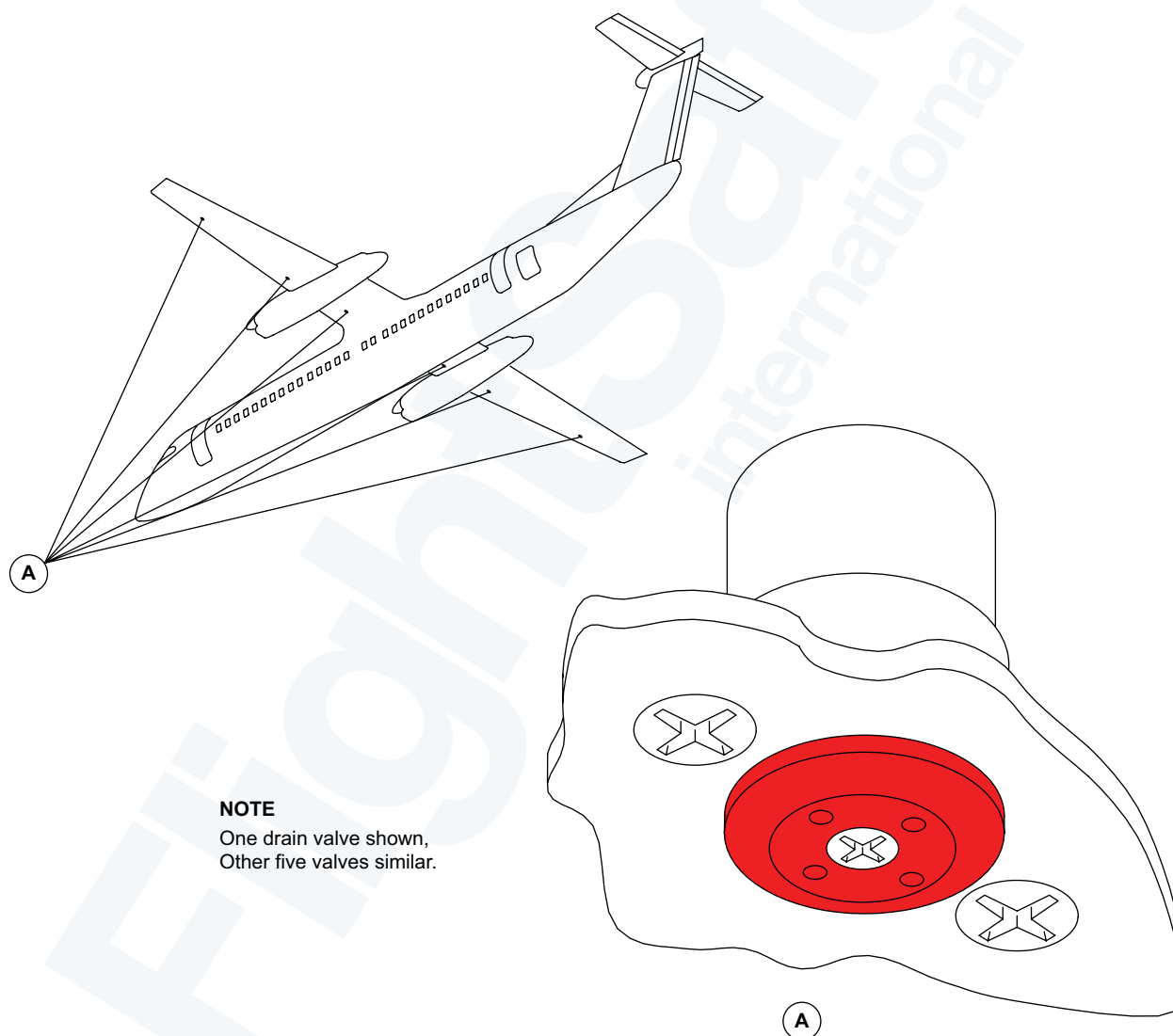
## Water Drain Valve

Refer to Figure 28-10. Water Drain Valve.

The water drain valves may be opened to drain any water accumulation in the tank. The valves are also used to drain residual fuel from the tanks when necessary.

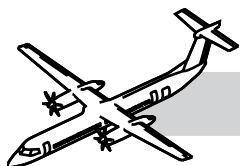
There are six water drain valves installed on the aircraft, three in each wing. They are at the lowest points of the surge bay, main tank and collector bay.

The drain valve is spring-loaded closed and may be locked in the open position. The secondary seal design allows the replacement of the primary sealing “O” ring with the fuel still in the tank. The valve is retained and held against the lower wing skin by a non-metallic nut inside the fuel tank. Electrical grounding of the aluminum valve is through the valve mounting flange and the wing skin.



**NOTE**  
One drain valve shown,  
Other five valves similar.

**Figure 28-10. Water Drain Valve**



## DRAIN THE WATER FROM THE WING TANKS AND THE DRY BAY

*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

The maintenance procedure that follows is for the draining of the water from the wing tanks and the dry bay. The water drain valves are on the bottom skin of the wings.

### **WARNING**

OBEY THE FUEL SAFETY PRECAUTIONS WHEN YOU DO WORK ON THE FUEL SYSTEM OR A FUEL SYSTEM COMPONENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

### **WARNING**

GROUND THE AIRCRAFT AND THE WORKSTAND BEFORE YOU DO FUEL SYSTEM MAINTENANCE. AN ELECTROSTATIC SPARK CAN CAUSE AN EXPLOSION OR A FIRE.

Drain the water from the wing tanks and the dry bay as follows:

- Put an approved fuel container directly below the water drain valve
- Push in and turn the core of the water drain valve clockwise with the fuel drain valve tool
- Drain the unwanted water/fuel into the approved fuel container
- Turn the core counterclockwise to close the water drain valve.

## WATER CONTAMINATION FUEL CHECK

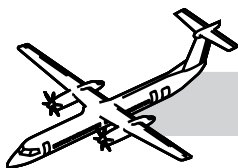
*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

The maintenance procedure that follows gives instructions to do the water contamination check for the fuel system.

- Make sure that you park the aircraft on level ground
- Ground the aircraft
- Obey the safety precautions when you do work on the aircraft fuel system
- Make sure the fuel in the tanks is stable.

Do the water contamination check for the fuel system as follows:

- Open each drain valve in the wing lower skin
- Collect a fuel sample in the bottle
- Close the drain valve
- Do a visual inspection of the fuel sample. Make sure that the fuel has no waterline or milky particles
- If you see contamination, drain the water from the tank until clean fuel flows
- Do the visual inspection of a fuel sample again.



## **FUNCTIONAL CHECK OF THE FUEL TANK COMPONENTS FOR ELECTRICAL BONDING**

*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

The components to be checked in this task are all those in the engine fuel feed system, fuel transfer system and APU fuel supply system. Also tested are the lines in the fuel vent, scavenge/motive flow, pressure relief, refuel/defuel, engine feed and APU plumbing lines.

This maintenance procedure is a fuel tank safety-critical item and is classified as an Airworthiness Limitation Item (ALI).

- Tools and equipment required:
  - Digital Low Resistance Ohmmeter
  - Loop Resistance Tester
  - Maintenance Stand.
- Lower the flaps to 35 degrees
- Obey all fuel safety precautions
- Pressure defuel the wing tanks
- Drain and vent the tanks
- Remove tank access panels
- Make sure the aircraft is de-energized and the aircraft is grounded
- Remove only enough anti-corrosion treatment or paint to enable the tests to be carried out
- Clean the testing areas.

Do a functional check of the components and lines for electrical bonding using the low resistance ohmmeter.

- Record the resistance values and ensure they do not exceed the limits

Do a functional check of the components and lines for electrical bonding with the loop resistance tester.

- Record the resistance values and ensure they do not exceed the limits
- Re-apply any paint or anti-corrosion that was removed or damaged during the testing
- Install the fuel tank access panels.

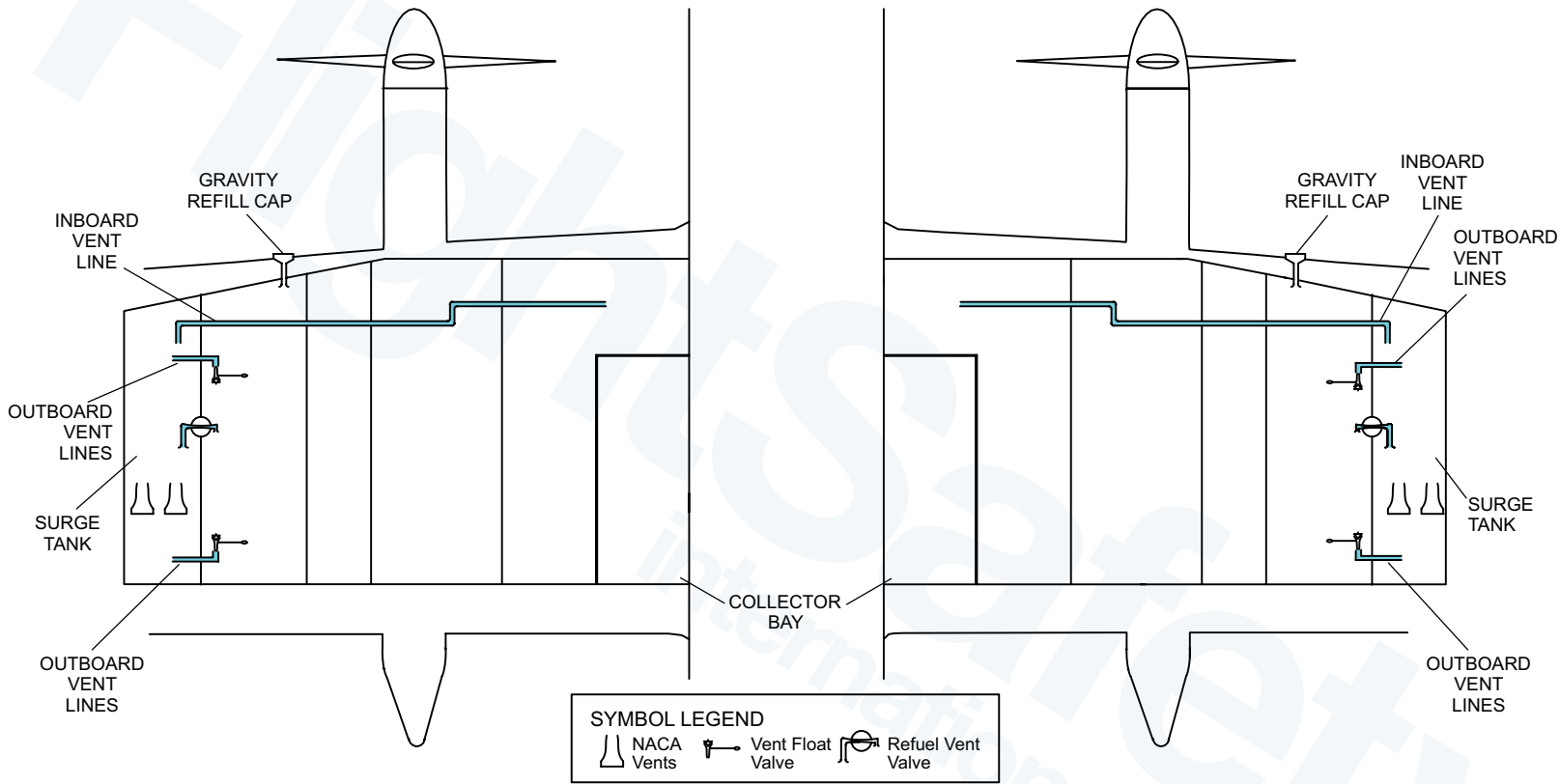
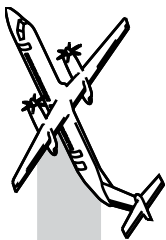
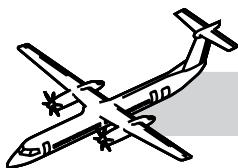


Figure 28-11. Vent System



## 28-12-00 VENTS

### GENERAL

Refer to Figure 28-11. Vent System.

The vent system lets air enter the fuel tanks as the fuel is supplied to the engines, or during aircraft defueling. This prevents structural damage to the tanks from an excessively low internal pressure. The vents also let air out of the tanks when the aircraft is being refueled to prevent the tanks from over-pressurizing.

### SYSTEM DESCRIPTION

The vent system includes these components:

- Inboard vent line
- Outboard vent line
- Vent float valves.

### COMPONENT DESCRIPTION

#### Inboard Vent Line

Refer to Figure 28-11. Vent System.

An unobstructed 1.0 in. (2.54 cm) diameter inboard vent line runs immediately below the wing top skin from the surge bay along the forward wing spar to the inboard part of the main fuel tank.

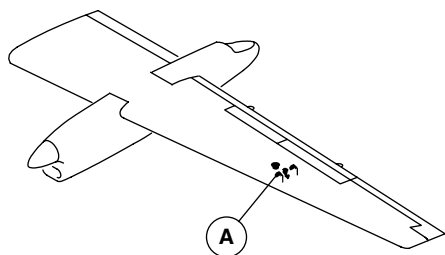
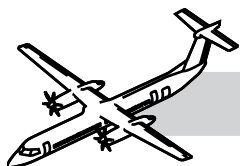
The outboard end of the vent line extends through the tank closure rib at WS 407 and down to within 0.5 in. (12.7 mm) of the bottom of the surge bay. Due to the slight pressurization of the surge bay excess fuel in the bottom of the bay is scavenged back into the main tank.

#### Outboard Vent Line

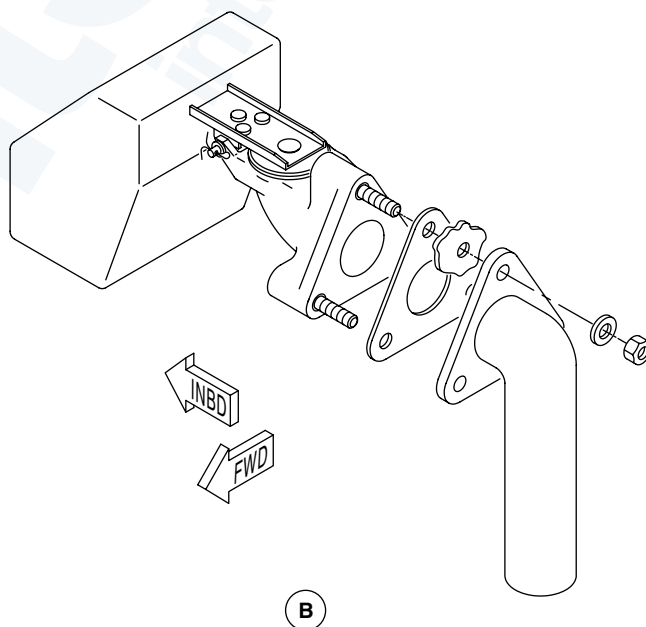
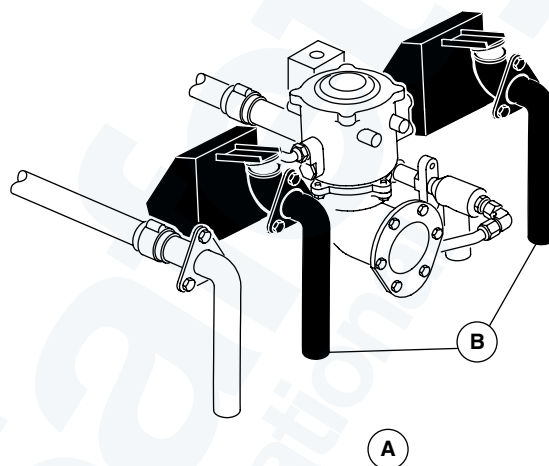
Refer to Figure 28-11. Vent System.

There are two outboard vent lines in each fuel tank connecting the main fuel tank to the surge bay. Each vent line is connected to a vent float valve through its own 1.0 in. (2.54 cm) hole in the tank closure rib at WS 407. The vent lines extend down to within 0.5 in. (12.7 mm) of the bottom of the surge bay. As the surge bay is pressurized in-flight, through the NACA vents, any fuel in the surge bay is scavenged by the outboard vent lines back into the main tank.



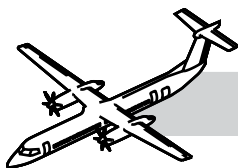


**NOTE**  
Left wing shown,  
right wing similar.



**Figure 28-12. Vent Float Valve and Standpipe**





## Vent Float Valves

Refer to:

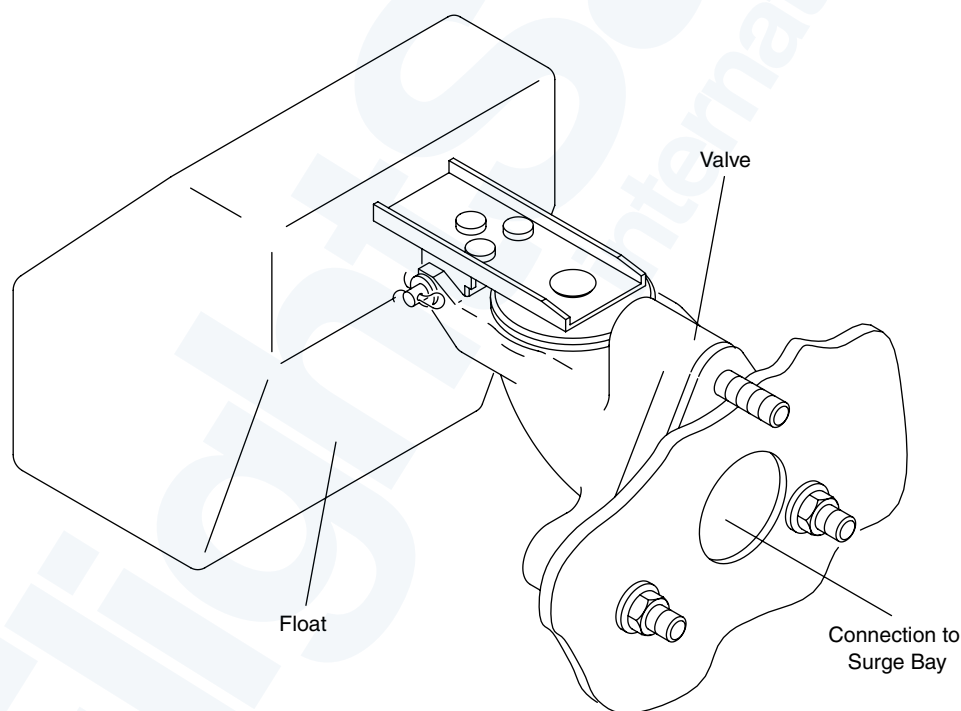
- Figure 28-12. Vent Float Valve and Standpipe.
- Figure 28-13. Vent Float Valve.

The purpose of the vent float valve is to prevent fuel from flowing into the surge bay.

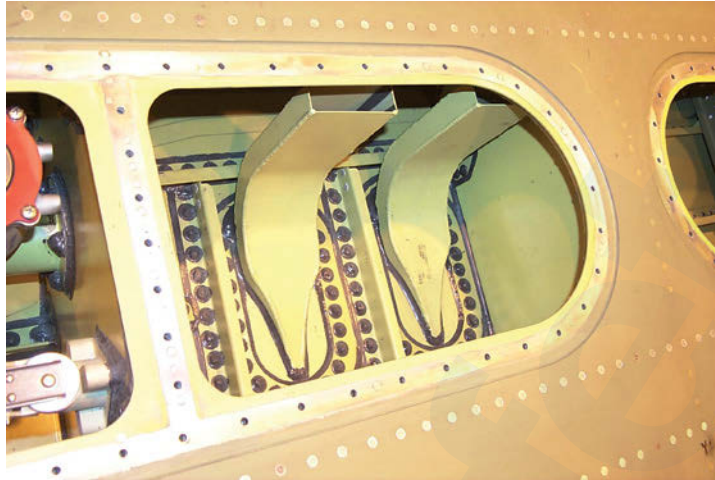
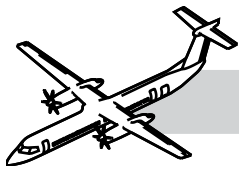
The vent float valve has a polyurethane foam float which closes the outboard vent line if the fuel level in the tank rises above the opening of the valve.

Two vent float valves are near the top of the rib dividing the main tank from the surge bay (WS 407).

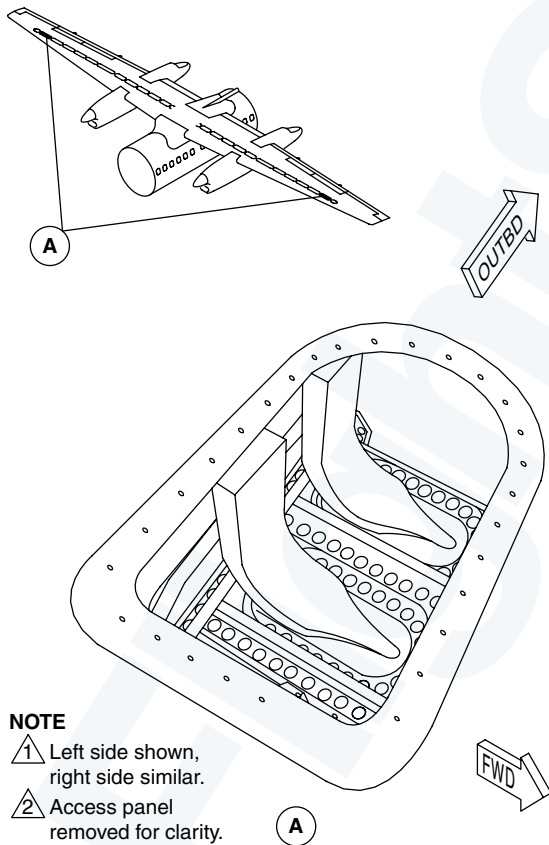
The vent float valve, on the forward part of the rib at WS 407, supplies venting during a climb. The vent float valve on the aft part of the rib at WS 407 supplies venting during a descent. The two vent float valves and the inboard vent line provide adequate venting for most flight attitudes.



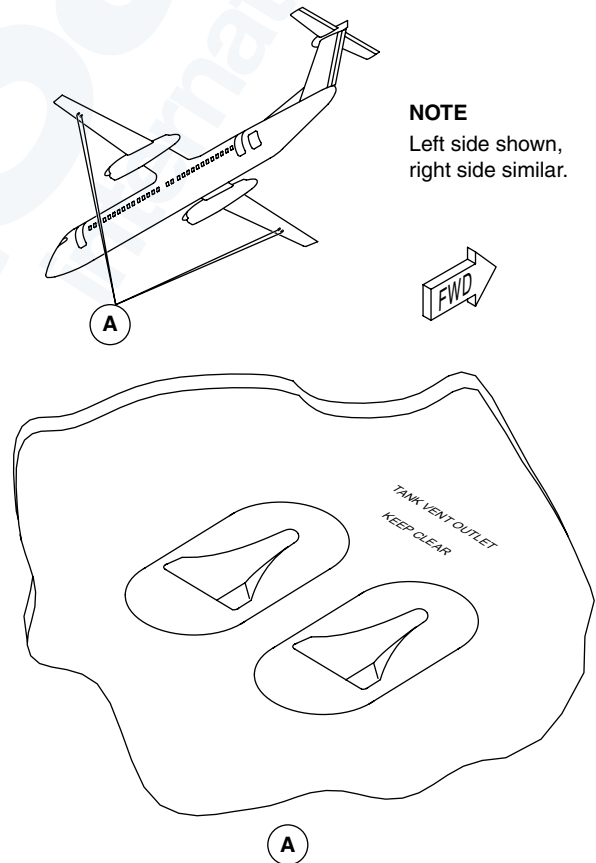
**Figure 28-13. Vent Float Valve**



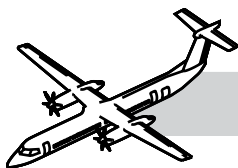
**Figure 28-14. Surge Bay NACA Vents**



**Figure 28-15. Surge Bay NACA Vents (Interior)**



**Figure 28-16. Surge Bay NACA Vents (Exterior)**



## OPERATION

## NOTES

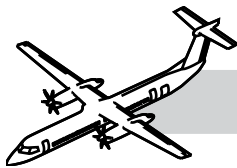
Refer to:

- Figure 28-11. Vent System.
- Figure 28-14. Surge Bay NACA Vents.
- Figure 28-15. Surge Bay NACA Vents (Interior).
- Figure 28-16. Surge Bay NACA Vents (Exterior).

The surge bay itself is vented through two NACA vents. The ram air vents keep a slight positive pressure in the fuel bay during flight. Pressurization of the surge bay (in-flight) scavenges fuel from the bay back into the main tank. The NACA vent standpipes prevent fuel in the surge bay from spilling overboard.

The refuel vent valves are opened with pressure fuel during refueling. If the tank overfills during refuel, the open vent valve dumps fuel into the surge bay. The refuel vent valve also releases tank pressure when it is more than 3.0 psi (20.7 kPa) above atmospheric pressure.

A refuel vent valve position switch sends a ground or open circuit signal to the refuel/defuel control panel for the DUMP VALVE OPEN (TANK1 or TANK2) indication and for refuel solenoid valve control during pressure refueling.



## 28-21-00 ENGINE FUEL FEED

### GENERAL

Refer to Figure 28-17. Engine Fuel Feed System.

The engine fuel feed system provides pressurized fuel from the tanks to the engines.

Each engine feed system consists of a primary ejector pump, an AC powered auxiliary pump and an emergency fuel shut-off valve.

The left and right engine feed systems operate independently of each other so that failure of one system does not cause the loss of the other system.

### SYSTEM DESCRIPTION

Each engine fuel feed system includes these components:

- Primary Ejector Pump
- Engine Feed Lines
- Inlet Strainer
- Engine Feed Shut-off Valve
- Auxiliary Pump
- Auxiliary Pump Pressure Switch.

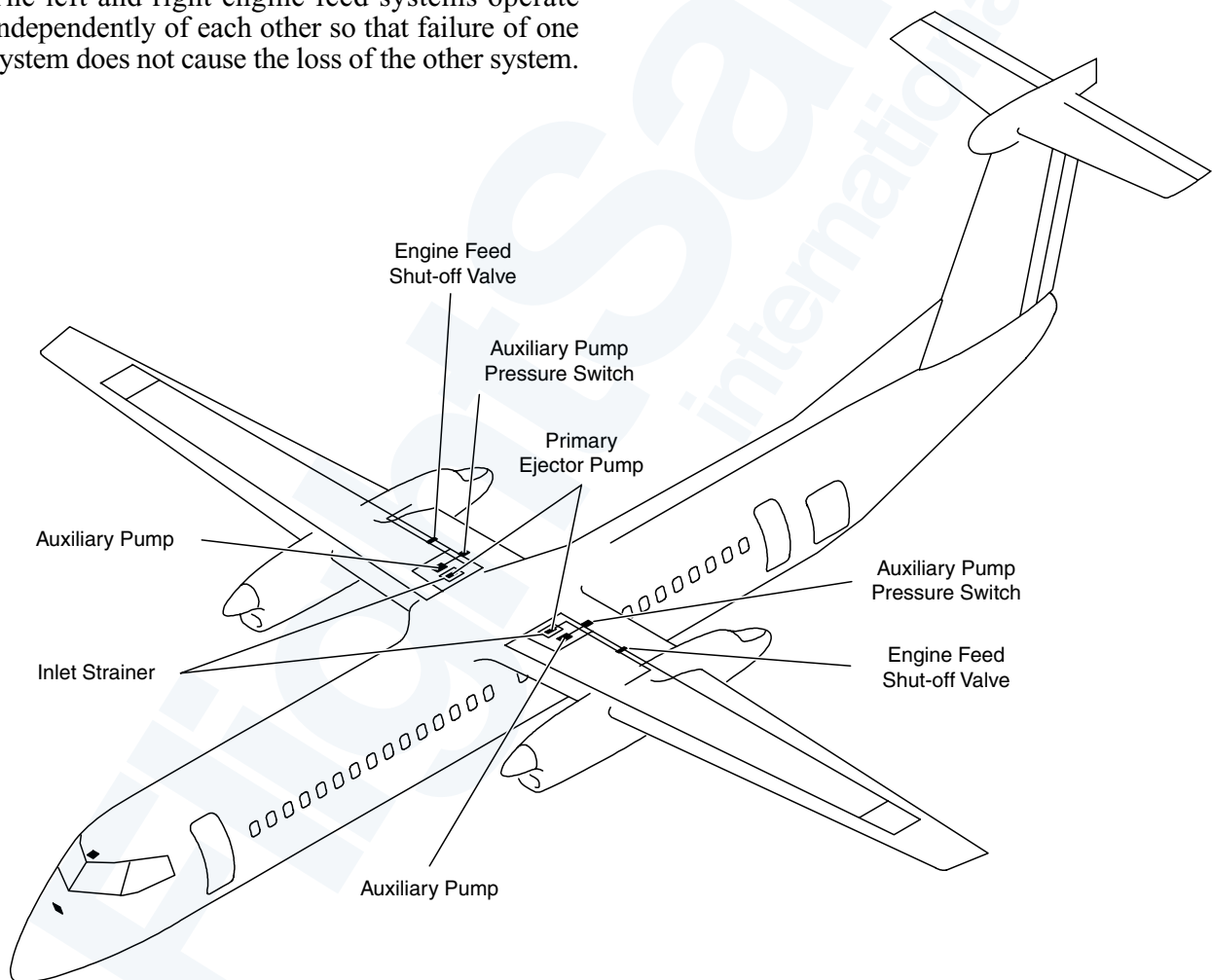
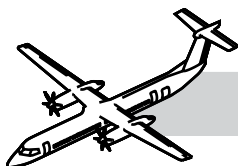


Figure 28-17. Engine Fuel Feed System



## COMPONENT DESCRIPTION

### Primary Ejector Pump

Refer to Figure 28-18. Primary Ejector Pump and Inlet Strainer.

There is one primary ejector pump in each collector bay.

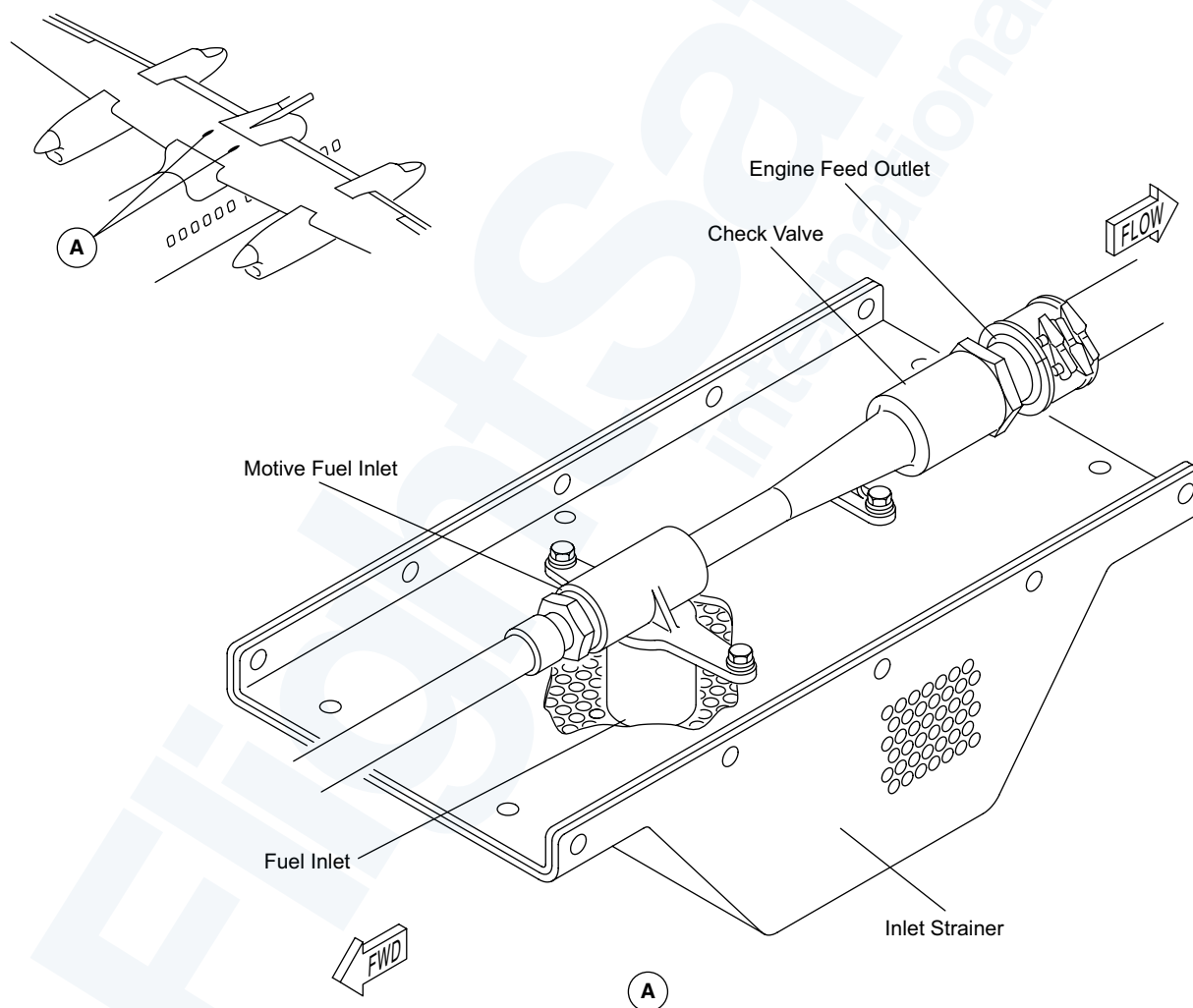
The function of the primary ejector pump is to make sure that there is positive pressure to the engine-driven pump to prevent the possibility of pump cavitation.

### Inlet Strainer

Refer to Figure 28-18. Primary Ejector Pump and Inlet Strainer.

The inlet strainers help prevent foreign objects from entering the fuel flow.

There are two inlet strainers; one for each primary ejector pump. Each inlet strainer is installed around the fuel inlet port of its related primary ejector pump.



**Figure 28-18. Primary Ejector Pump and Inlet Strainer**

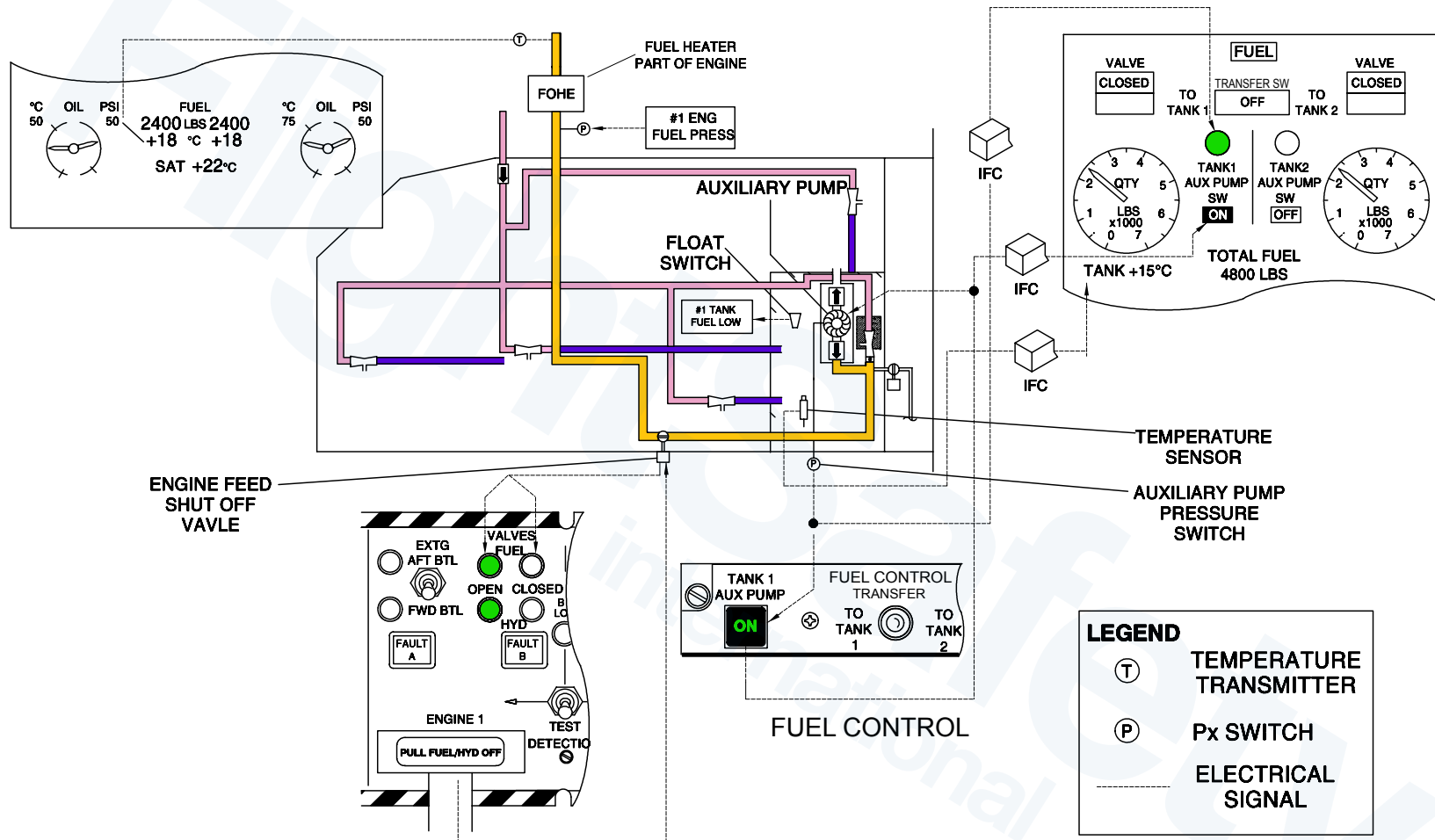
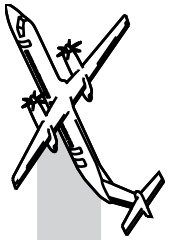
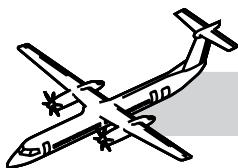


Figure 28-19. Engine Fuel Feed System



## Engine Feed Lines

Refer to Figure 28-19. Engine Fuel Feed System.

The engine feed lines transfer the fuel from the primary ejector pumps and/or the auxiliary pumps to the engines.

## Engine Feed Shut-Off Valve

Refer to Figure 28-20. Engine Feed Shut-Off Valve.

The valve is installed inside the tank on the rear spar.

The battery bus supplies the 28 VDC to the motor actuator which operates a ball valve in the engine fuel feed line. The electrical motor is installed on the dry side of the spar on a spline to the valve. The motor actuator has

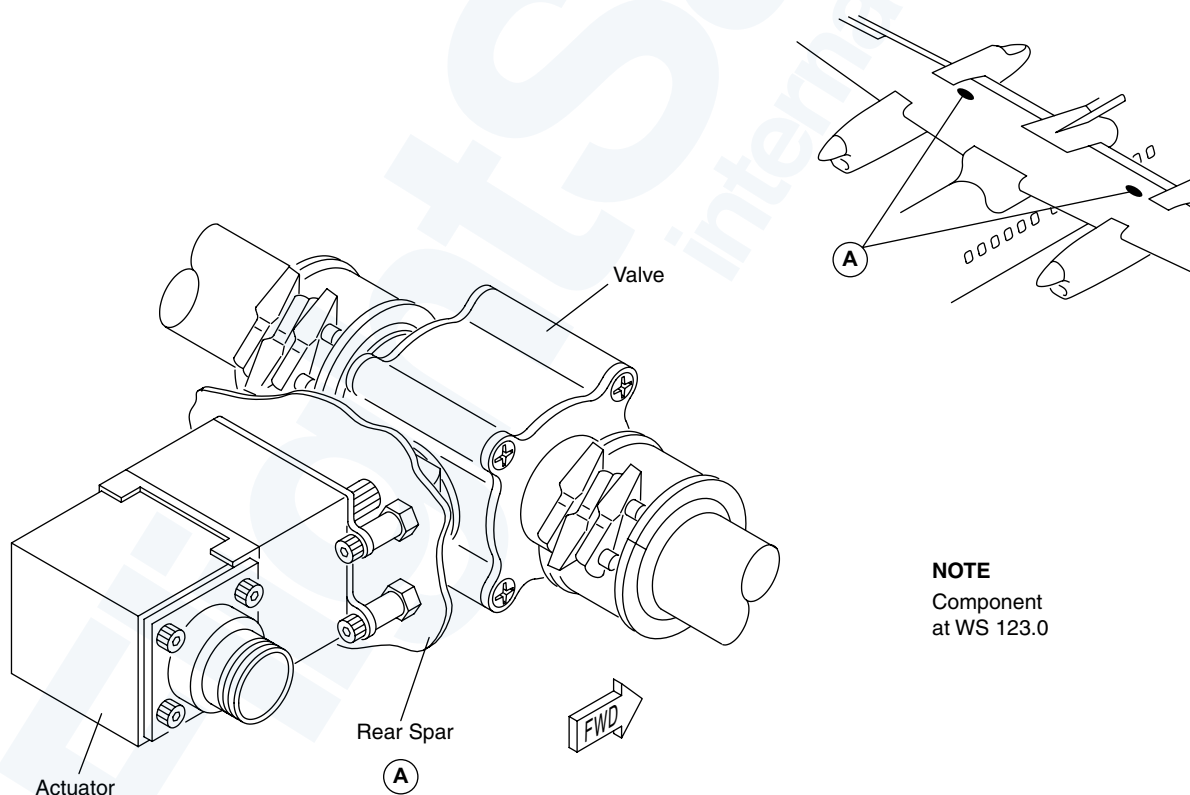
physical and electrical position indication. It is replaced without changing the valve or draining the fuel tank.

There are two thermal relief valves which open between 65 and 70 psig to protect the system from over pressurization when the valve is closed.

The engine feed shut-off valve is controlled and operated by the PULL FUEL/HYD OFF handle on the engine fire protection panel.

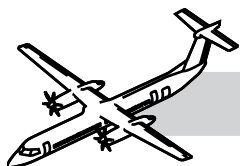
The valve position switches illuminate the green (OPEN) and white (CLOSED) FUEL VALVE lights on the Fire Protection Panel (FPP).

The red position lever on the drive shaft of the valve operates the position switches for the green and white lights on the FPP.



**Figure 28-20. Engine Feed Shut-Off Valve**





## Auxiliary Pump

Refer to Figure 28-21. Auxiliary Pump.

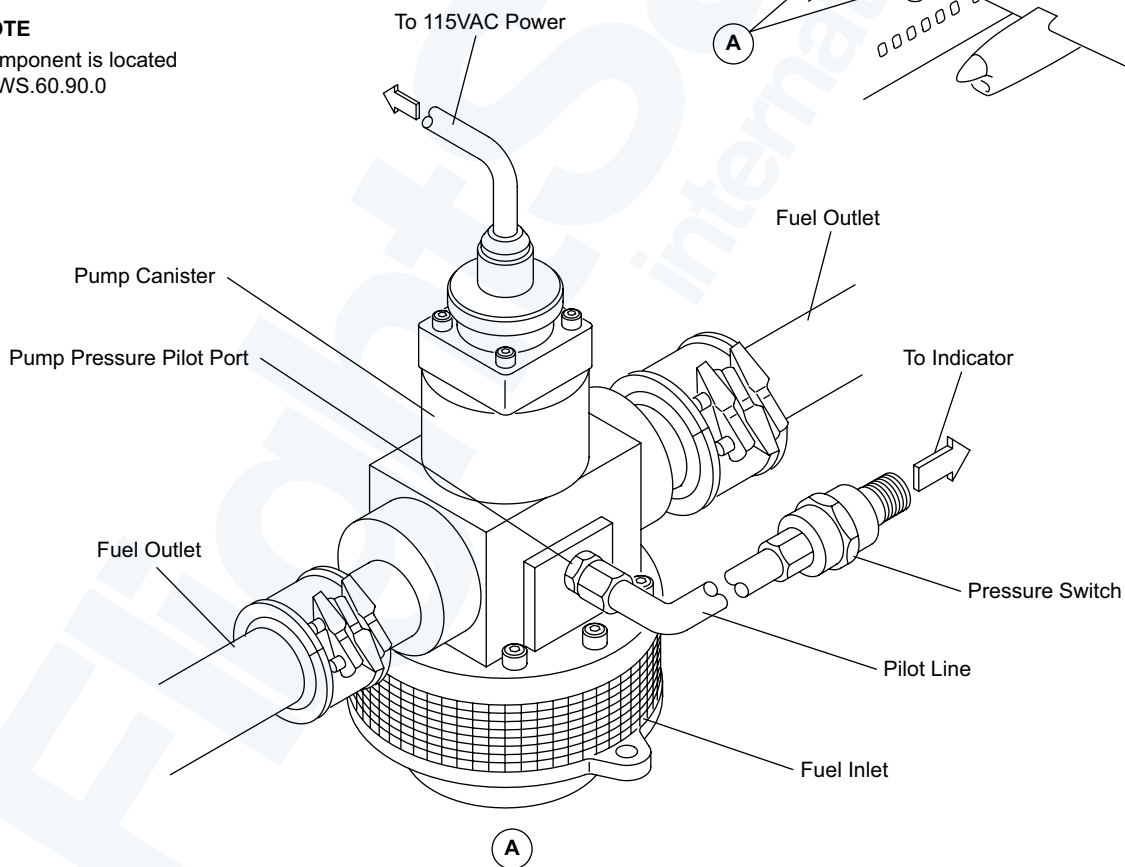
The 115 VAC auxiliary pump is installed on the wing lower skin inside the collector bay of each main tank, adjacent to the primary ejector pump.

The function of the auxiliary pump is to transfer fuel and to serve as a backup source for pressurized engine fuel feed if a primary ejector pump does not function properly.

The auxiliary pump is a submerged, electrically operated, centrifugal pump. It is connected in parallel with the primary ejector pump.

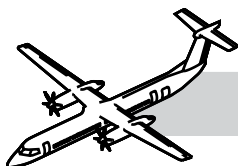
### NOTE

Component is located at WS.60.90.0



**Figure 28-21. Auxiliary Pump**





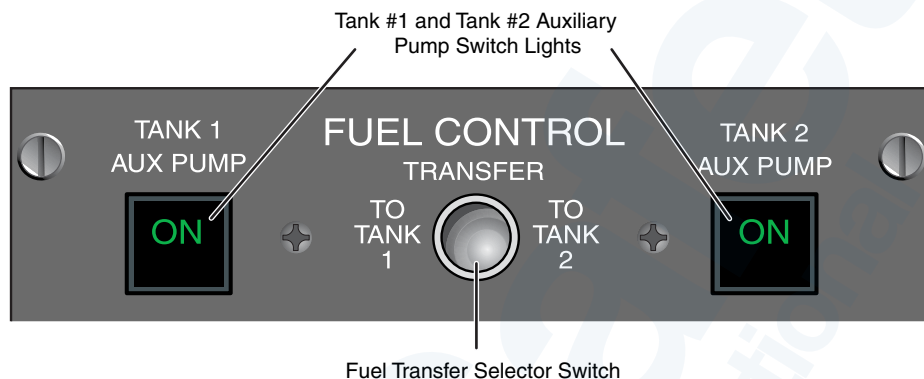
## Auxiliary Pump Pressure Switch

Refer to:

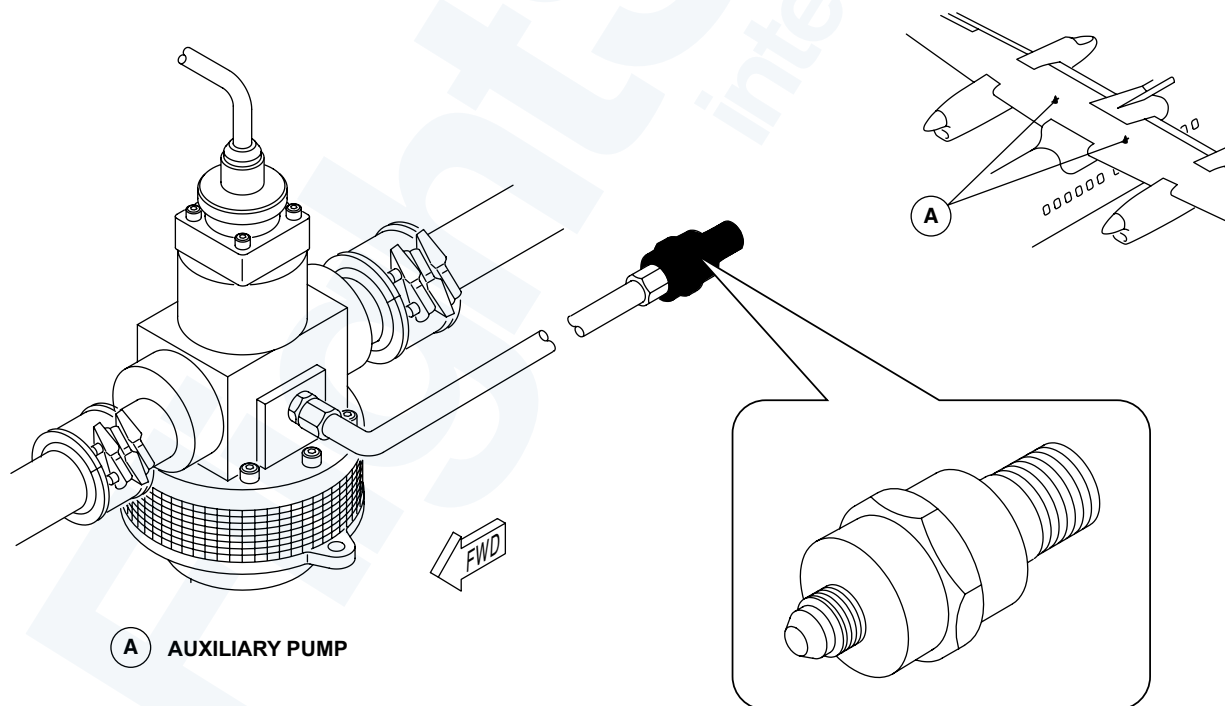
- Figure 28-22. Fuel Control Transfer Panel.
- Figure 28-23. Auxiliary Pump Pressure Switch.

The Auxiliary Pump Pressure Switch sends a signal to the FUEL CONTROL TRANSFER panel to indicate that auxiliary pump is operating, and to the FDPS for display on the fuel system page.

There are two auxiliary pump pressure switches, one connected to each auxiliary pump. They are on the wing rear spar.



**Figure 28-22. Fuel Control Transfer Panel**



**Figure 28-23. Auxiliary Pump Pressure Switch**

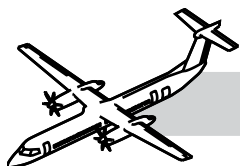
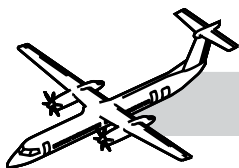


Figure 28-24. MFD Fuel Page



Refer to Figure 28-24. MFD Fuel Page.

## NOTES

The auxiliary pump pressure switches, through the IFC, will also display pump operation on the MFD Fuel page, in the form of green round lights.

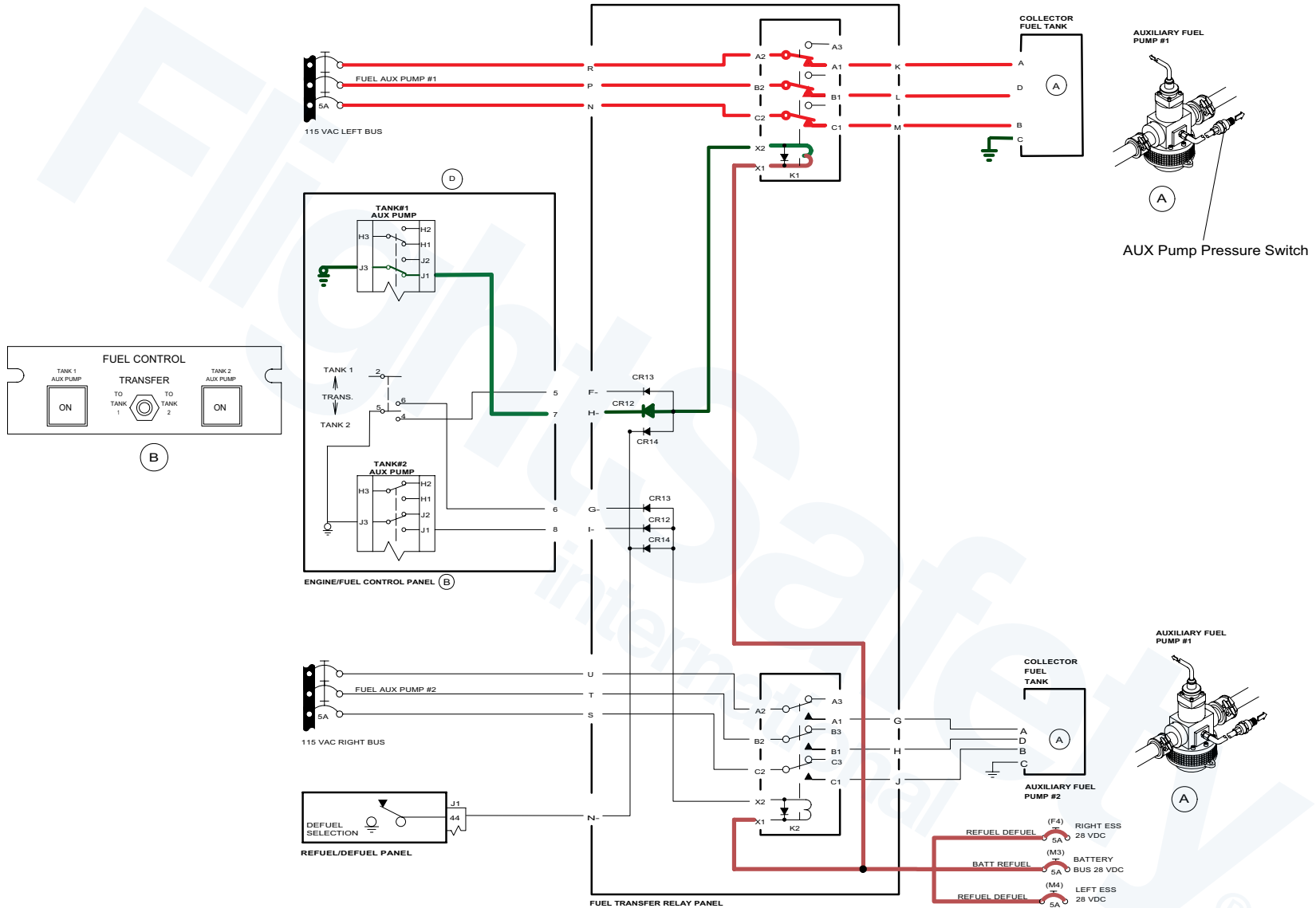
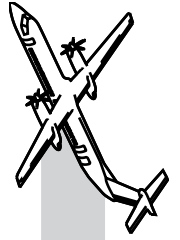
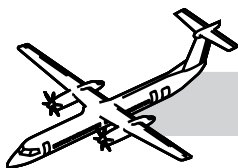


Figure 28-25. Engine Fuel Feed System Operation



## OPERATION

Refer to Figure 28-25. Engine Fuel Feed System Operation.

The Engine Fuel Feed system is supplied by gravity before engine start, then from the primary ejector pump during the engine run. The 115 VAC Auxiliary (AUX) pump can supplement the engine fuel supply when selected on.

### Primary Ejector Pump Operation

The primary ejector pump operates with motive flow fuel from an operating engine Fuel Metering Unit (FMU).

The engine high pressure pump maximum capacity is 5500 pounds per hour at 1200 psi (8274 kPa). The engine fuel consumption is between 125 pph to 2875 pph. The excess fuel is directed as the Motive Flow fuel to the wing fuel tank ejector pumps.

The primary ejector venturi receives high pressure-low volume fuel:

- The high pressure fuel through the venturi creates suction and draws in fuel from the tank
- The ejector pumps out low pressure-high volume fuel through the Engine Feed SOV directly to the engine.

### Auxiliary Pump Operation in Engine Feed Mode

The left fuel AUX tank operates with:

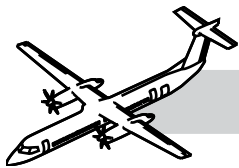
- A selection of the TANK 1 AUX PUMP switchlight to the ON position, providing a ground for relay K1
- Energizing K1 provides power from Left 115 VAC bus to the auxiliary pump.

The relay K1 at pin X1 is hot at all times from the three REFUEL circuit breakers (F4, M3 and M4).

## OPERATIONAL TEST OF THE ENGINE FEED SHUT-OFF VALVE

*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

- Energize the aircraft electrical system
- Pressurize No.1 hydraulic system
- Lower the flaps to 35 degrees
- Remove hydraulic pressure
- Pull the two PULL FUEL/HYD OFF handles on the FPP
- On the FPP make sure the white FUEL VALVES CLOSED lights come ON and the green OPEN lights go OFF
- At the wings rear spars check that the red indicator levers on the valves, point to CLOSED
- Push the PULL FUEL/HYD handles in to their normal positions
- Check that green FUEL VALVES OPEN lights are ON
- The white FUEL VALVE CLOSED lights are OFF
- The indicator levers on the valves point to the CLOSED position.



## 28-22-00 APU FUEL FEED

### GENERAL

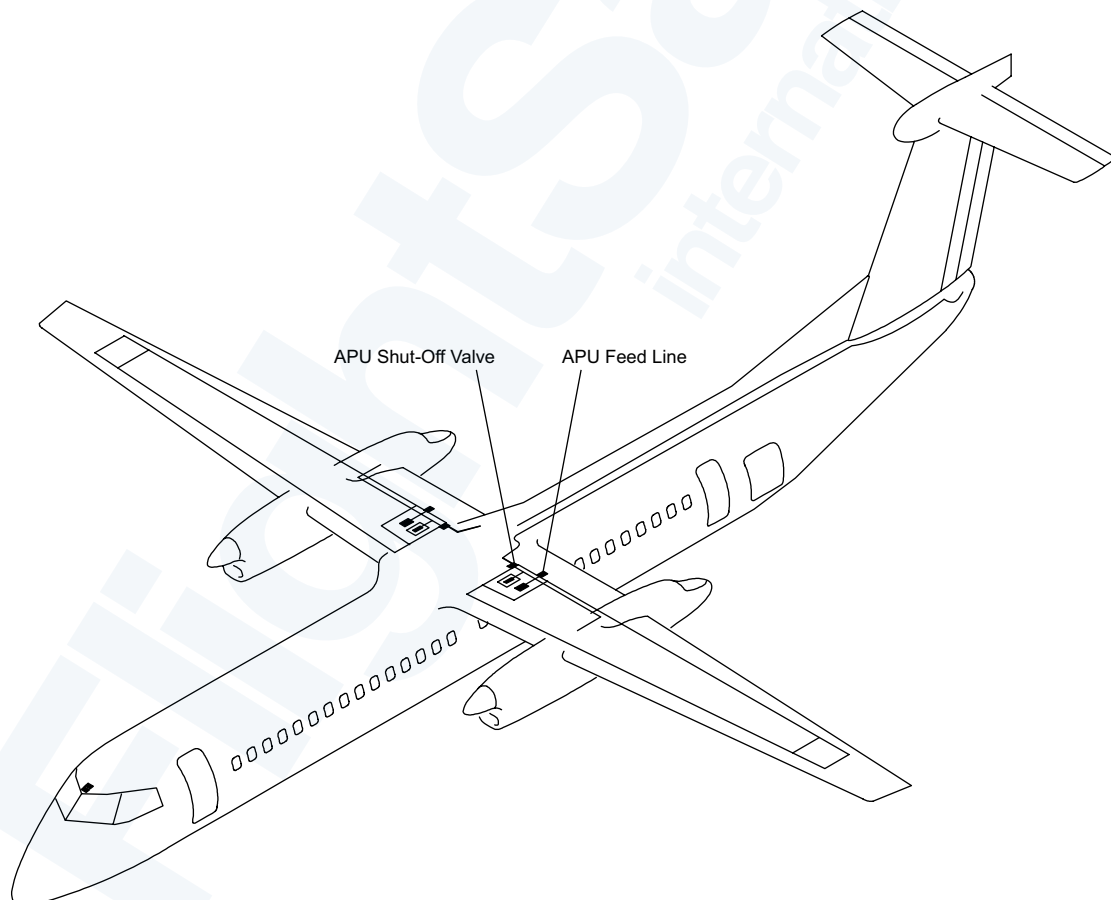
The APU fuel feed system supplies a flow of fuel from the left (No.1) fuel tank to the optional APU.

### SYSTEM DESCRIPTION

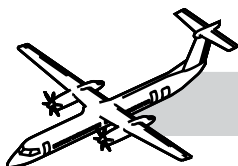
Refer to Figure 28-26. APU Fuel Feed Locator.

The APU fuel feed system includes these components:

- APU shut-off valve
- APU feed line.



**Figure 28-26. APU Fuel Feed Locator**



## COMPONENT DESCRIPTION

### APU Shut-Off Valve

Refer to Figure 28-27. APU Shut-Off Valve.

The APU shut-off valve gives a positive fuel stop in the supply line to the APU.

The APU shut-off valve closes and stops the fuel flow to the APU under these conditions:

- The APU PWR switchlight is selected OFF
- Fire is sensed in the APU/tailcone area
- The APU fire extinguisher switch (EXTG) on the FPP is selected ON
- The Weight on Wheels (WOW) switch is open (aircraft off ground).

The valve itself is an open/close ball type valve enclosed in a tee-shaped fitting inside the fuel tank.

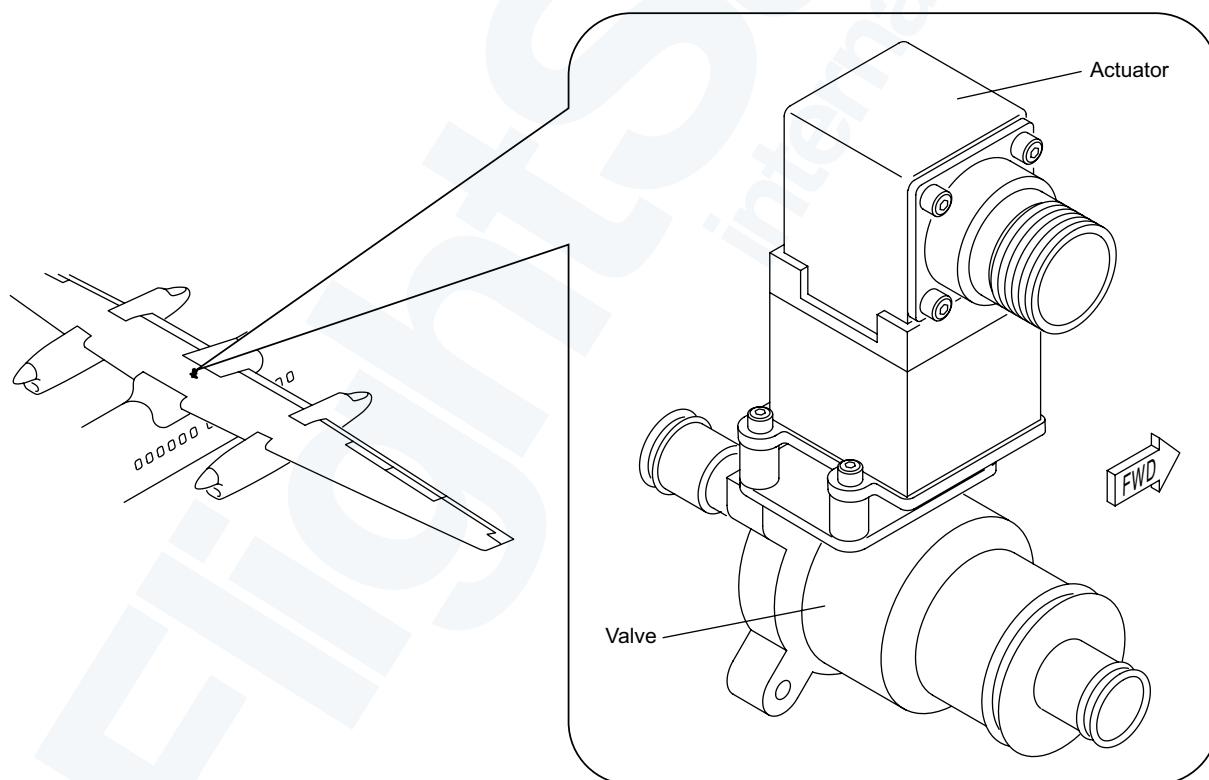
#### CAUTION

APU will shut down if the fire test button is pushed when APU is running.

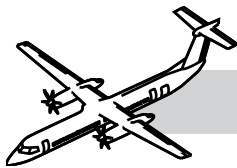
## OPERATION

Refer to:

- Figure 49-1 (49-30-00) APU Relay Panel in Q400 MSM
- Figure 49-2 (49-30-00) APU Relay Panel (K5 Energized) in Q400 MSM.



**Figure 28-27. APU Shut-Off Valve**



## **OPERATIONAL CHECK OF THE APU FUEL FEED SHUT-OFF VALVE**

*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

Energize the aircraft electrical system

For Method No.1 do the check as follows:

- Open and tag CB LEFT DC ESSENTIAL L3 APU FUEL SOV/IND
- Try to start the APU . If the APU fails to start then the APU FEED SOV is closed and operates properly

### **NOTE**

The APU may spin up momentarily, until the residual fuel in the manifold is consumed, then shut down. The APU FADEC may log a fault for the uncommanded shutdown/unsuccessful start.

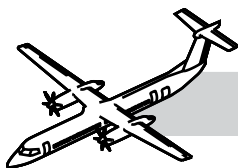
- Reset the CB.

For Method No.2 do the check as follows:

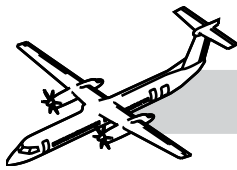
- Remove the access panel for the left center wing dry bay
- Push the PWR switchlight on the APU control panel
- In the left center wing dry bay make sure the red pointer on the APU fuel feed SOV is in the open position
- Make sure the APU fuel valve open green light on the FPP is on
- Push and hold the fire test switch on the FPP
- Make sure the red pointer on the sov moves to the closed position

- Make sure the APU FUEL VALVE CLOSED white light on the FPP is ON
- Release the FIRE TEST switch on the FPP
- Make sure the red pointer on the SOV is in the OPEN position
- Make sure the APU FUEL VALVE OPEN green light is ON
- Make sure the APU FUEL VALVE CLOSED white light is OFF
- Push the PWR switchlight on the APU control panel
- Re-install the access panel.





PAGE INTENTIONALLY LEFT BLANK



## 28-23-00 FUEL TRANSFER

### GENERAL

The fuel transfer system provides a means to transfer fuel from one wing tank to the other. The primary purpose of the system is to correct lateral imbalance between the two wing tanks.

In case of an in-flight shutdown, it allows the crew to use the remaining fuel from both tanks for sustained single engine operation. The system can empty one fuel tank for maintenance without defueling the aircraft.

### NOTE

Only tank to tank transfer is available: there is no engine crossfeed capability.

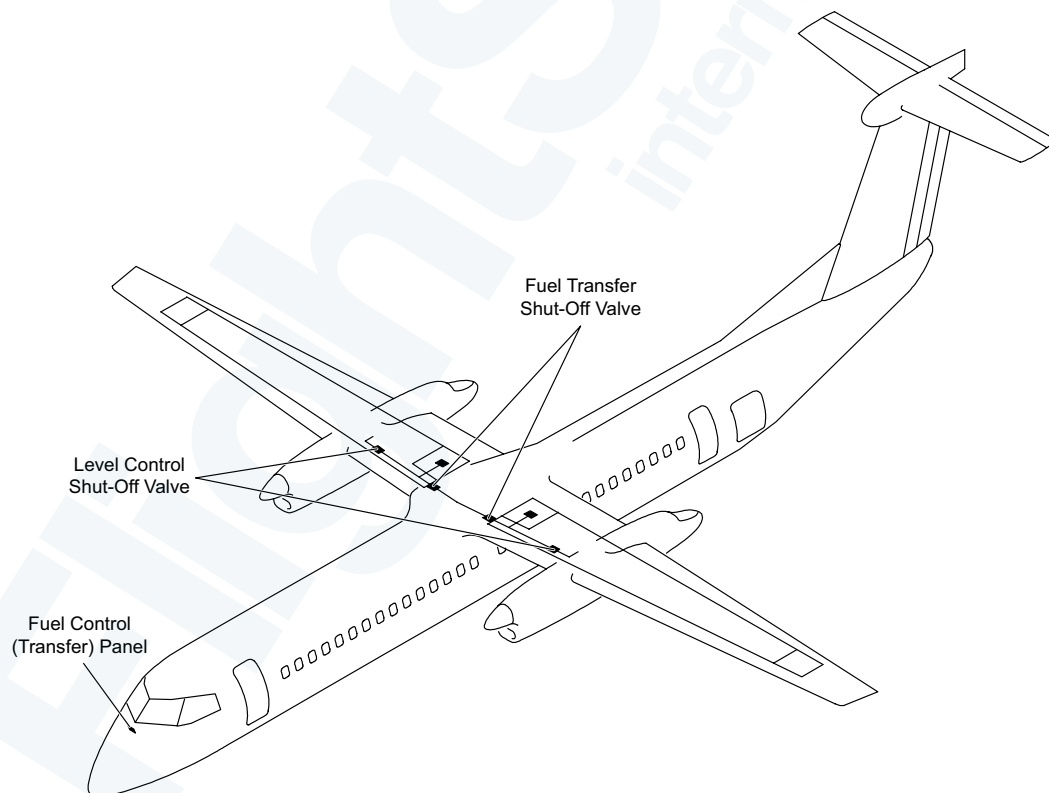
### SYSTEM DESCRIPTION

Refer to Figure 28-28. Fuel Transfer System.

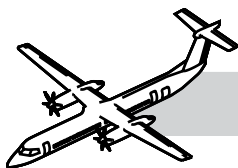
The fuel transfer system includes these components:

- Fuel Transfer Shut-Off Valves
- Level Control Shut-Off Valves
- Level Control Solenoids
- Fuel Control Transfer Panel.

The fuel transfer system is controlled by a three position TRANSFER switch on the FUEL CONTROL TRANSFER panel in the flight compartment.



**Figure 28-28. Fuel Transfer System**



## COMPONENT DESCRIPTION

### Refuel/Defuel and Transfer Shut-Off Valve

Refer to Figure 28-29. Refuel/Defuel and Transfer Shut-Off Valve.

The shut-off valves open to allow refueling, defueling or fuel transfer.

There is one valve in each wing. The valve is on the tank closure rib at WS 42.

The valve consists of two parts:

- The valve body
- The actuator.

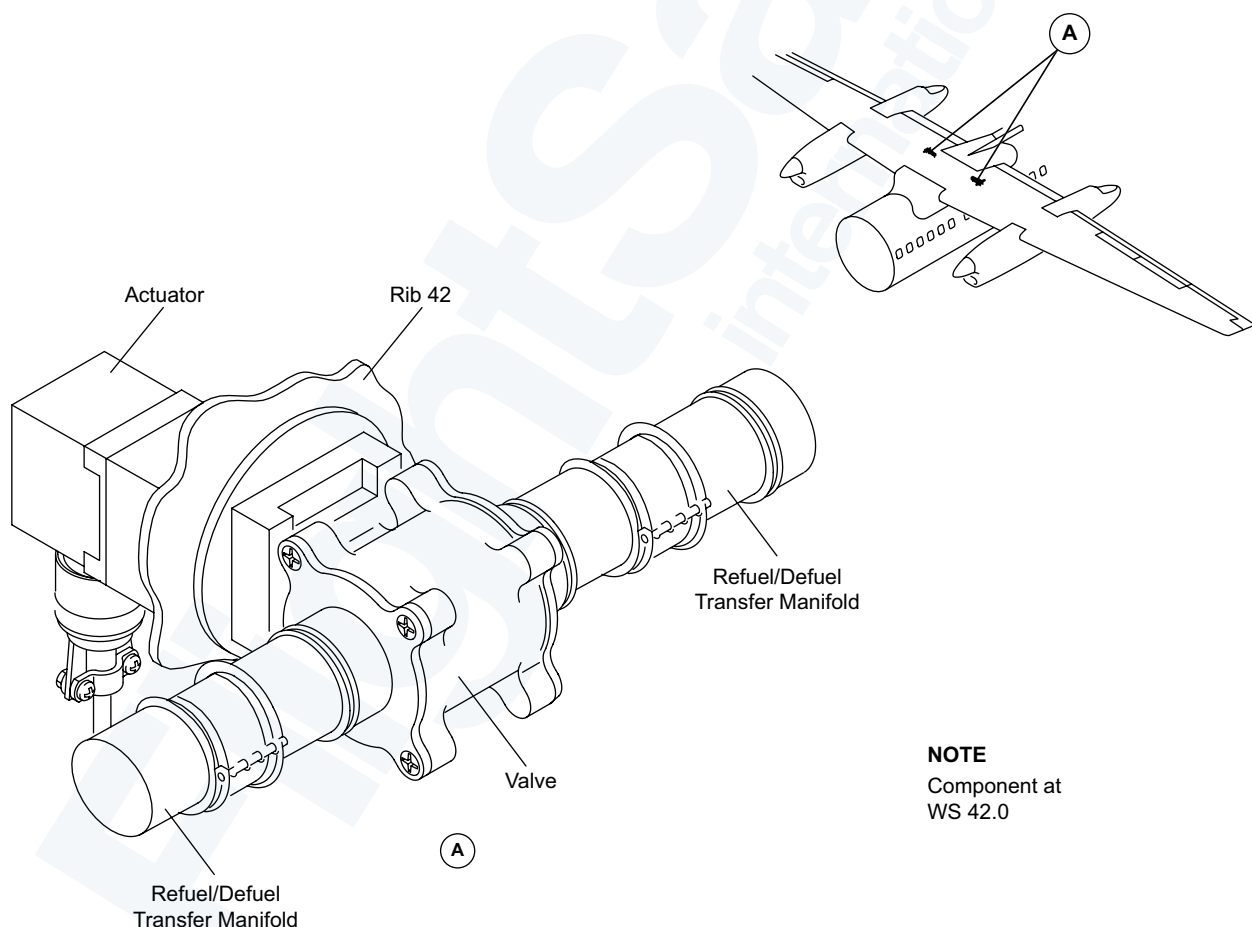
The actuator opens and closes the valve and the valve itself is an open/close ball type valve enclosed in a T-shaped fitting inside the fuel tank.

### Level Control Shut-Off Valve

The level control shut-off valve lets fuel flow during transfer and refuel operations. It also acts as the primary shut-off device for fuel flow into the tank.

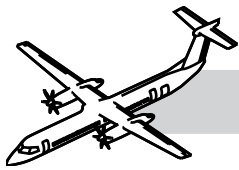
There is one level control shut-off valve in each wing.

The valves are hydro-mechanically controlled by the level control solenoids.



**NOTE**  
Component at  
WS 42.0

**Figure 28-29. Refuel/Defuel and Transfer Shut-Off Valve**

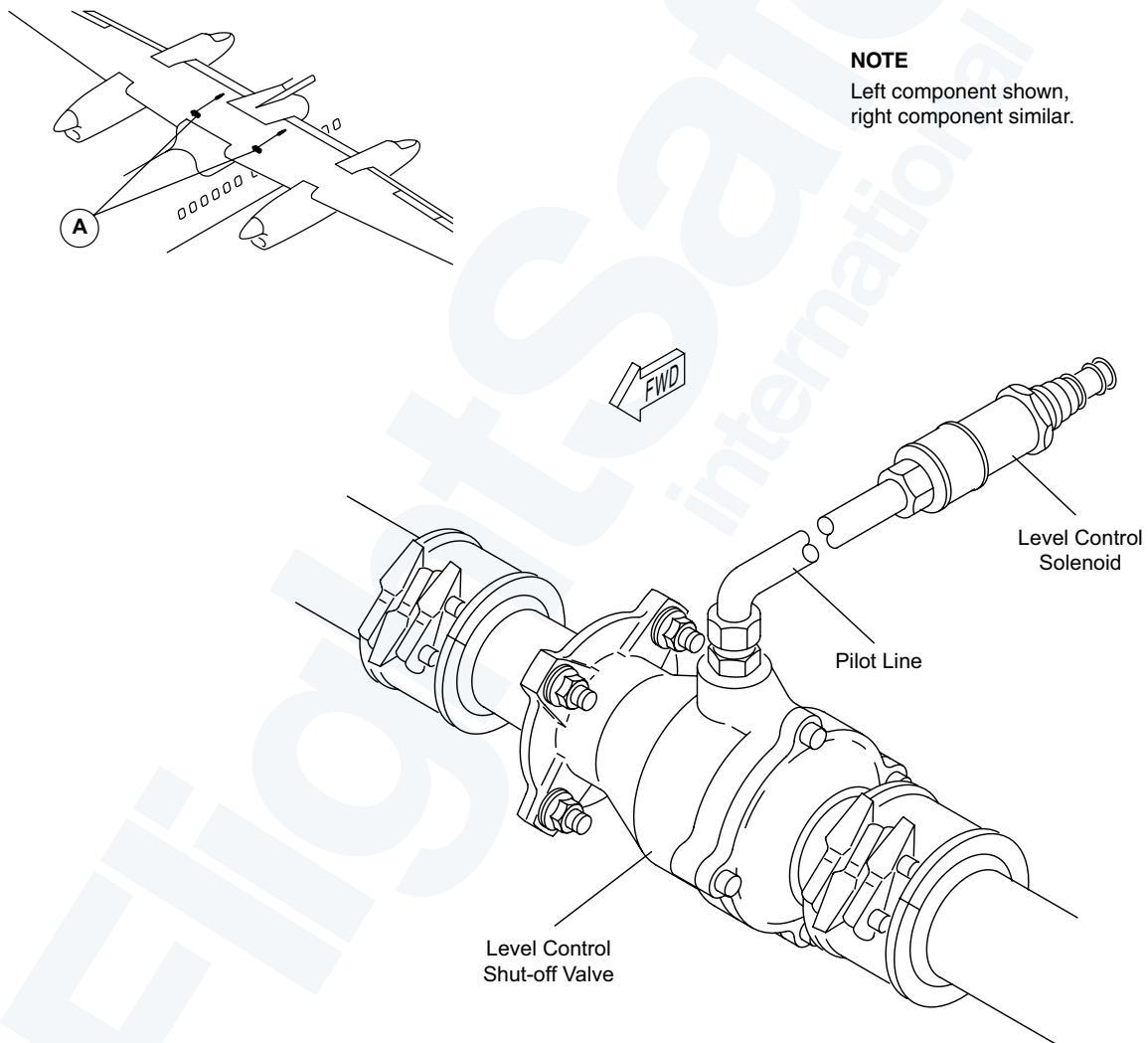


## Level Control Solenoid

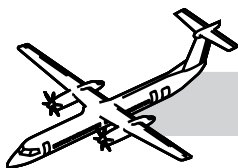
Refer to Figure 28-30. Level Control Solenoid.

The level control solenoid controls the opening and closing of the level control Shut-Off Valve (SOV).

There is one level control solenoid in each wing. It is on the rear wing spar.



**Figure 28-30. Level Control Solenoid**



## Fuel Transfer Panel

Refer to Figure 28-31. Fuel Transfer Schematic.

The FUEL CONTROL TRANSFER panel is in the flight compartment on the engine instrument panel.

The FUEL CONTROL TRANSFER panel has a TRANSFER switch and two switchlights. The TRANSFER switch is used to set the direction of fuel transfer (TO TANK 1 or TO TANK 2). A pressure signal from an operating auxiliary pump causes the ON switchlight segment to turn green. Fuel transfer indications are shown on the FUEL CONTROL TRANSFER panel and the Fuel Page of the MFD.

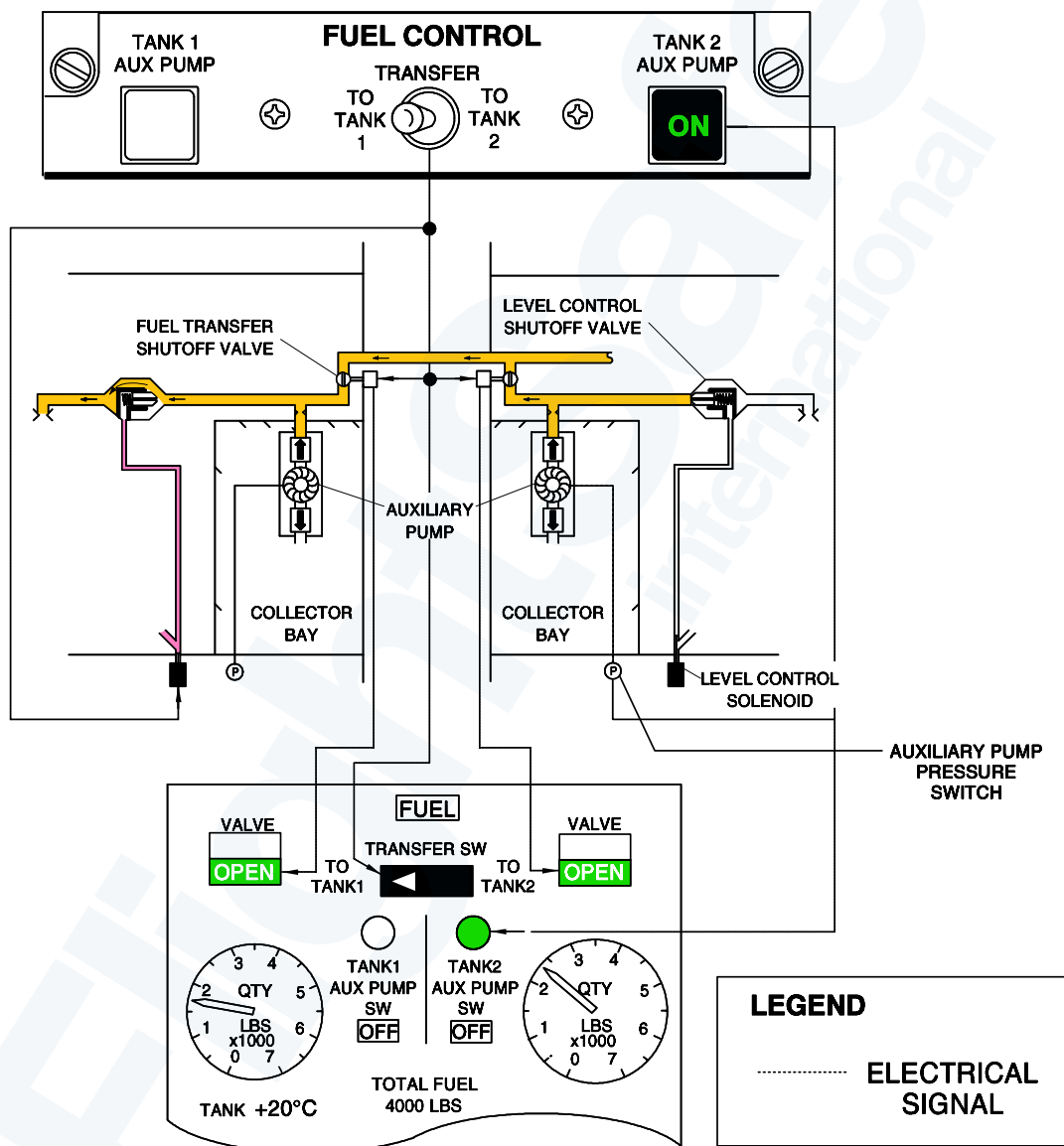
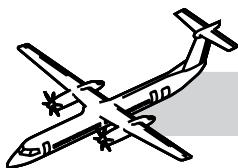


Figure 28-31. Fuel Transfer Schematic





## OPERATION

Refer to:

- Figure 28-32. Fuel Transfer Operation (Sheet 1 of 3).
- Figure 28-33. Fuel Transfer Operation (Sheet 2 of 3).
- Figure 28-34. Level Control Valve Operation.
- Figure 28-35. Fuel Transfer Operation (Sheet 3 of 3).

Fuel transfer in the aircraft is controlled through the FUEL TRANSFER panel and the High Level Control Unit (HLCU). The fuel transfer system shares some of its components with the refuel/defuel system. The fuel is transferred by an Auxiliary (AUX) pump in the donor tank. (The operation of fuel transfer to tank 1 is described; the operation of fuel transfer to tank 2 is similar).

The selection of the TRANSFER switch to the TO TANK 1 position on the FUEL CONTROL panel provides ground signals from the TRANSFER switch to close the relays in the Fuel Transfer relay panel that follow:

- K2 (which energizes No.2 Aux pump), shown on Sheet 3
- K3 (which causes the Level Control solenoid to open)
- K5 (which opens the No.1 Refuel/Defuel/Transfer Shut-Off Valve), shown on Sheet 1
- K6 (which opens the No.2 Refuel/Defuel/Transfer Shut-Off Valve), shown on Sheet 1
- K7, (which energizes the level control solenoid valve, from HLCU, shown on Sheet 2
- K8 (which energizes No.2 Aux pump), shown on Sheet 2
- K10 (which energizes HLCU), shown on Sheet 1.

The relay K2 receives 28 VDC from the left essential bus and a ground signal from the TRANSFER switch through diode CR16. The closed relay sends three phases AC from the right AC bus to the No.2 Aux Pump which causes the Aux Pump to operate.

The rise in Aux Pump output fuel pressure closes the auxiliary pump pressure switch. The closed auxiliary pump pressure switch supplies a ground signal to the IOM No.1 and the Advisory Control Unit (ACU) (Refer to 28-21-00 Engine Fuel Feed).

The ground signal to the IOM No.1 causes:

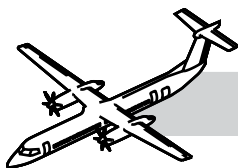
- The green TANK 2 AUX PUMP indicator on the ESID fuel page to illuminate
- The No.2 ENG FUEL PRESS caution light to extinguish.

The ground signal to the ACU causes:

- The right Aux Pump ON light on the FUEL TRANSFER panel to illuminate.







The relay K3 receives 28 VDC from the FQC and a ground signal from the TRANSFER switch which energizes the relay closed. The closed relay sends 28 VDC from the FQC to the tank 2 Level Control solenoid which causes the solenoid to energize open. The open solenoid allows the Level Control valve back pressure to be bled out which allows the Aux pump pressure to open the Level Control valve and transfer the fuel.

The closed relay K7 supplies the ground signal from the TRANSFER switch to energize the relay K6 closed. The relay K5 receives 28 VDC from the left essential bus and a ground signal from the TRANSFER switch to energize the relay K5 closed.

The closed relays K6 and K5 send 28 VDC (valve open) signals from the Fuel Quantity Computer (FQC) to the two Refuel/Defuel/Transfer SOV pins "D" which energize open the SOV and allows the fuel to transfer.

Relay K10 receives 28 VDC from the left essential bus and a ground signal from the TRANSFER switch. The closed K10 relay provides 28 VDC from the left essential bus to energize the HLCU at pin 3. The not-full signal from the HLCU supplies ground signals to close the relays K7 and K8. (Refer to 28-24-00 Refuel/Defuel).

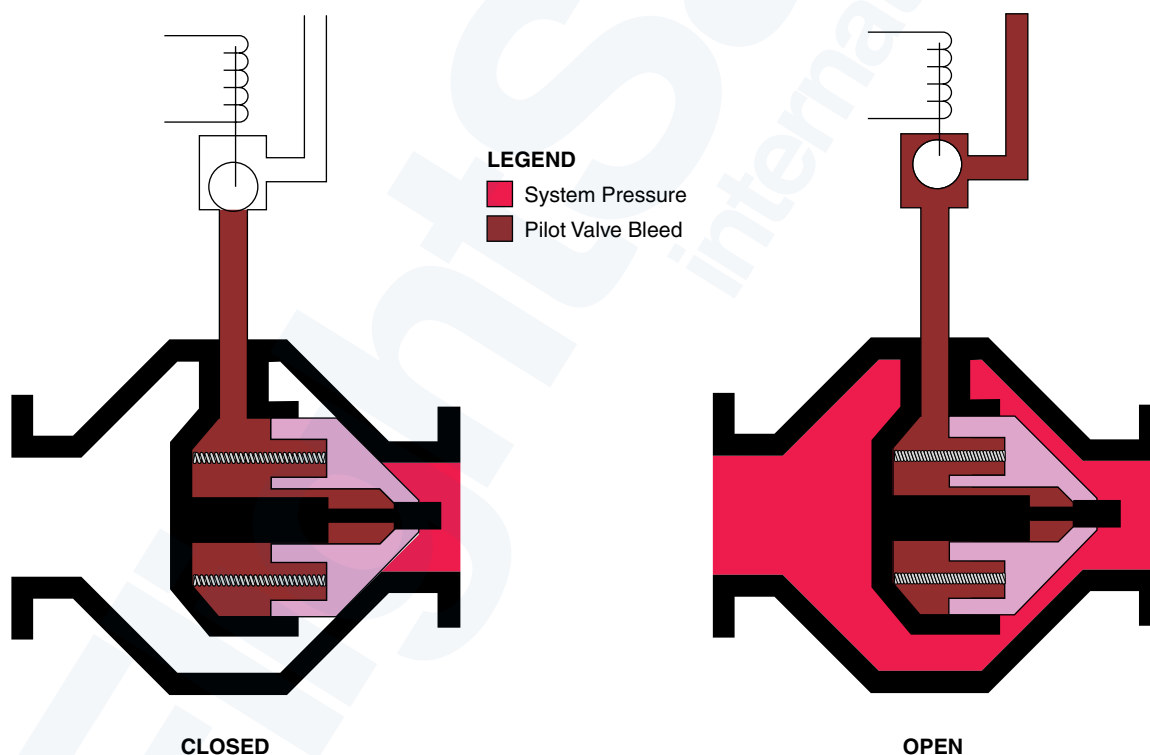


Figure 28-34. Level Control Valve Operation

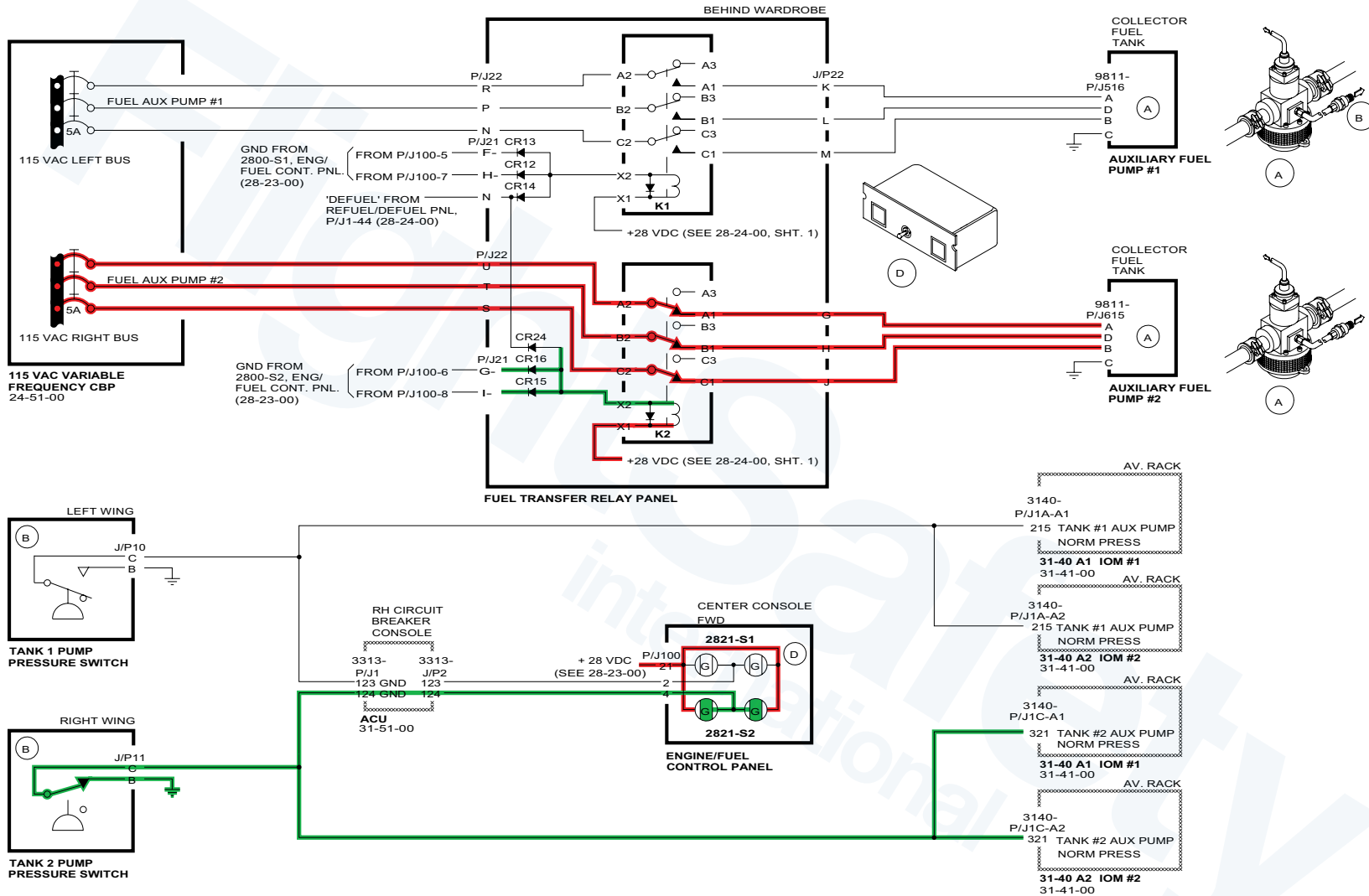
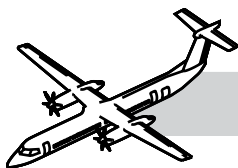


Figure 28-35. Fuel Transfer Operation (Sheet 3 of 3)



The fuel transfer will continue until the TRANSFER switch is set to the center position or the High Level Control Unit in the receiver tank senses a full condition.

If the tank becomes full, the high level sensor removes the ground signal from the relays K7. This de-energizes the Fuel Solenoid Valve which causes the receiving tank Level Control Valve to close and stop the flow.

When K7 de-energizes this also removes the ground from another relay K6, this relay controls the Refuel/Defuel/Transfer SOV. So the Refuel/Defuel/Transfer SOV closes, as well as the Level Control Solenoid Valve, when the HLCU senses high level in that tank.

## **OPERATIONAL CHECK OF THE FUEL TRANSFER SYSTEM**

*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

Obey all the fuel system safety precautions

Energize the aircraft DC and AC electrical systems

Make sure there is 500 to 1000lbs of fuel in each tank

Do the operational check as follows:

- Select the MFD to the FUEL page
- Select the TRANSFER switch on the FUEL CONTROL panel to the TO TANK 1 position.

Check as follows:

- The fuel transfer valves indications on the FUEL page change from CLOSED to OPEN
- The green ON indication comes on in the TANK 2 AUX PUMP switchlight

- The solid green circle for No.2 Aux Pump on the MFD FUEL PAGE comes ON
- The fuel quantity increases in tank No.1
- The fuel quantity decreases in tank No.2
- Transfer approximately 200 lbs of fuel from tank No.2 to tank No.1
- Select the TRANSFER switch on the FUEL CONTROL panel to the OFF position.

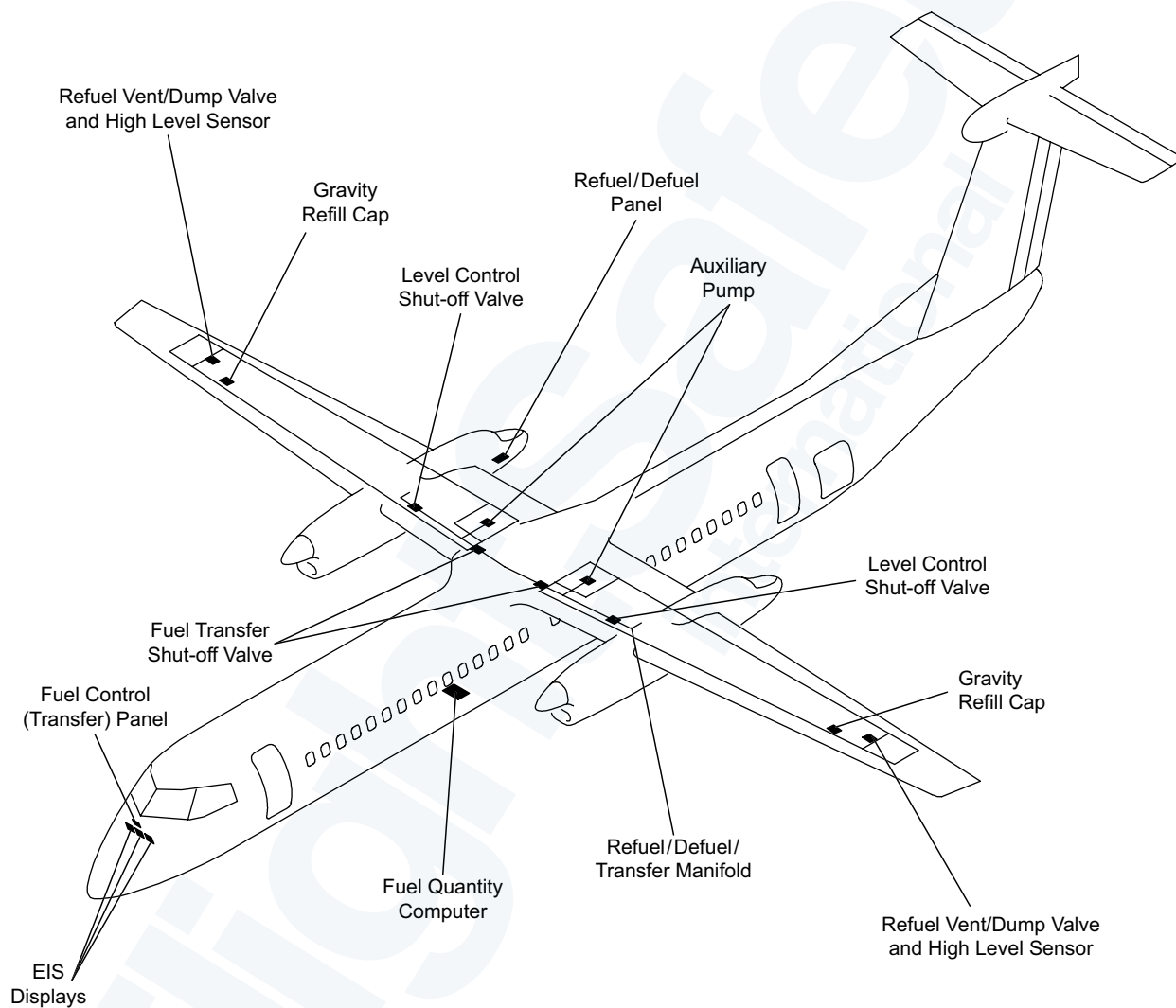
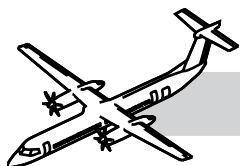
Check as follows:

- The fuel transfer valves positions on the FUEL page change from OPEN to CLOSED
- The TANK 2 AUX PUMP switchlight goes blank
- The solid green circle for No.2 Aux pump on the MFD changes to a white circle outline
- The fuel transfer stops.

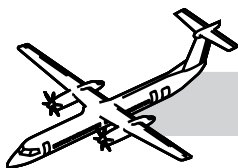
**REPEAT THE ABOVE STEPS FOR TRANSFER FROM TANK No.1 TO TANK No.2**

On completion make sure the fuel is evenly distributed between the tanks.

De-energize the aircraft electrical systems.



**Figure 28-36. Refuel/Defuel System**



## 28-24-00 REFUEL/DEFUEL

NOTES

### GENERAL

The refuel/defuel system is used for pressure refueling or defueling through a single point adapter. Gravity refuel of an individual tank is also possible.

### SYSTEM DESCRIPTION

Refer to Figure 28-36. Refuel/Defuel System.

The aircraft is refueled or defueled through the single point pressure refueling/defueling adapter in the aft portion of the right engine nacelle.

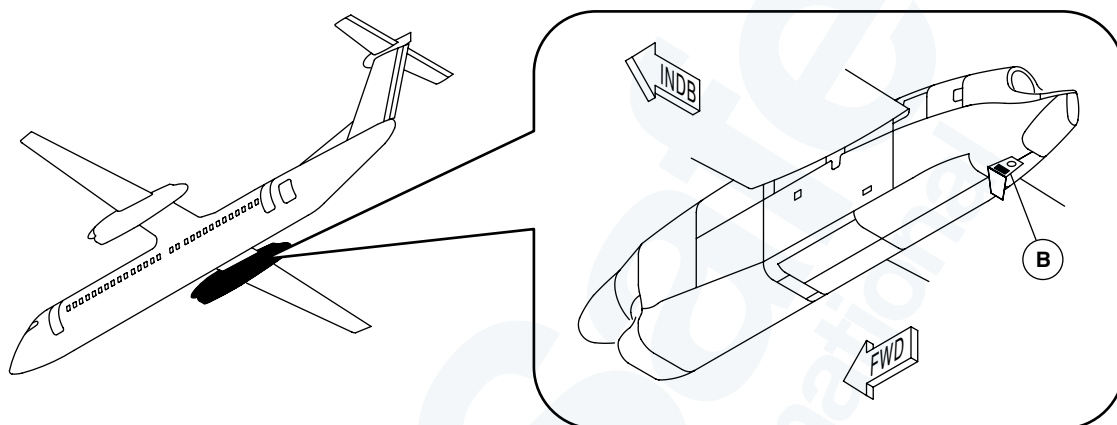
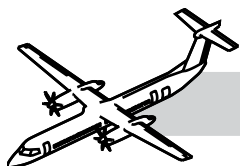
Both automatic and manual refueling/defueling modes are available. Selected and actual fuel tank quantities are displayed on the refuel/defuel control panel in the right engine nacelle.

The FQC controls the logic of the automatic refueling and defueling process.

Defueling can be done by external suction pressure, or by use of the auxiliary pumps in each collector bay.

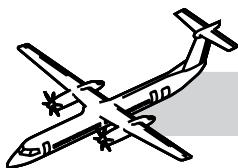
The refuel/defuel system has these components:

- Refuel/Defuel Control Panel
- Refuel/Defuel Indicator
- Refuel/Defuel Shut-Off Valve
- Refuel/Defuel Shut-Off Valve Actuator
- Refuel/Defuel Adapter
- No-Flow Pressure Switch
- Refuel Vent Valve
- High Level Sensor
- FQC.



**(B)**

**Figure 28-37. Refuel/Defuel Control Panel**



## COMPONENT DESCRIPTION

## NOTES

### Refuel/Defuel Control Panel

Refer to Figure 28-37. Refuel/Defuel Control Panel.

The purpose of the refuel/defuel control panel is control pressure refueling and defueling operations.

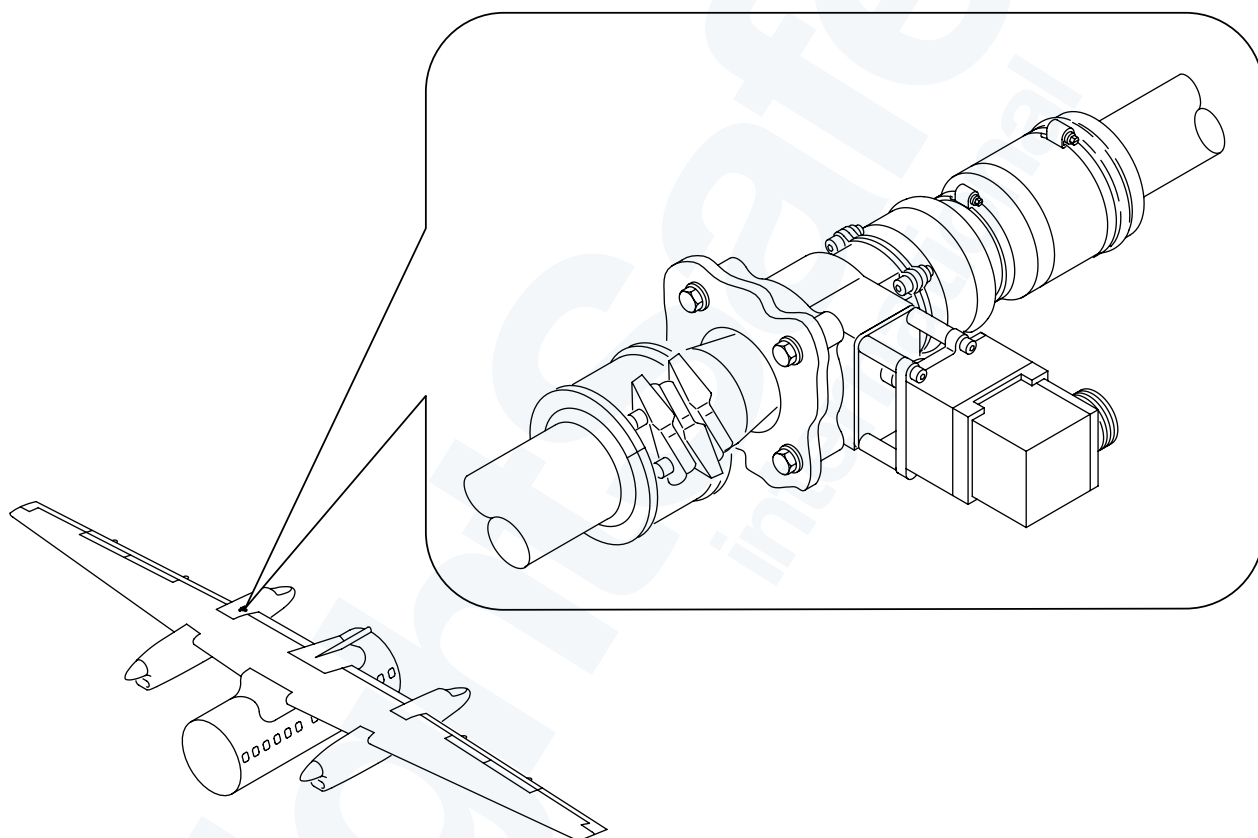
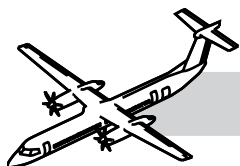
### Refuel/Defuel Indicator

Refer to Figure 28-37. Refuel/Defuel Control Panel.

The Refuel/Defuel Indicator (RDI) is on the refuel/defuel control panel.

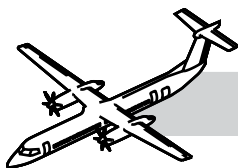
The Refuel/Defuel indicator shows the quantity of fuel:

- In each wing
- Selected in the automatic mode for refueling
- Selected for each individual wing in the manual mode for refueling



**Figure 28-38. Refuel/Defuel Shut-Off Valve**





## Refuel/Defuel Shut-Off Valve

## NOTES

Refer to Figure 28-38. Refuel/Defuel Shut-Off Valve.

The refuel/defuel shut-off valve is installed in the fuel inlet manifold. The valve is on the right side aft wing spar. The motor actuator can be replaced without changing the SOV or draining the fuel tank. A thermal relief valve is installed in the SOV to prevent excessive pressure in the fuel lines.

The essential battery bus supplies the 28 VDC to the motor actuator which operates the valve.

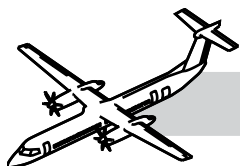
The valve opens when the aircraft is refueled or defueled and is controlled by the five position rotary selector switch on the refuel/defuel panel.

When the selector switch is in the OFF position, the valve is closed.

The valve is open when the selector switch is in any of the other four positions.

### NOTE

The valve will close when the refuel/defuel panel access door is closed, regardless of the position of the rotary selector switch.

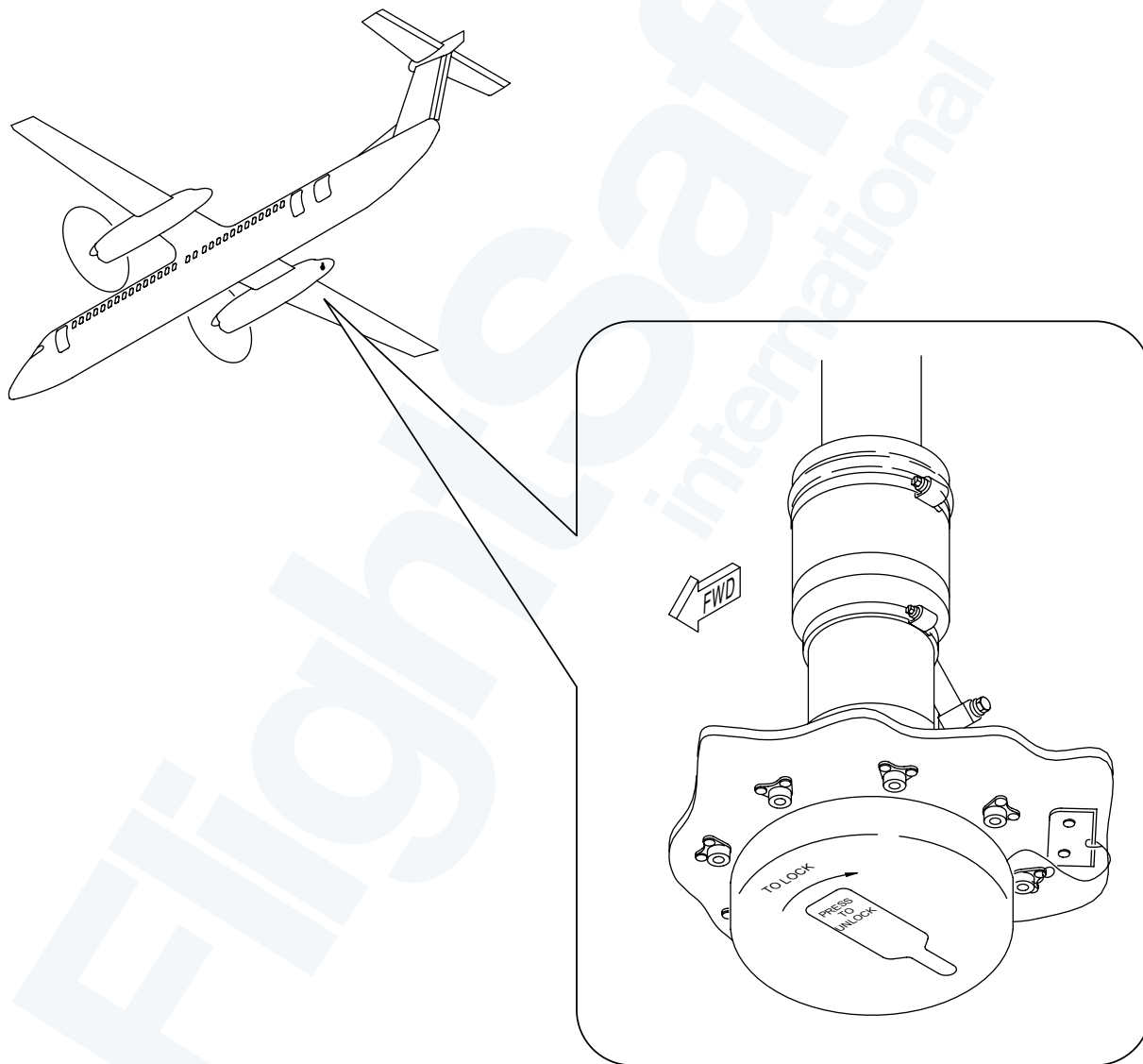


## Refuel/Defuel Adapter

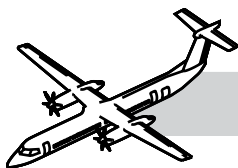
Refer to Figure 28-39. Refuel/Defuel Adapter.

The refuel/defuel adapter gives single point pressure refuel/defuel access for the aircraft.

The adapter has a metal body that has a spring-loaded poppet valve which closes after the fuel nozzle is removed. A circular cap protects the poppet valve from damage and contamination.



**Figure 28-39. Refuel/Defuel Adapter**

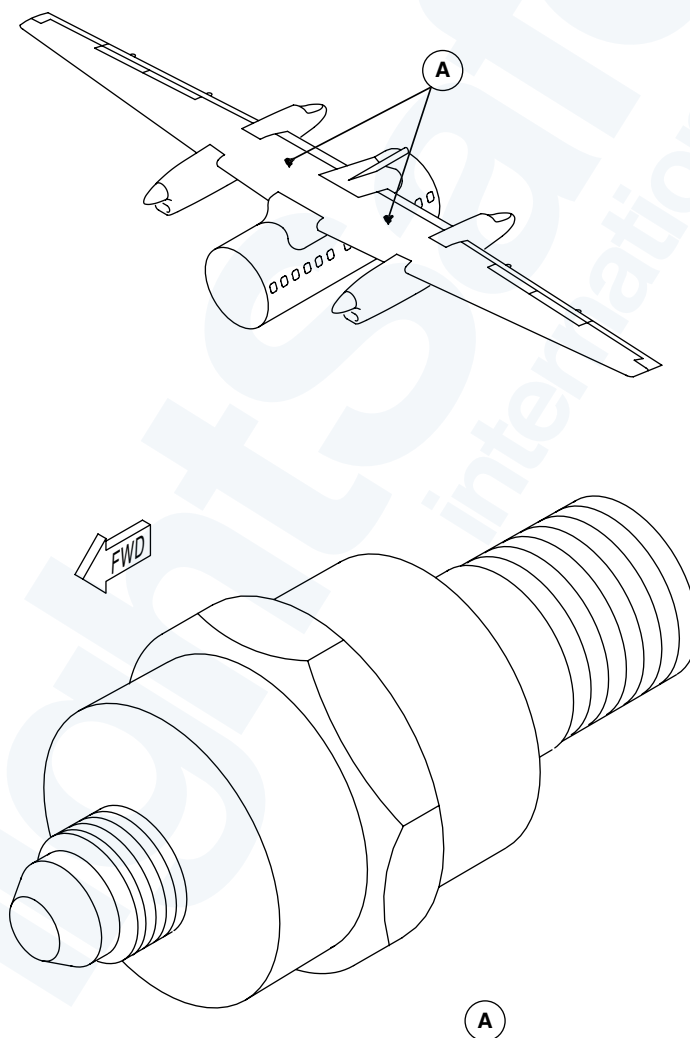


## No-Flow Pressure Switch

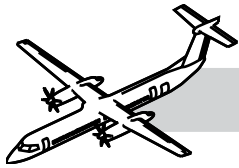
Refer to Figure 28-40. No-Flow Pressure Switch.

The no-flow pressure switch sends a signal to the refuel/defuel control panel to indicate that the refueling process has stopped.

There are two no-flow pressure switches in the refuel/defuel system. The switches are installed downstream of each level control SOV.



**Figure 28-40. No-Flow Pressure Switch**



## Refuel Vent Valves

Refer to Figure 28-41. Refuel Vent Valves.

There is a refuel vent (dump) valve installed in each wing tank on the closure rib at WS 407. The valve is kept open by refueling pressure.

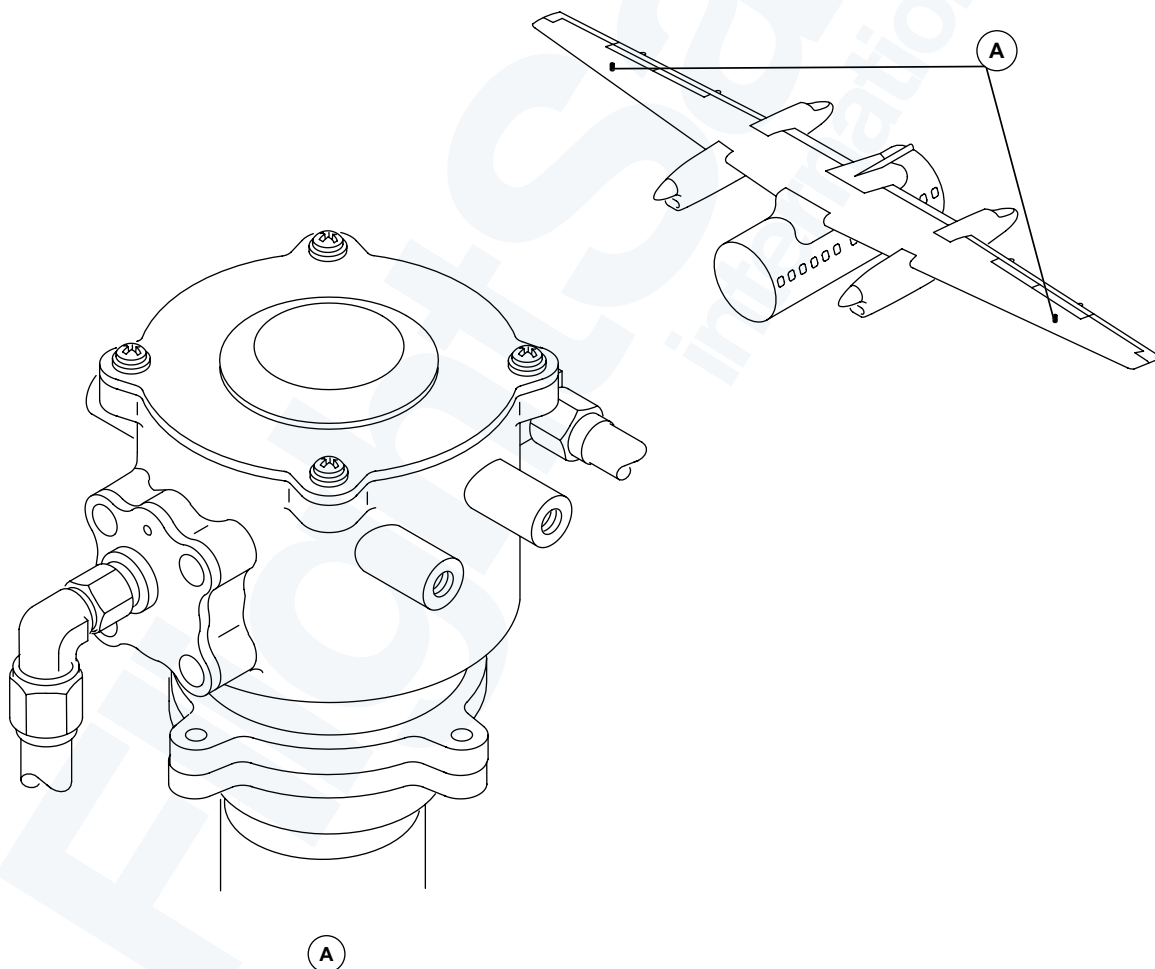
If the tank overfills due to a malfunction of the pressure refueling shut-off system, the open valve dumps fuel into the surge bay. If the surge bay fills up, fuel will spill overboard through the NACA vents.

The refuel vent/dump valve also releases pressure from the tank if it is more than  $3.0 \pm 0.25$  psi ( $20.7 \pm 1.72$  kPa) above atmospheric pressure.

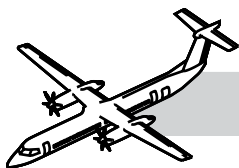
A microswitch on the valve senses the valve position and sends the information to the refuel/defuel control panel.

The DUMP VALVE OPEN (TANK 1 or TANK 2) indicator light on the refuel/defuel control panel comes on when the related refuel vent/dump valve opens.

If the valve does not open during refueling, the refuel/defuel control panel will stop the pressure refueling by sending a signal to the level control solenoid to close the level control shut-off valve.



**Figure 28-41. Refuel Vent Valves**

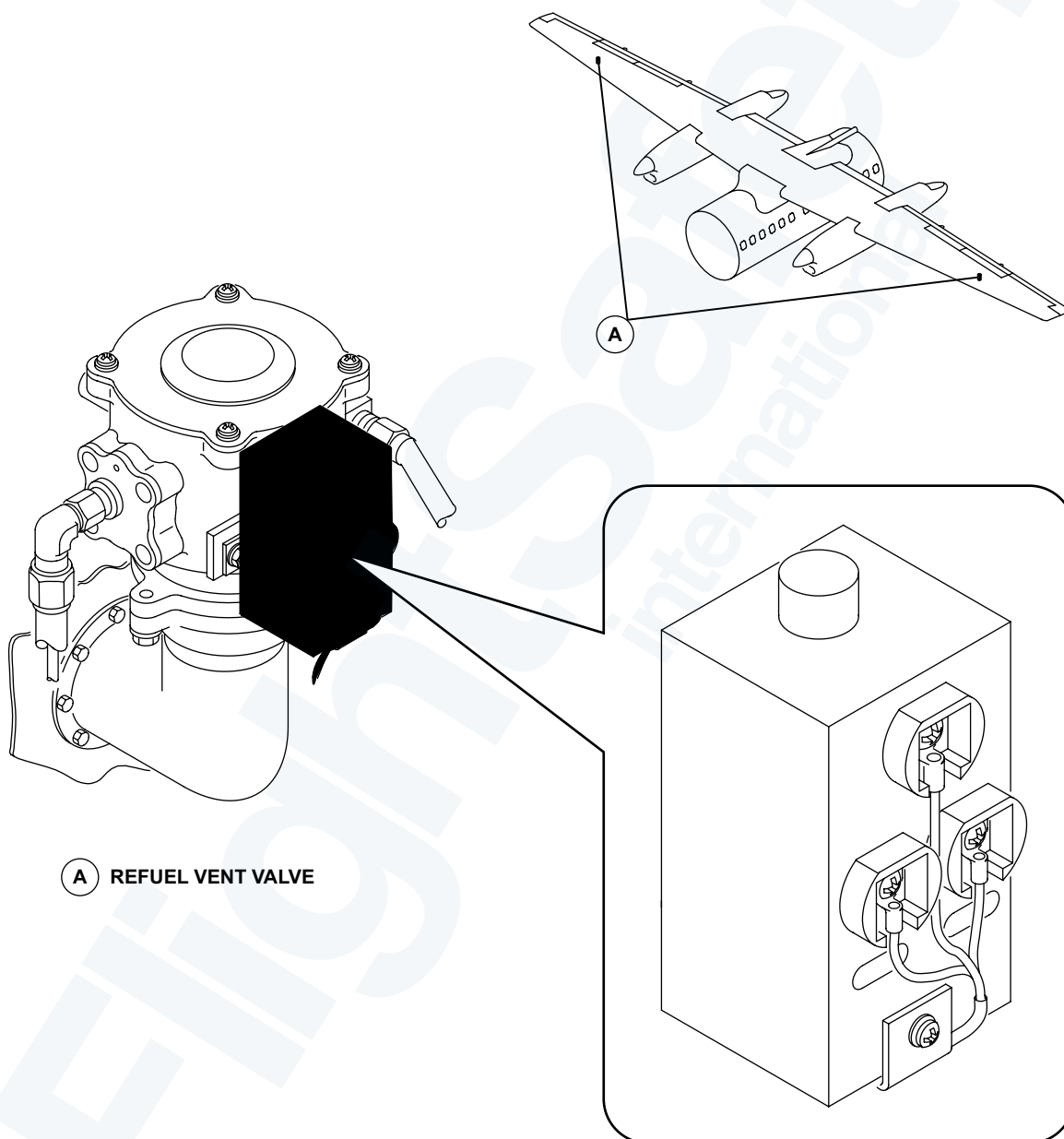


## High Level Sensor

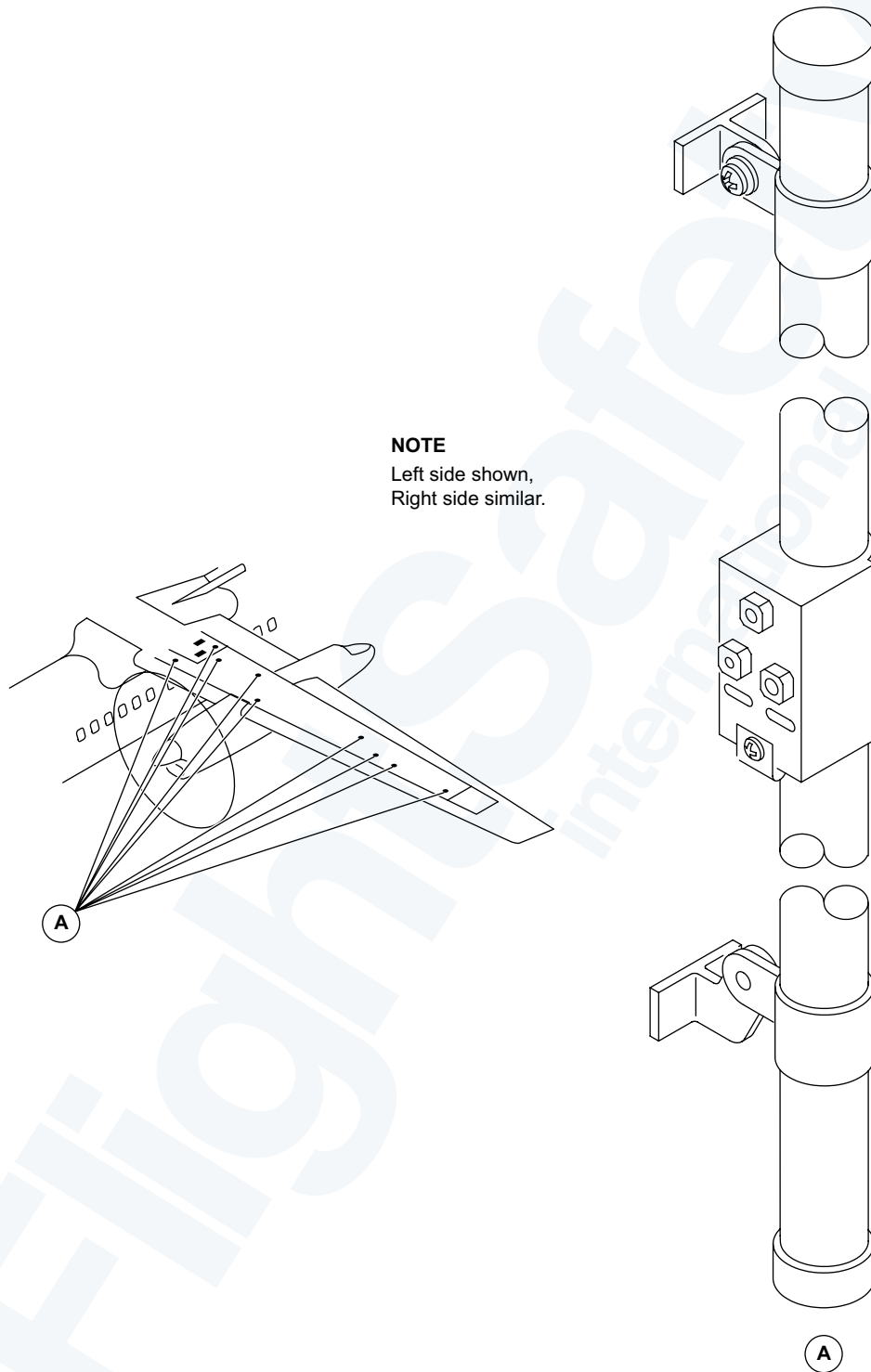
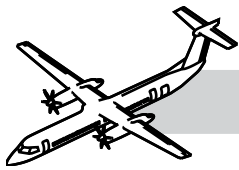
Refer to Figure 28-42. High Level Sensor.

The high level sensor signals the HLCU to stop the refuel or fuel transfer operation when the tank has reached its maximum allowable capacity.

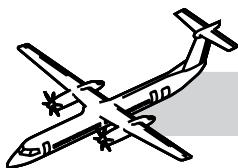
A high level sensor is attached to each refuel vent (dump) valve at the outboard end of each wing tank. It is a dual thermistor bead type. One bead gives a reference temperature while the other senses the fuel level.



**Figure 28-42. High Level Sensor**



**Figure 28-43. Fuel Quantity Probes**



## Fuel Quantity Probes

## NOTES

Refer to Figure 28-43. Fuel Quantity Probes.

The fuel quantity probes sense the fuel level in the tanks and transmit the information to the FQC.

There are 18 fuel quantity probes, nine in each wing. The probes contain two concentric metal cylinders, a terminal block, and two mounting brackets. The cylinders form the capacitor elements (plates); the inner plate is the high-impedance element and the outer plate is the low-impedance element.

Changes in fuel levels around each probe cause corresponding changes in the effective capacitance of each probe, a change in capacitance shows as a change in fuel quantity on the MFD. The probes are monitored by the FQC.

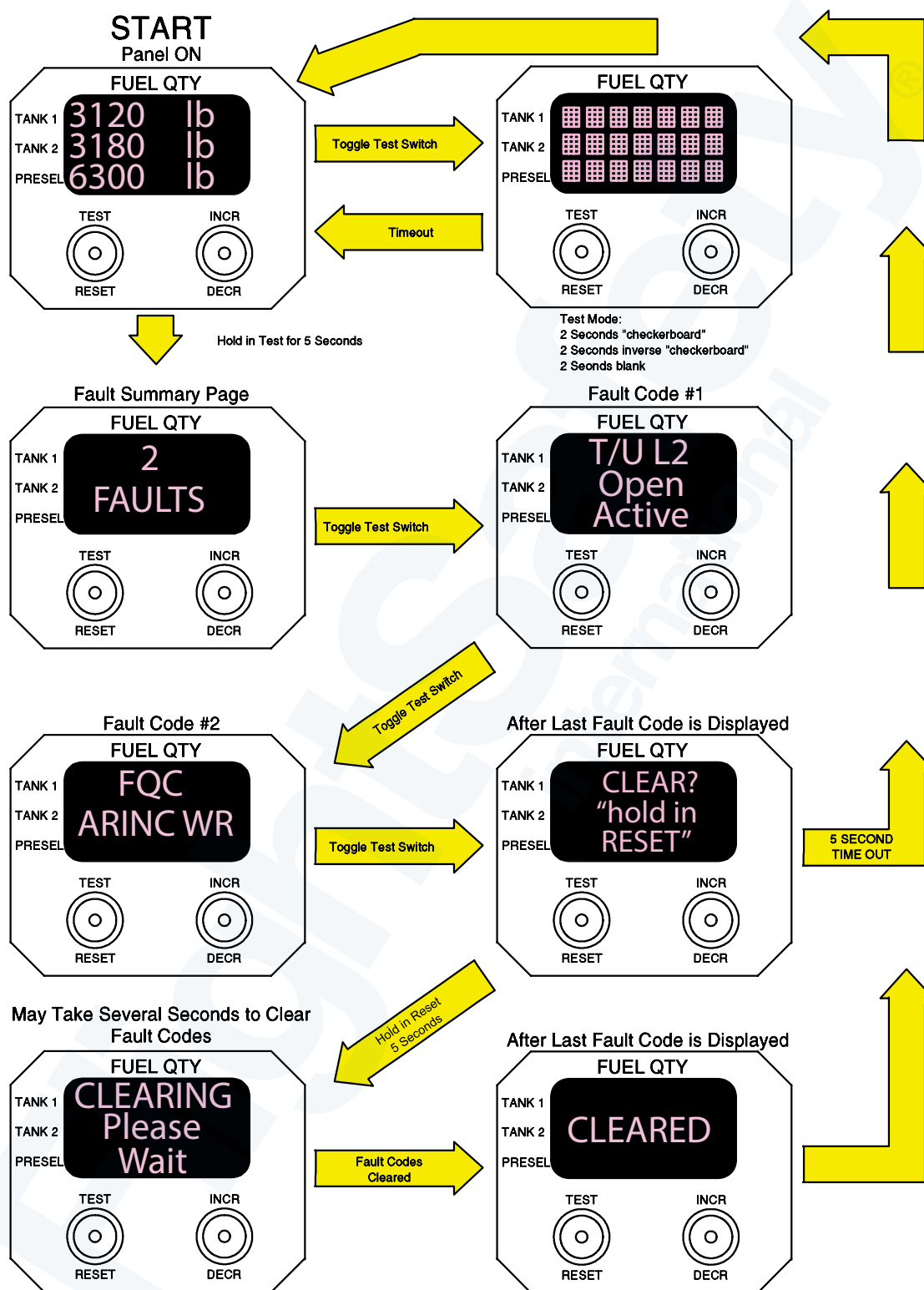
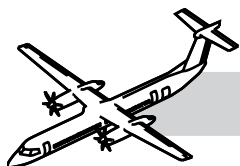
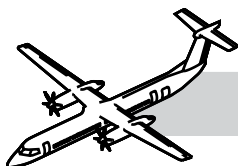


Figure 28-44. Fuel System Fault Codes





## CONTROLS AND INDICATIONS

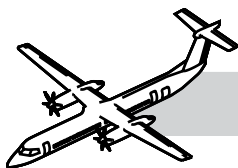
## NOTES

Refer to Figure 28-44. Fuel System Fault Codes.

The RDI on the refuel/defuel panel records fault information from the FQC. The TEST / RESET toggle switch on the RDI shows and clears fault codes.



**Figure 28-45. Preselect Refueling (1 of 2)**



## OPERATION

## NOTES

Refer to Figure 28-45. Preselect Refueling (1 of 2).

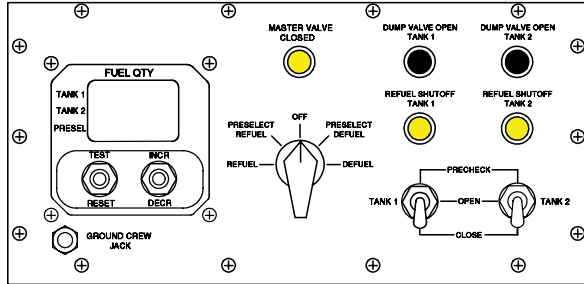
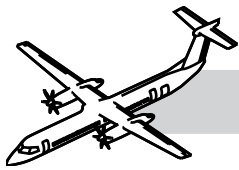
The INCR/DECR toggle switch on the RDI sets the desired amount of fuel for an automatic refueling or defueling operation.

When the INCR/DECR switch is toggled to the INCR or DECR position, the current PRESEL display value on the RDI will increase or decrease by 10 lbs (or 10 kg).

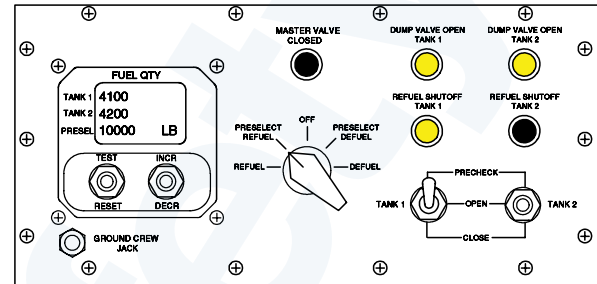
After this has been repeated 10 times, the PRESEL display value will increase or decrease by 100 lbs (or 100 kg).

The PRESEL display value will never fall below zero or exceed 5800 kg (13780 lbs).

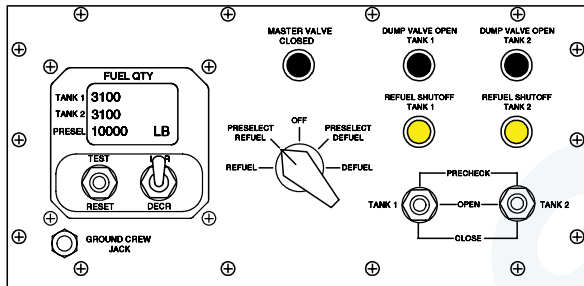
Four seconds after the INCR/DECR switch is returned to the center position, the refueling or defueling process will start. If the INCR/DECR switch is moved out of the center position, the refueling or defueling process will stop and will be restored after four seconds.



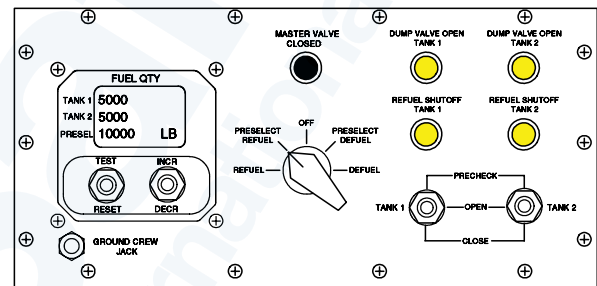
**INITIAL CONFIGURATION**



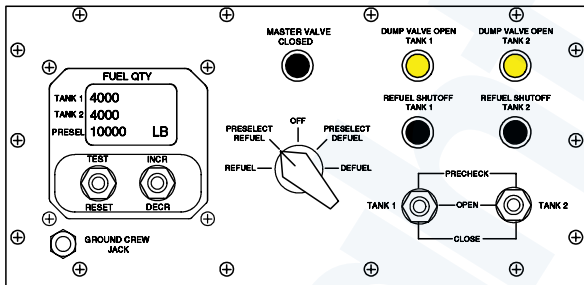
**PRE-CHECK TEST (No. 1 TANK) DURING REFUELING**



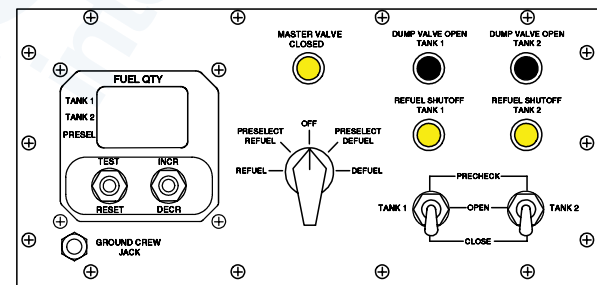
**PANEL SET TO INITIATE REFUELING**



**TANKS FULL**



**REFUELING IN PROGRESS**

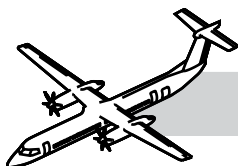


**REFUELING COMPLETE**

**LEGEND**

- Light Off.
- Light On

**Figure 28-46. Preselect Refueling (2 of 2)**



Refer to Figure 28-46. Preselect Refueling (2 of 2).

## NOTES

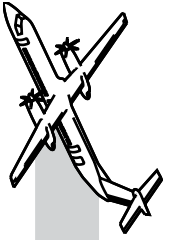
In the automatic PRESELECT refueling mode, the desired fuel quantity is selected using the INCR/DECR switch on the RDI.

The PRESEL segment of the RDI displays the current total quantity of both tanks before the desired quantity is selected. When the desired quantity has been set, the refueling process starts four seconds after the INCR/DECR switch is returned to the center position.

The preselected total fuel quantity is distributed equally between the two tanks. If the tank quantities reach an imbalance of 600 lbs (273 kg), the PRESEL segment will flash the word BALANCE.

Refueling to the heavy tank is stopped until the imbalance is less than 350 lbs (227 kg).

During pressure refueling (automatic or manual), the REFUEL SHUTOFF (TANK 1 and TANK 2) advisory lights will extinguish to confirm that the tanks are being refuelled (refer to No-Flow Pressure Switch).



28-68

FOR TRAINING PURPOSES ONLY

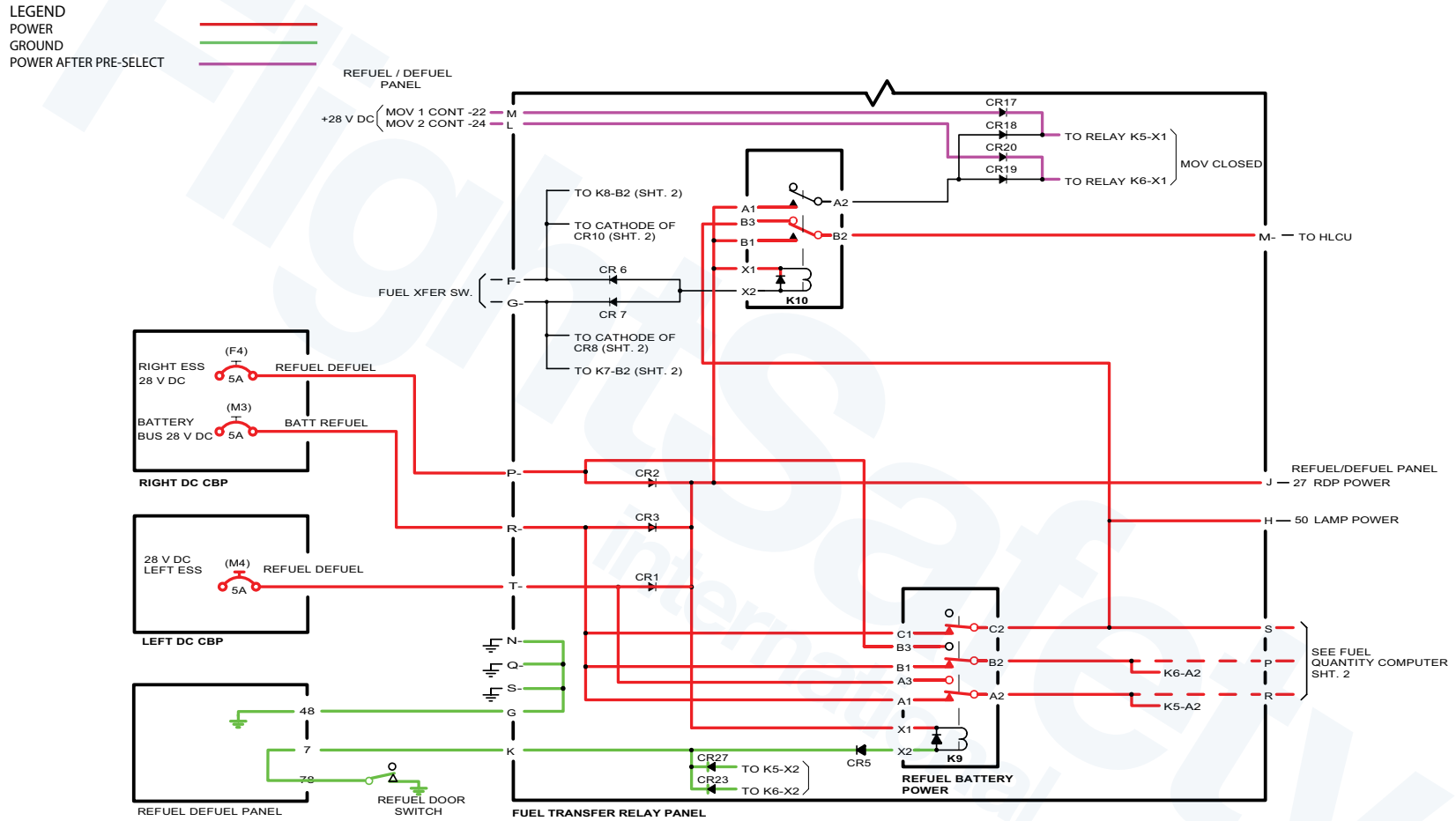
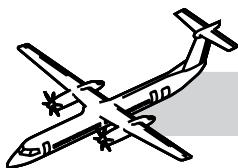


Figure 28-47. Refuel/Defuel Operation (Sheet 1 of 2)



Refer to:

- Figure 28-47. Refuel/Defuel Operation (Sheet 1 of 2).
- Figure 28-48. Refuel/Defuel Operation (Sheet 2 of 2).
- Figure 28-49. Refuel/Defuel System Schematic.

## Refuel/Defuel Operations

The automatic and manual refuel/defuel operations are through the fuel adapter in the aft right nacelle. The FQC controls the preset Refuel and Defuel processes. The HLCU monitors the high level sensors and will stop the refuel operation when the wing fuel tank level is high.

Defueling of the wing tanks is completed with external suction and/or the auxiliary pump operating in each collector bay.

(The Refuel/Defuel/Transfer Motorized Shut-Off Valve (R/D/T SOV) can be indicated as MOV in the schematic drawings).

## Automatic Refuel

(Refer to AMM TASK 12-10-28-650-801 Pressure Refueling).

The RDP pin 27 is powered at all times from the three REFUEL circuit breakers.

When the RDP door is opened, the RDP door switch supplies a ground signal to:

- The RDP flood light from the RDP pin 61 which causes the light to illuminate (see Sheet 2).
- The CAWP connector 3312 P5 pin 18 from the RDP pin 14 which causes the FUELING ON light to illuminate (see Sheet 2).
- To the RDP pin 7 to the fuel transfer relay panel pin “K” energizes the relay K9 closed (see Sheet 1).

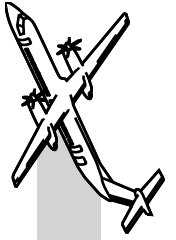
The closed relay K9 supplies 28 VDC to:

- Through pin “R” to the FQC pin 49 (see Sheet 2).
- Through pin “P” to the FQC pin 55 (see Sheet 2).
- Relay K5-A2 which supplies power to energize open the Tank 1 R/D/T SOV (see 28-23-00 Figure 5).
- Relay K6-A2 which supplies power to energize open the Tank 2 R/D/T SOV (see 28-23-00 Figure 6).

Contact C2 supplies 28 VDC power:

- Through pin “H” to the RDP pin 50 to supply power to the five indicator lights (see Sheet 1).
- The RDP on pin 27 and the (open door) ground on pin 62 to energize the Refuel/Defuel SOV closed. The closed Refuel/Defuel SOV sends a (valve closed) ground signal to the RDP pin 41 which causes the MASTER VALVE CLOSED light to come on.
- Through pin “S” to the FQC pins 41, 42, and 52 (see Sheet 2).
- To relay K10-B3 this supplies power to the HLCU (see Sheet 1).

The fuel nozzle is connected to the RDP adapter and pressurized fuel (20 psi min. to 50 psi max.) is supplied to the RDP Refuel/Defuel SOV.



28-70

FOR TRAINING PURPOSES ONLY

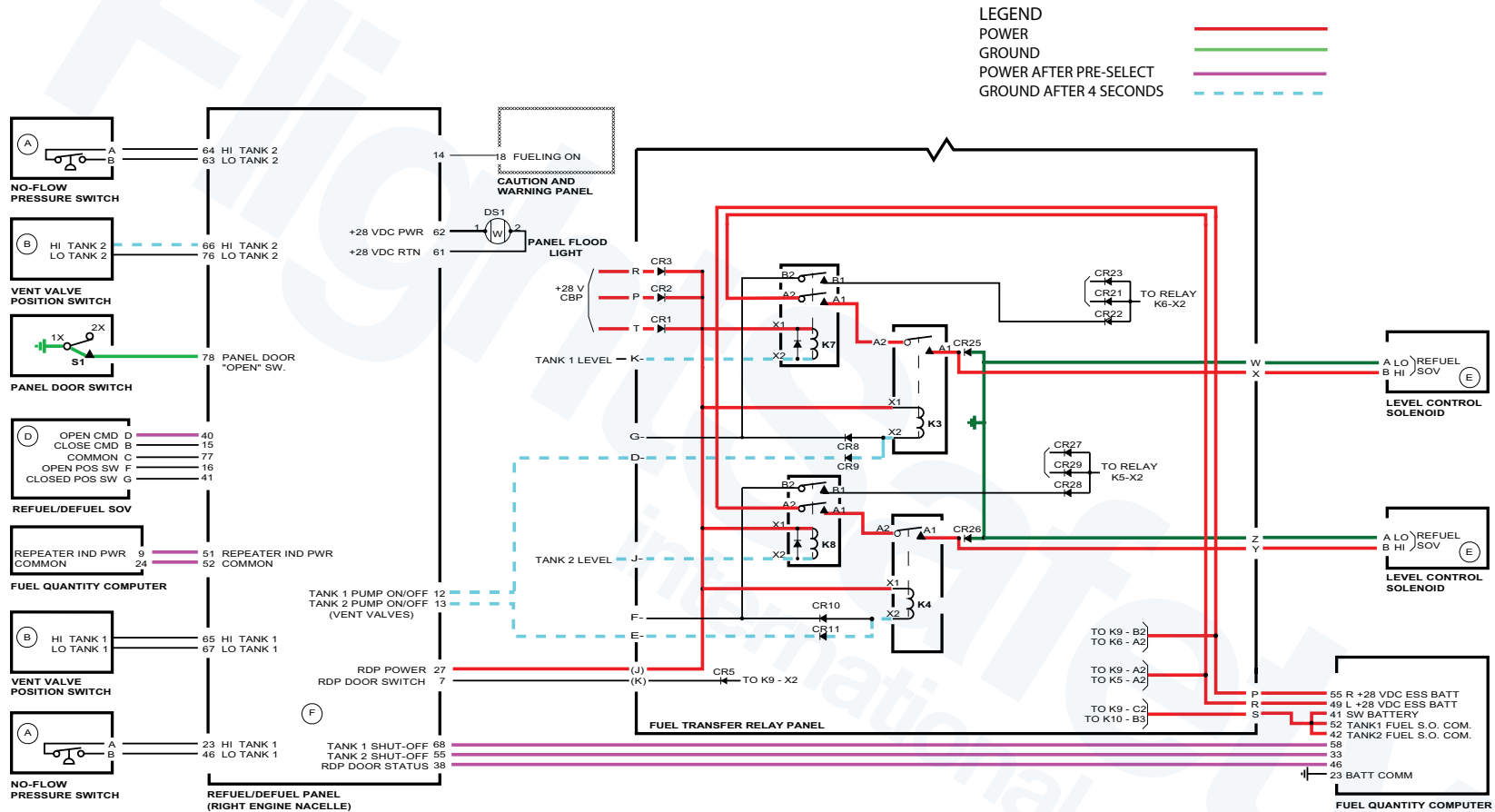
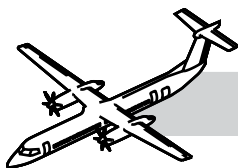


Figure 28-48. Refuel/Defuel Operation (Sheet 2 of 2)





The fuel selector is set to the PRESELECT REFUEL position. This action causes:

- The FQC pin 46 to receive an RDI enable signal (RDP Door Status) from the RDP pin 38.
- The FQC sends an indication power to the FQC pin 9 to the RDP pin 51 and an indication common to the FQC pin 24 to the RDP pin 52. These signals cause the Refuel/Defuel Indicator (RDI) to come on.

The INCR/DECR switch is held to the INCR position to select the fuel load.

- This sends a load request signal to the FQC which calculates the fuel load to the two fuel tanks
- Four seconds after the INCR/DECR switch is released, the FQC sends a 28 VDC (valve open) signals to the RDP pins 55 and 68.
- The RDP sends the 28 VDC (valve open) signals from pins 22 and 24 to the fuel transfer relay panel pins “L” and “M” which causes the relays K5 and K6 to energize closed to energize the two R/D/T SOV open.
- The RDP also sends a 28 VDC (valve open) signal from pin 40 directly to the Refuel/Defuel SOV which causes the valve to open and allows pressurized fuel into the fuel manifold. The open Refuel/Defuel SOV removes the ground signal from the RDP pin 41 which causes the MASTER VALVE CLOSED light to extinguish.
- The pressurized fuel opens the two vent valves (one in each tank) which send two ground signals to the RDP pins 65 and 66 and causes the DUMP VALVE OPEN 1 and 2 lights to illuminate.
- The two (vent valve open) ground signals from the RDP pins 12 and 13 to the fuel transfer relay panel pins “D-” and “E-” energize the relays K3 and K4 closed.

- The tank 1 and 2 High Level Sensor (not full) ground signals keep the relays K7 and K8 energize closed.
- The closed relays K7 and K3 energize the tank 1 Refuel Solenoid Valve open.
- The closed relays K8 and K4 energize the tank 2 Refuel Solenoid Valve open.
- The two open refuel solenoid valves allow the back-pressure fuel to be bled off from the two level control SOV which open the two level control SOV and allow the fuel to enter the fuel tanks.
- The no-flow pressure switches sense 2 psi and open, causing the REFUEL SHUT-OFF lights to extinguish.

When the FQC senses the pre-selected level, the FQC stops the refuel operation to each tank. The FQC removes the 28 VDC (valve open) signals to the RDP pins 55 and 68.

- The RDP removes the ground signals from pins 12 and 13 to the fuel transfer relay panel pins “D-” and “E-” which causes the relays K3 and K4 to de-energize open.
- The open relays K3 and K4 remove 28 VDC from pin “X” and “Y” which causes the Tank 1 and 2 refuel solenoid valves to de-energize closed which causes the level control valves to close and stop the fuel flow.
- The RDP also removes the ground (valve close) signals from pins 22 and 24 to the fuel transfer relay panel pins “L” and “M” which causes the relays K5 and K6 to de-energize open to energize the two R/D/T SOV closed and stop the fuel flow.
- The decrease in fuel pressure causes the no-flow pressure switches to close which supplies ground signals to the RDP pins 23 and 64 and cause the REFUEL SHUT-OFF TANK 1 and 2 indicator lights to illuminate.

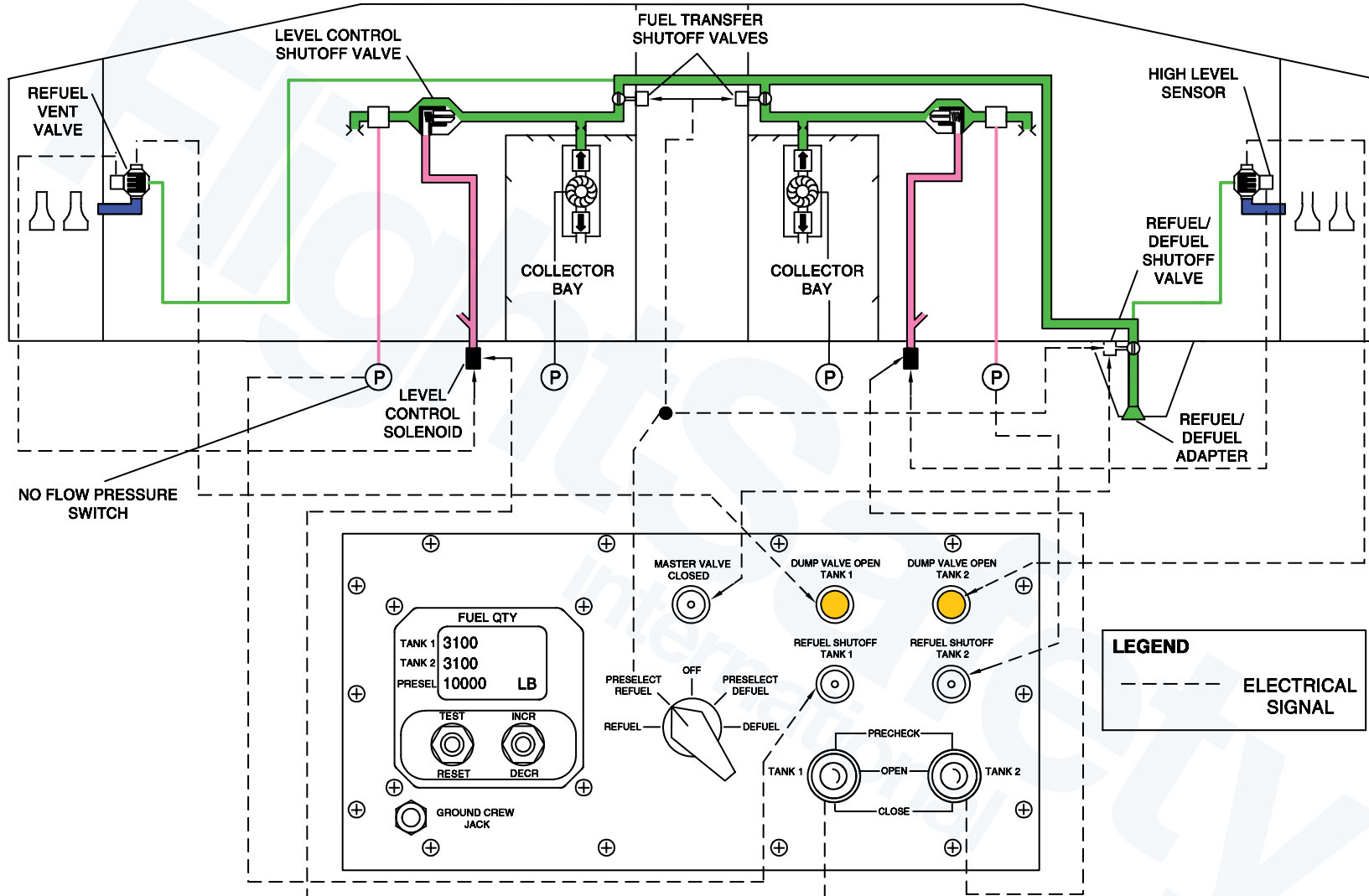
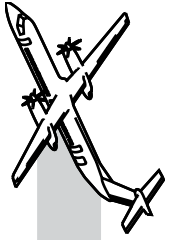
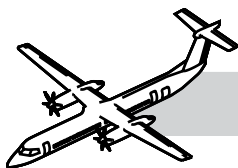


Figure 28-49. Refuel/Defuel System Schematic



The DUMP VALVE OPEN TANK 1 and 2 lights stay on due to the pressure in the fuel manifold.

If the FQC does not automatically stop the refueling and the tanks become full, the high level sensors remove the ground signals from the relays K7 and K8. This de-energizes the Fuel Solenoid Valves which causes the two Level Control Valves to close and stop the fuel flow.

## Precheck Test

During the refuel operation the TANK 1 switch is set to the PRECHECK position. This action tests the automatic shutoff operation of the HLCU.

- The PRECHECK (open circuit) signal from RDP pin 32 is sensed by the HLCU on pin 4.
- The HLCU removes the ground signal to the fuel transfer relay panel pin “K-” which causes the relay K7 to de-energize open. The open relay K7 de-energizes the level control solenoid valve open which causes the level control valve to close and stop the fuel flow.

When the switch is released to the OPEN position, the relay K7 re-energizes closed and opens the level control solenoid valve which opens the level control valve and restarts the fuel flow.

## Manual Refuel

The manual mode of refuel uses the same relays and valves as the automatic mode but with no control from the FQC. The TANK 1 and TANK 2 switches and the High Level Sensors are used to control the fuel flow in manual mode.

When the TANK 1 switch is set to the CLOSE position the RDP removes the ground signal from pin 12 to the fuel transfer relay panel pin “D-”.

- The open circuit signal from pin “D-” causes the relay K3 to de-energize.

- The open relay K3 removes 28 VDC from pin “X” to de-energize closed the Tank 1 refuel shut-off solenoid valve which closes the level control valve to close and stops the fuel flow.

When the TANK 2 switch is set to the CLOSE position, the RDP removes the ground signal from pin 13 to the fuel transfer relay panel pin “E-” which causes the relay K4 to de-energize open.

- The open relay K4 removes 28 VDC from pin “Y” to de-energize closed the Tank 2 refuel shut-off solenoid valve which closes the level control valve to close and stops the fuel flow.

The High Level Sensor senses the fuel is high in the left fuel tank; the sensor removes the ground signal from the relay K3.

- The open relay K3 removes 28 VDC from pin “X” to de-energize closed the Tank 1 refuel shut-off solenoid valve which closes the level control valve to close and stops the fuel flow.

This action is similar on the right fuel tank.

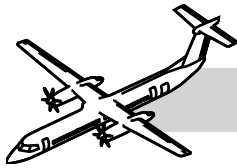
## Automatic Defuel

(Refer to AMM TASK 12-10-28-650-802).

The FQC stops the defueling when the pre-selected fuel level is reached. The fuel tank auxiliary pump operation must be stopped when the fuel quantity nears 50 lbs to prevent damage to the pumps. To stop the auxiliary pumps during defuel, pull the FUEL AUX PUMP No.1 and FUEL AUX PUMP No.2 circuit breakers.

The automatic Defuel mode uses similar operations to automatic refuel mode with the exceptions that follow:

- RDP indicator lights indications
- Automatic auxiliary pump operation with 115 VAC power available.
- Fuel nozzle suction is applied (Max. 10 in-Hg (254 mm Hg)).



The fuel selector is set to the PRESELECT DEFUEL position. This action causes:

- The RDI to illuminate.
- The RDP sends a ground signal out from pin 44 to the refuel transfer relay panel P/J21 pin “N” which causes the relays K1 and K2 to energize closed. The closed relays supply 115 VAC power to the two fuel tank auxiliary pumps.

The INCR/DECR switch on the RDI is held to the DECR position to select the fuel load.

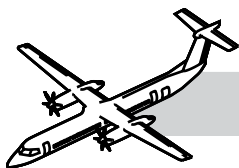
- The FQC receives the load request signal and calculates the fuel load to the two fuel tanks.
- Four seconds after the INCR/DECR switch is released, the FQC sends 28 VDC (valve open) signals to the RDP pins 55 and 68 which energize relays K5 and K6 closed. The relays K5 and K6 energize the two R/D/T SOV open.
- The RDP also sends 28 VDC (valve open) signals from the pin 40 to the two Refuel/Defuel SOV which cause the valves to open. The open Refuel/Defuel SOV removes the ground signal from the RDP pin 41 which causes the MASTER VALVE CLOSED light to go out. The open Refuel/Defuel SOV allows defueling.

The two refuel vent valves stay closed (due to no fuel pressure) which:

- Removes the ground signals from the RDP pin 65 and 66 which causes the two DUMP VALVE OPEN lights to stay off.
- The open circuit signals from the RDP pins 12 and 13 to the fuel transfer relay panel pins “D-” and “E-” de-energize the relays K3 and K4 open which causes the two refuel shut-off solenoid valves to stay closed and keeps the two Level Control SOV closed.

When the FQC senses the selected fuel quantity it removes the 28 VDC (open) signal to the RDP pins 55 and 68. (Refer to 28-23-00 Fuel Transfer Operation).

- The RDP removes the ground signal from pins 22 and 24 to the fuel transfer relay panel pins “L” and “M” which causes the relays K5 and K6 to de-energize open. The open relays send 28 VDC to pins “B-” and “U” which causes the Tank 1 and 2 R/D/T valves to energize closed and stop the fuel flow.



PAGE INTENTIONALLY LEFT BLANK

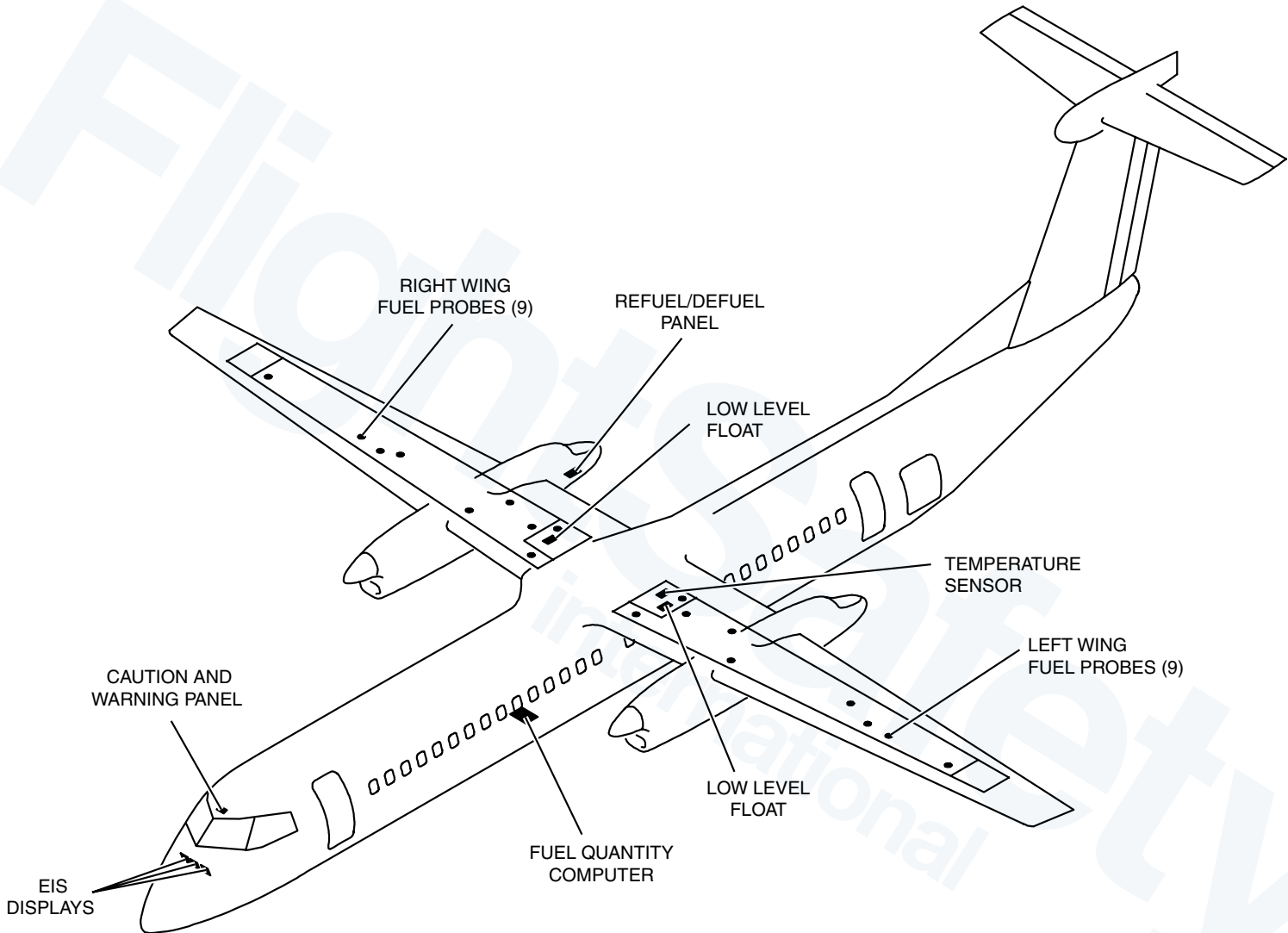
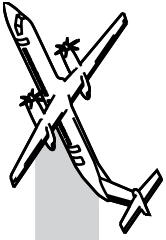
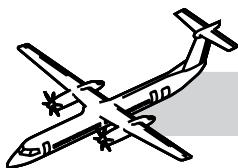


Figure 28-50. Fuel Indicating System



## 28-40-00 INDICATING

NOTES

### GENERAL

The fuel indicating system supplies information on fuel quantity, low level detection and fuel temperature for display in the flight compartment. Fuel quantity is also shown on the refuel/defuel control panel.

### SYSTEM DESCRIPTION

Refer to Figure 28-50. Fuel Indicating System.

The fuel indicating system provides an accurate measure of the fuel quantity in the wing tanks.

Nine DC capacitance type fuel probes in each wing send inputs to the FQC to calculate the aircraft fuel quantity.

The EIS in the flight compartment displays the fuel quantity information to the flight crew.

The fuel quantity information is also displayed on the RDI, in the refuel/defuel panel.

A magnetic dipstick in each wing provides an independent mechanical indication of fuel tank quantity.

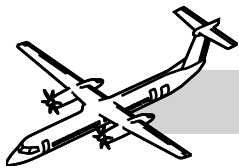
Low fuel level is sensed by a float switch in each collector bay and is indicated by a caution light on the CAWP.

Fuel tank temperature is measured by a RTD sensor in the left collector bay and is displayed in the flight compartment.

The fuel indicating system has these components:

- Fuel quantity computer
- High level control unit
- High level switch
- Float switch
- Temperature sensor
- Magnetic dipstick
- Fuel quantity probe.





## COMPONENT DESCRIPTION

### Fuel Quantity Computer (FQC)

Refer to Figure 28-51. Fuel Quantity Computer (FQC).

The FQC is in the avionics bay below the passenger compartment floor.

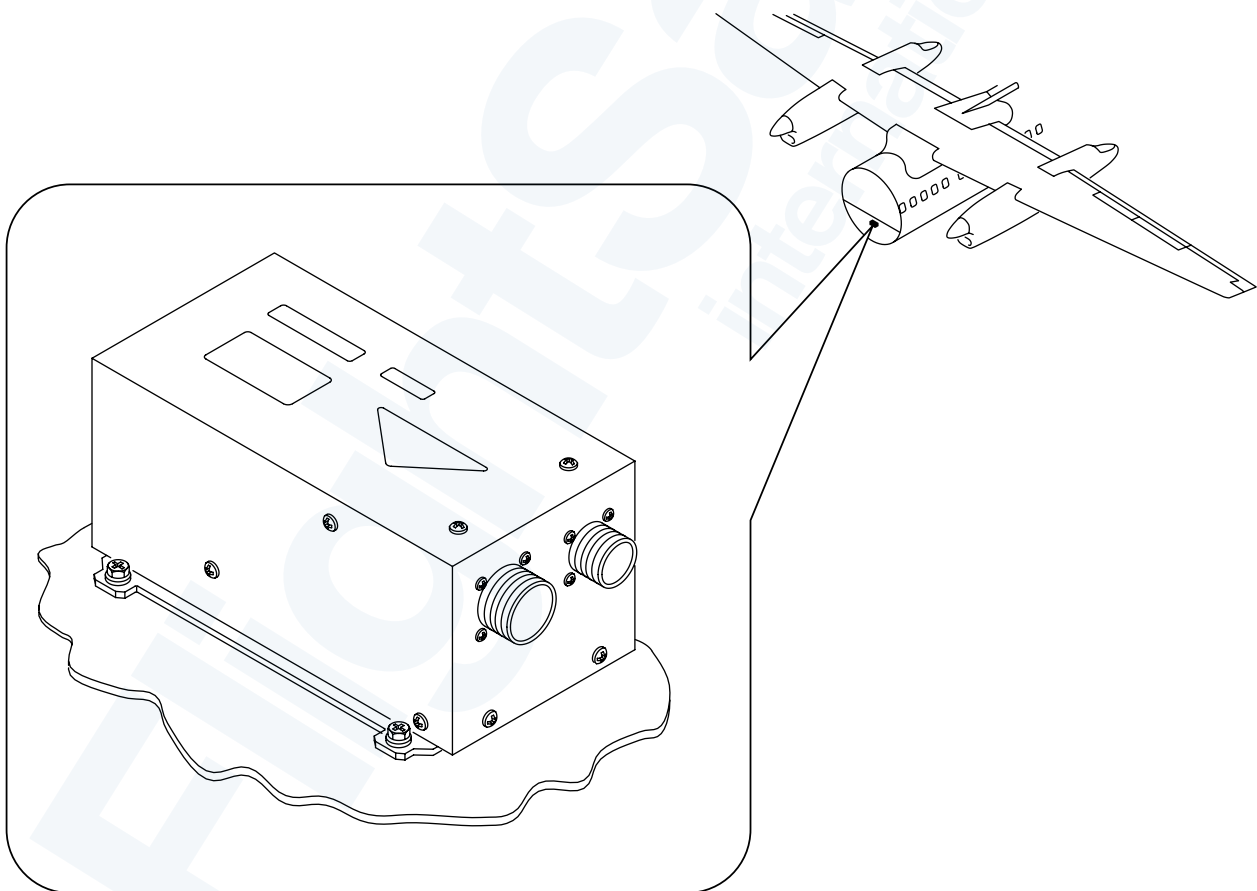
The FQC has two independent processors. Each of these processors:

- Calculates the fuel quantities in each tank independently
- Controls the level control solenoid for pre-select refueling

- Performs a BIT of the fuel gauging components
- Sends quantity, temperature compensation, and BIT data to the EIS displays.

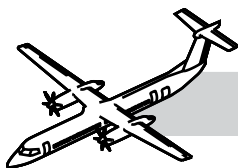
The right processor also calculates fuel temperature measurements and transmits the information to the RDI.

The FQC performs an initial BIT when it starts up and a continuous BIT when it is in operation. The FQC is self-calibrating and needs no adjustments.



**Figure 28-51. Fuel Quantity Computer (FQC)**



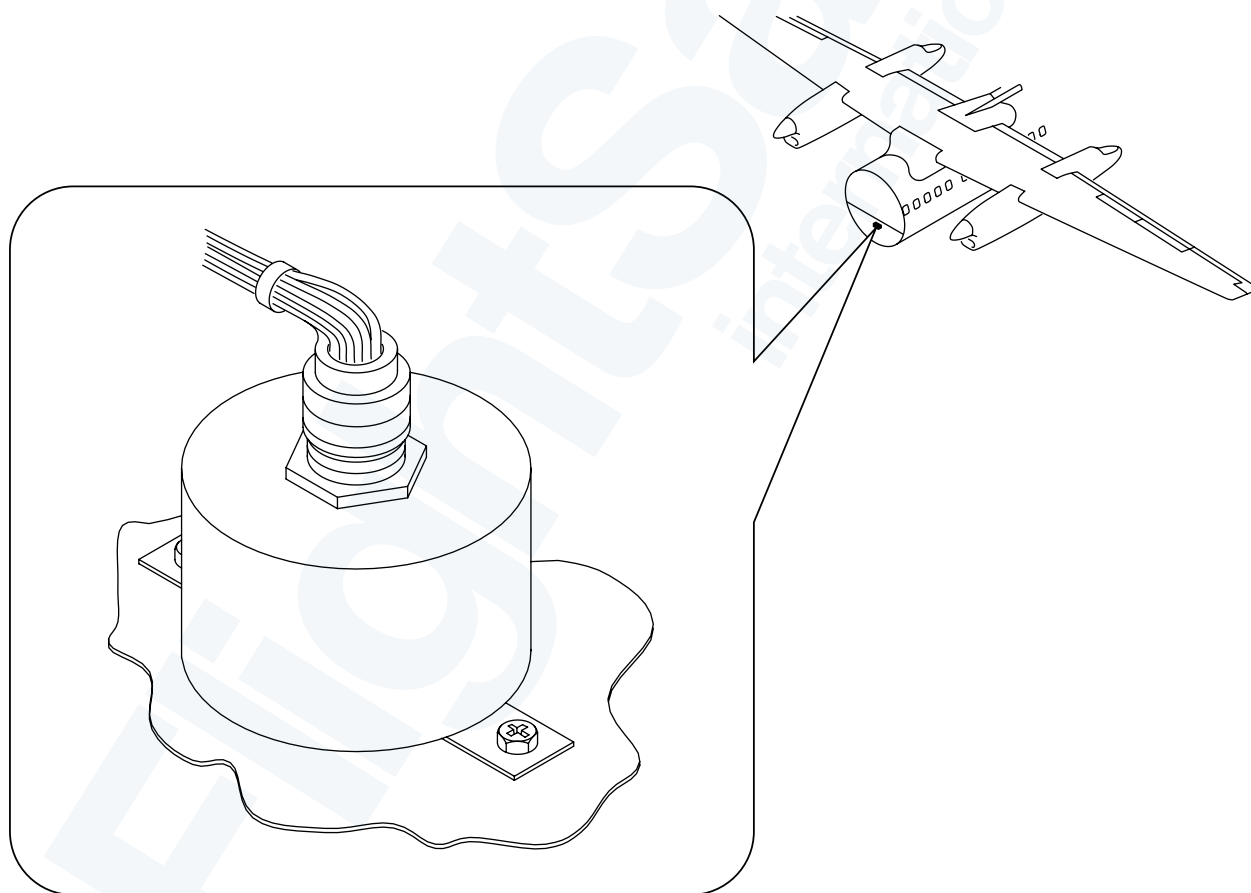


## High Level Control Unit (HLC)

Refer to Figure 28-52. High Level Control Unit.

The HLCU monitors the high level sensors and transmits a signal to prevent overfilling the tanks during refuel or fuel transfer operations. This unit also serves as a signal conditioner for the temperature sensor.

The high level control unit is below the cabin compartment floor. The pre-check test switch on the refuel/defuel control panel tests the automatic shut-off operation of the high level control unit.



**Figure 28-52. High Level Control Unit**

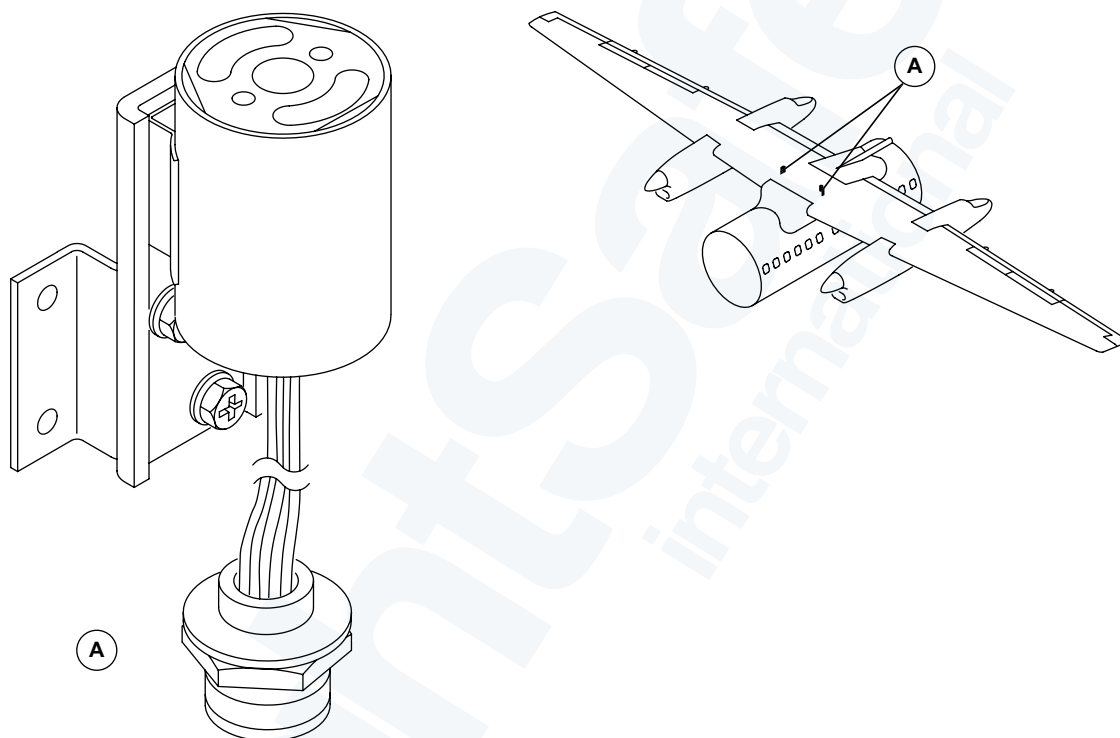
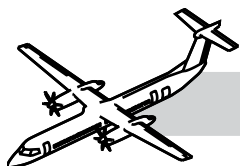
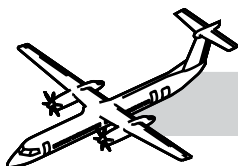


Figure 28-53. Float Switch



## Float Switch

## NOTES

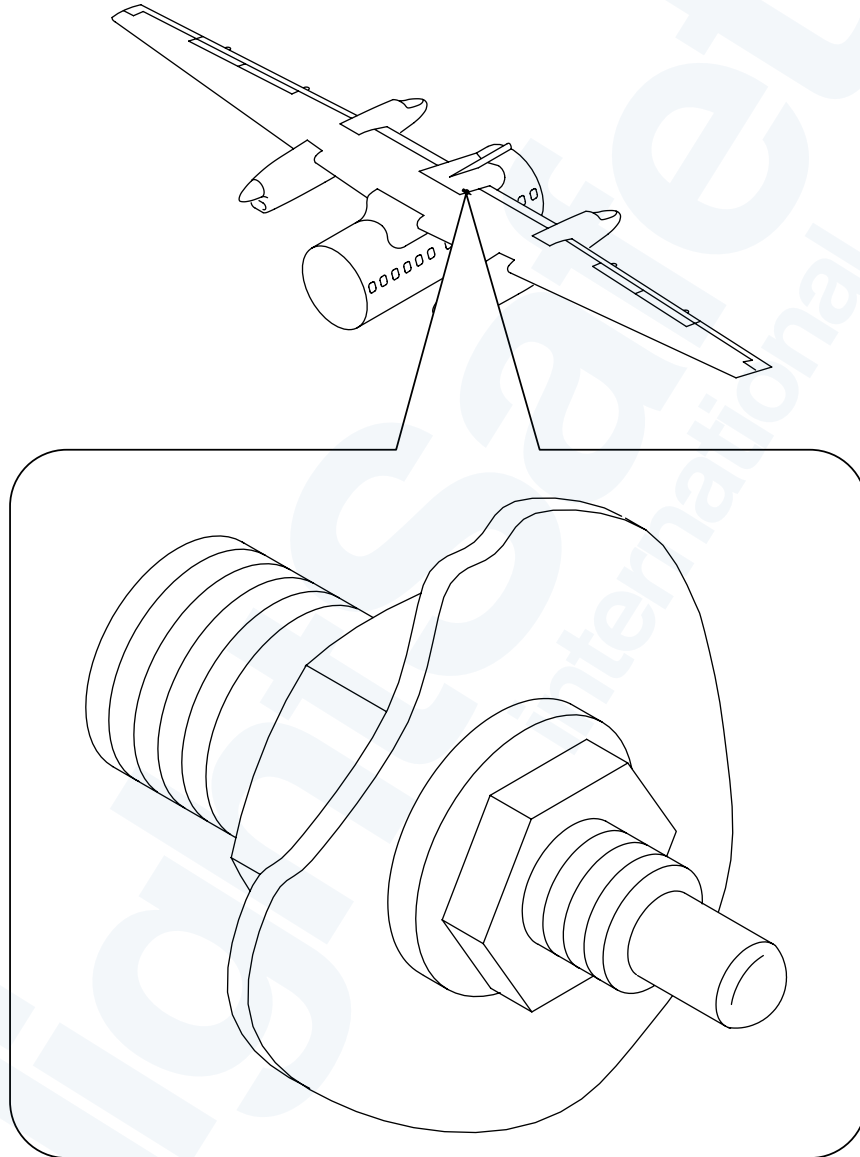
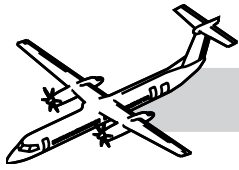
Refer to Figure 28-53. Float Switch.

The low level float switch sends a signal to the caution and warning panel to indicate a low fuel level in the collector bay.

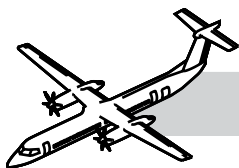
There are two low fuel level float switches, one installed in each collector bay. Each float switch is installed near the top of the collector bay on WS 60.9.

The control logic of this switch is independent of the Fuel Quantity Gauging System (FQGS). The switch will be activated when the fuel collector tank quantity drops to approximately 305 lbs and engine running with park brake OFF.

At this point the appropriate FUEL TANK LOW caution light on the caution and warning panel in the flight compartment will illuminate.



**Figure 28-54. Temperature Sensor**



## Temperature Sensor

## NOTES

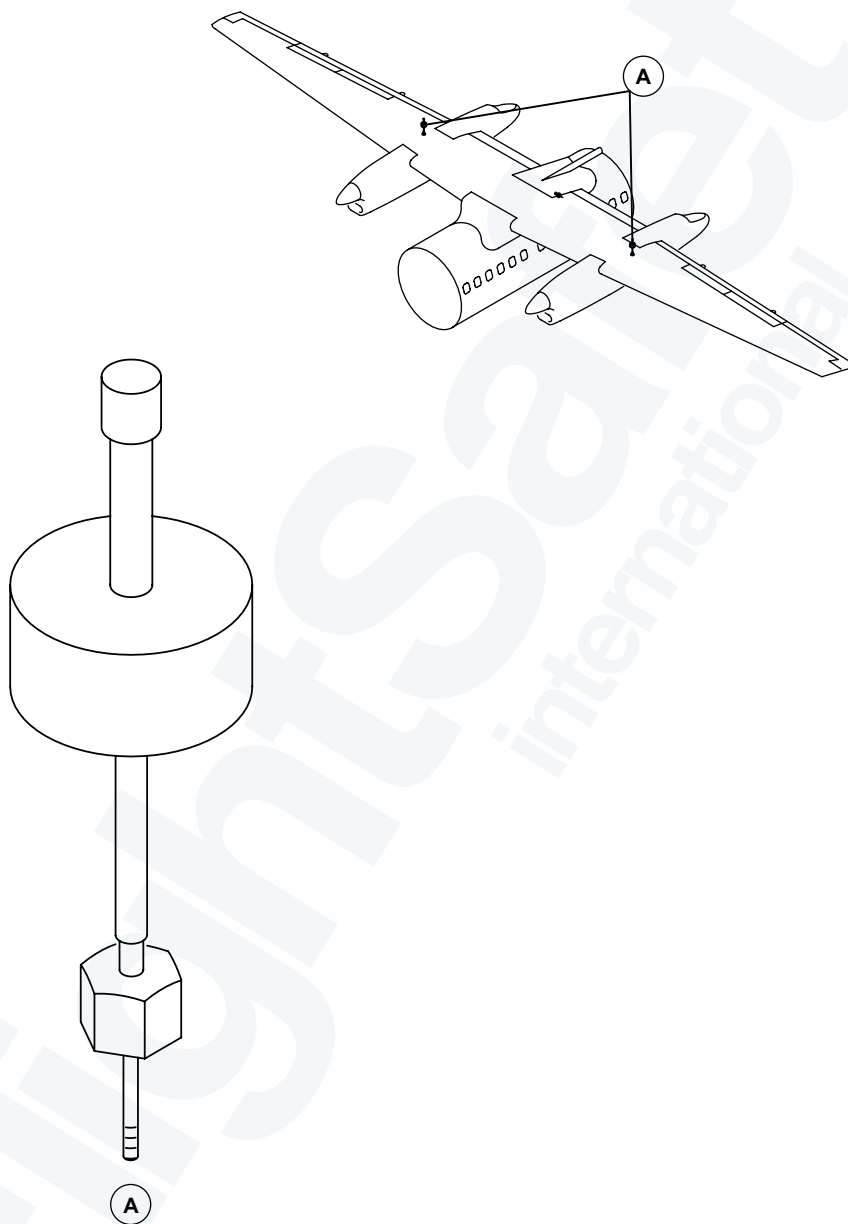
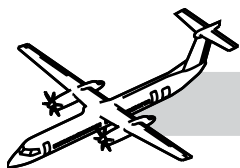
Refer to Figure 28-54. Temperature Sensor.

The temperature sensor senses the fuel temperature in the left tank and transmits the information for display in the flight compartment.

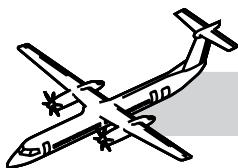
The tank temperature sensor is on the No.1 tank collector bay wall.

The sensor uses electrical resistance to monitor the fuel temperature. The analog output is sent to the HLCU for signal conditioning, and then to the EIS for display. The system is operational when electrical power is supplied to the aircraft.

The temperature is shown on the MFD Fuel Page and ranges from -70 to +75°C (-94 to 167°F).



**Figure 28-55. Magnastick**



## Magnastick

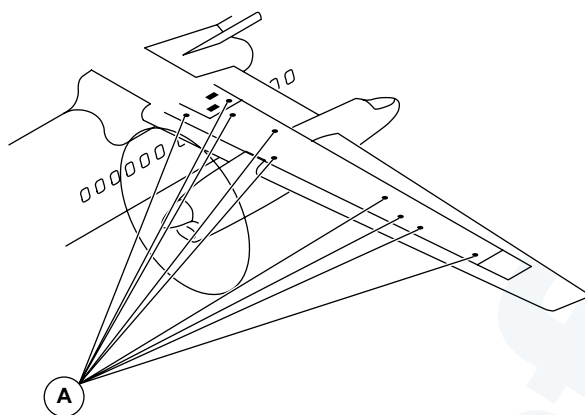
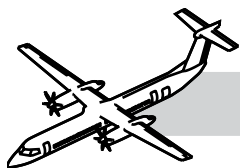
## NOTES

Refer to Figure 28-55. Magnastick.

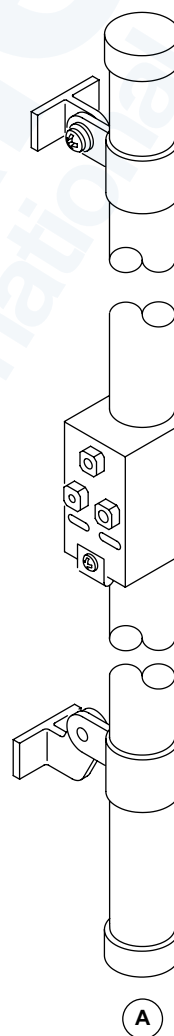
The magnetic dipsticks give an alternate means to measure the fuel quantity when the aircraft is on the ground.

One magnetic dipstick is installed in each main tank, outboard of the engines.

A magnastick is a calibrated rod with a magnet attached. It moves within a tube that extends vertically from the bottom of the fuel tank.

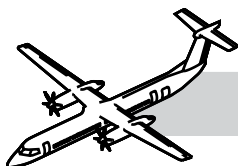


**NOTE**  
Left side shown,  
Right side similar.



**Figure 28-56. Fuel Quantity Probe**





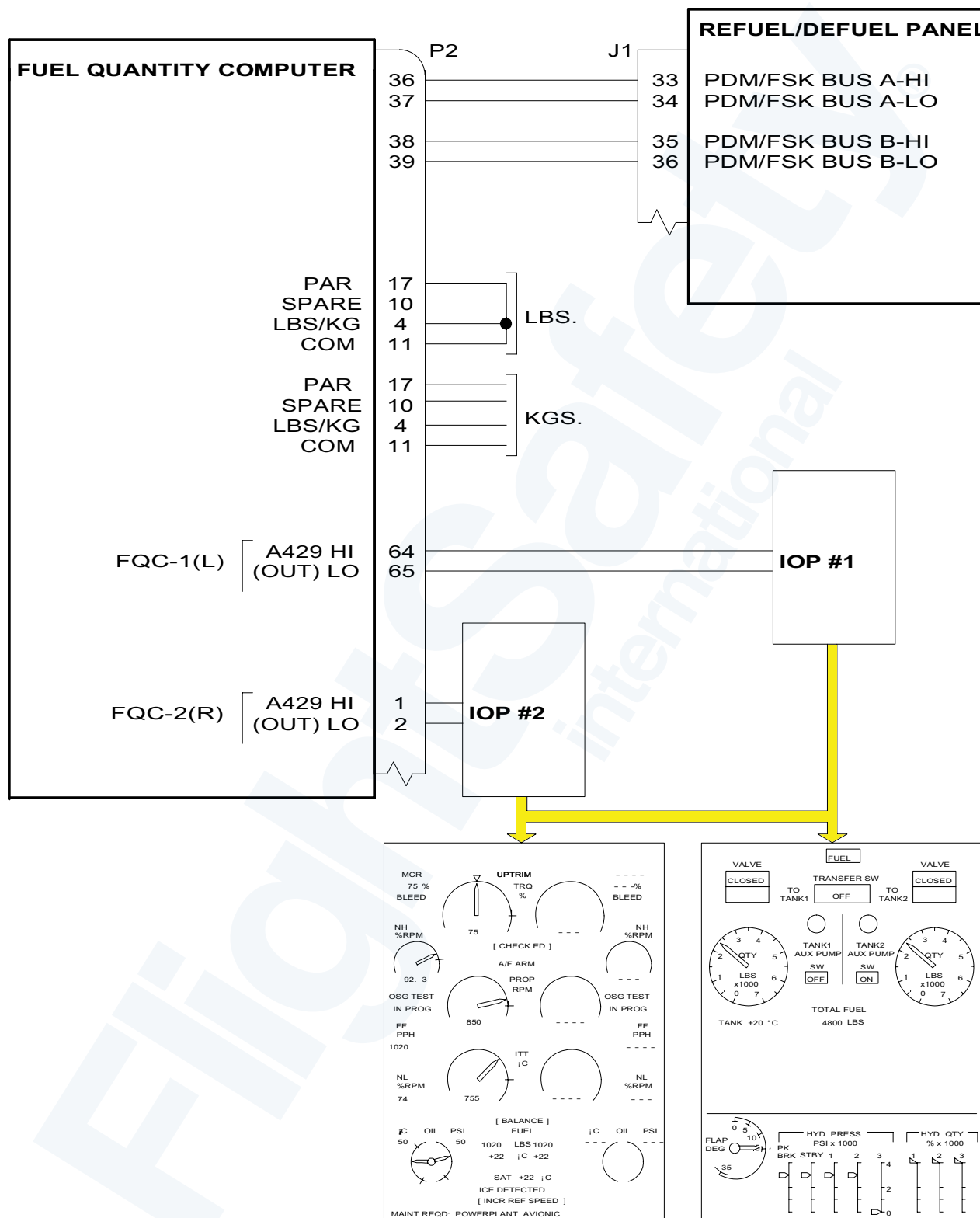
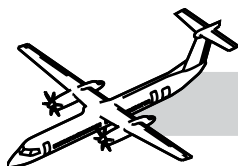
## Fuel Quantity Probe

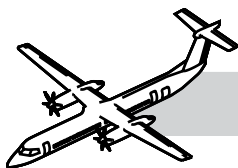
## NOTES

Refer to Figure 28-56. Fuel Quantity Probe.

The fuel quantity probes sense the fuel level in the tanks and transmit the information to the FQC.

There are 18 fuel quantity probes, nine in each wing. The probes contain two concentric metal cylinders, a terminal block, and two mounting brackets. The cylinders form the capacitor elements (plates); the inner plate is the high-impedance element and the outer plate is the low-impedance element.





## OPERATION

## NOTES

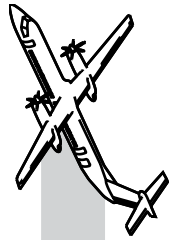
Refer to:

- Figure 28-57. Fuel Indication.
- Figure 28-58. Fuel Indicating System (Sheet 1 of 2).
- Figure 28-59. Fuel Indicating System (Sheet 2 of 2).
- Figure 28-60. Fuel System Synoptic.

### Fuel Indications

The fuel indicating system shows the fuel quantity in the flight compartment and on the refuel/defuel panel.

On the center ED, it shows the two fuel tank quantities and engine fuel inlet temperature (°C only).



28-90

FOR TRAINING PURPOSES ONLY

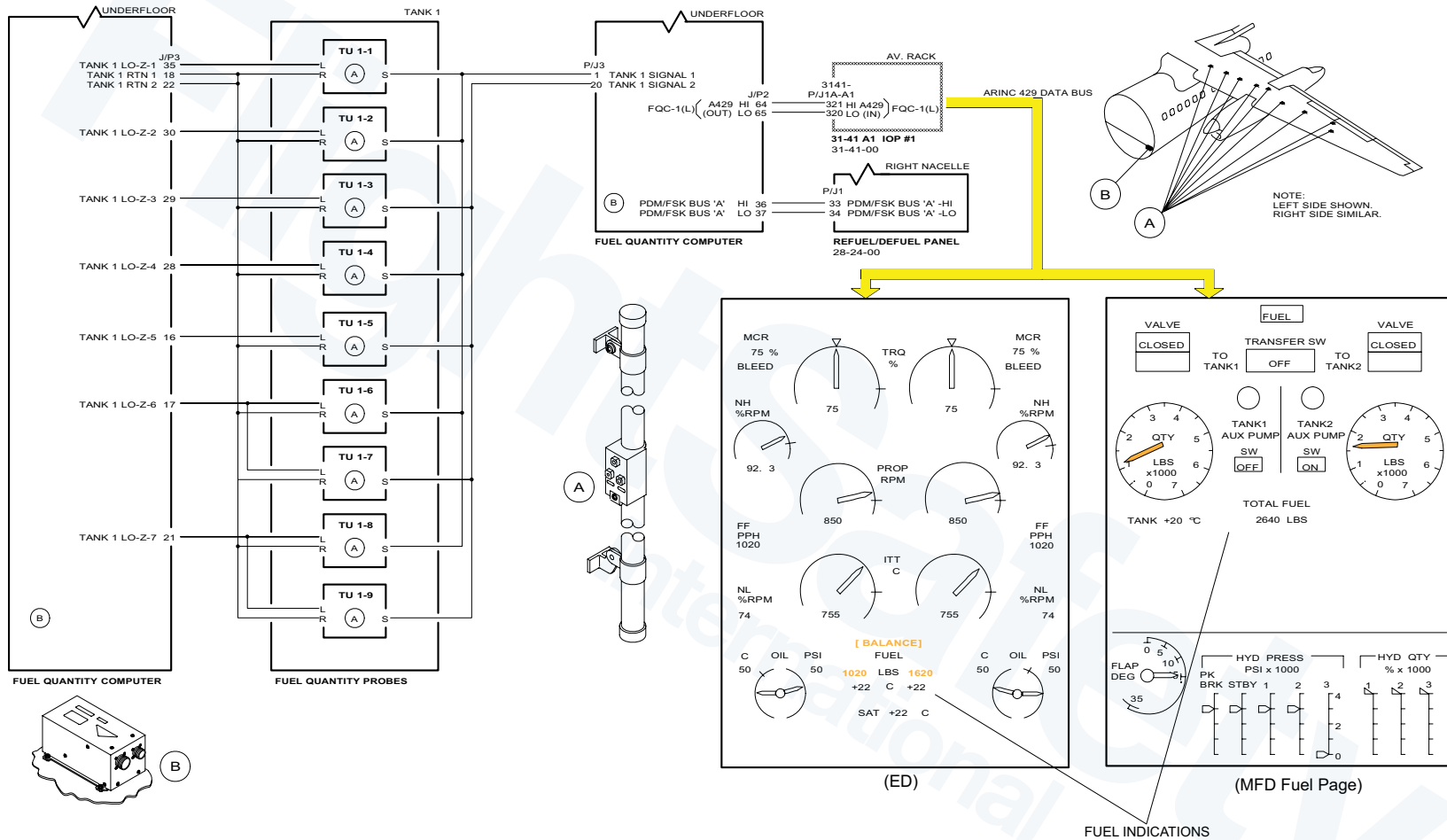
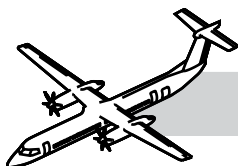


Figure 28-58. Fuel Indicating System (Sheet 1 of 2)



On the MFD Fuel Page, two analog gauges show the fuel tanks quantities. The left tank fuel temperature and the total fuel are shown in digital format.

The normal range fuel quantity and balance are shown as white indications. The indication will change to yellow when there is an imbalance greater than 600 lbs (272 kg). If the fuel data becomes invalid the indication is white dashes. The two tank fuel quantities are also shown on the RDI.

## NOTES

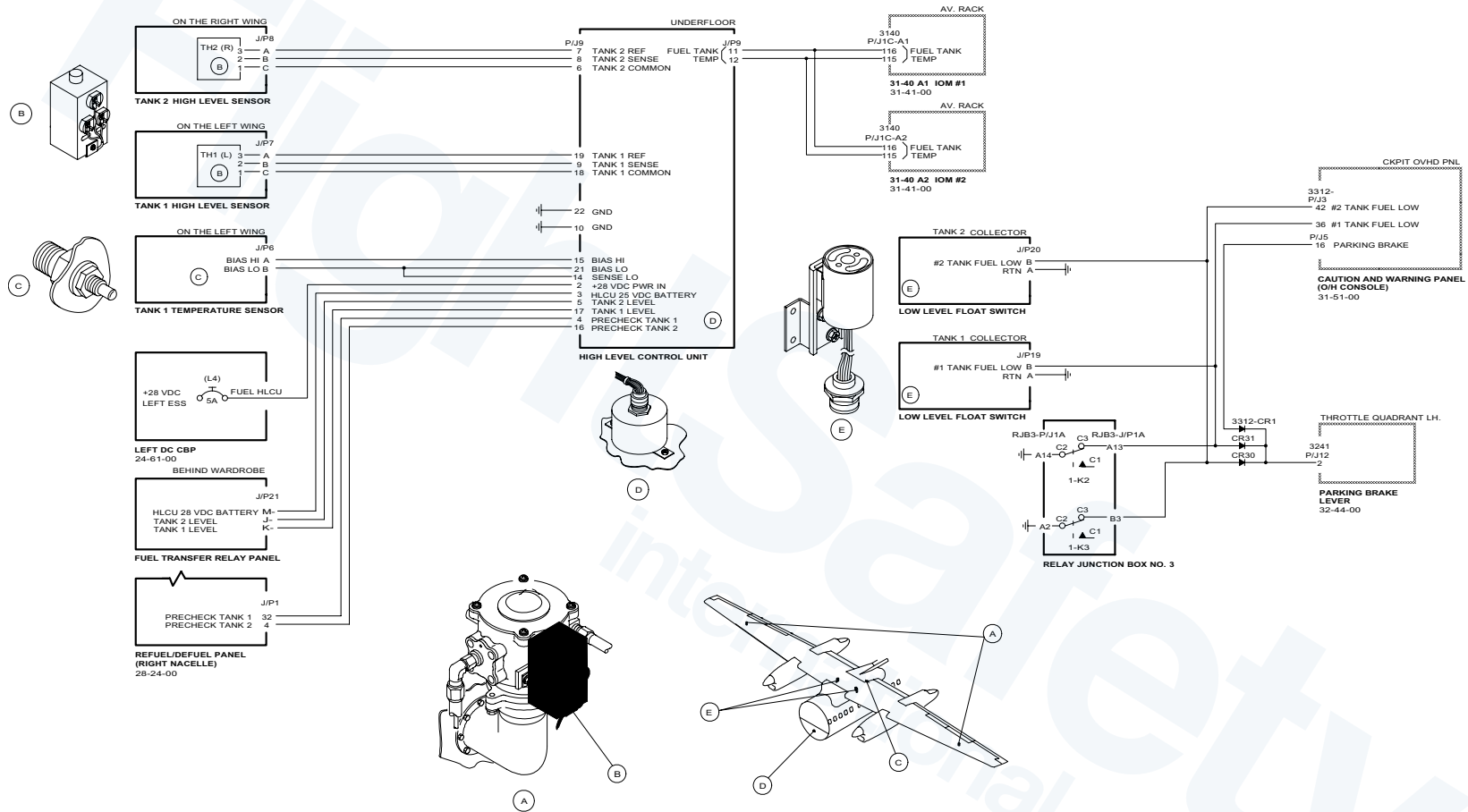
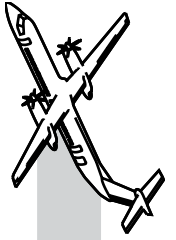
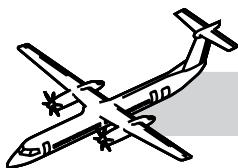


Figure 28-59. Fuel Indicating System (Sheet 2 of 2)



## Fuel Indication Operations

### NOTE

The left side is described, the right side is similar.

The FQC supplies a 1 kHz electrical excitation to the nine probes. The nine fuel quantity probes in each wing sense the dielectric (capacitance) value changes caused by the changing fuel levels. As the fluid level rises the overall capacitance rises proportionately.

The FQC measures this capacitance and converts the probe signals into a fuel weight (lbs or kg) ARINC signal.

The FQC sends the converted quantity data to the IFC Input/Output modules which converts the data for the EFIS. The EFIS converts the data into displays of fuel quantity the ED and MFD units.

The FQC also sends the converted quantity data directly to the refuel/defuel panel for quantity indication.

The fuel quantity can be shown in kilograms (kg) or pounds (lbs). This is accomplished by changing the pin programming on the FQC connector 2800-P2 pins 4, 10, 11 and 17.

Figure 28-57. Fuel Indication.

The high level sensors also sense the fuel tank temperature and send the analog output to the HLCU for signal conditioning. The IFC then receives the HLCU conditioned signal and the left wing tank temperature sensor resistance. The IFC processes the data and sends it to the EFIS for display. The temperature is shown on the MFD Fuel Page (-70 to +75 °C).

On the Engine Display (ED) the fuel indications are:

- The quantity of each fuel tank (digital).
- The fuel inlet temperatures to the engines (digital)
- The out of [BALANCE] fuel quantity indication.

On the MFD FUEL page the fuel indications are:

- A gauge for each fuel tank (analog)
- The total fuel quantity (digital)
- The fuel temperature of the left wing tank (digital)
- The Aux fuel pumps and the transfer valves (analog).

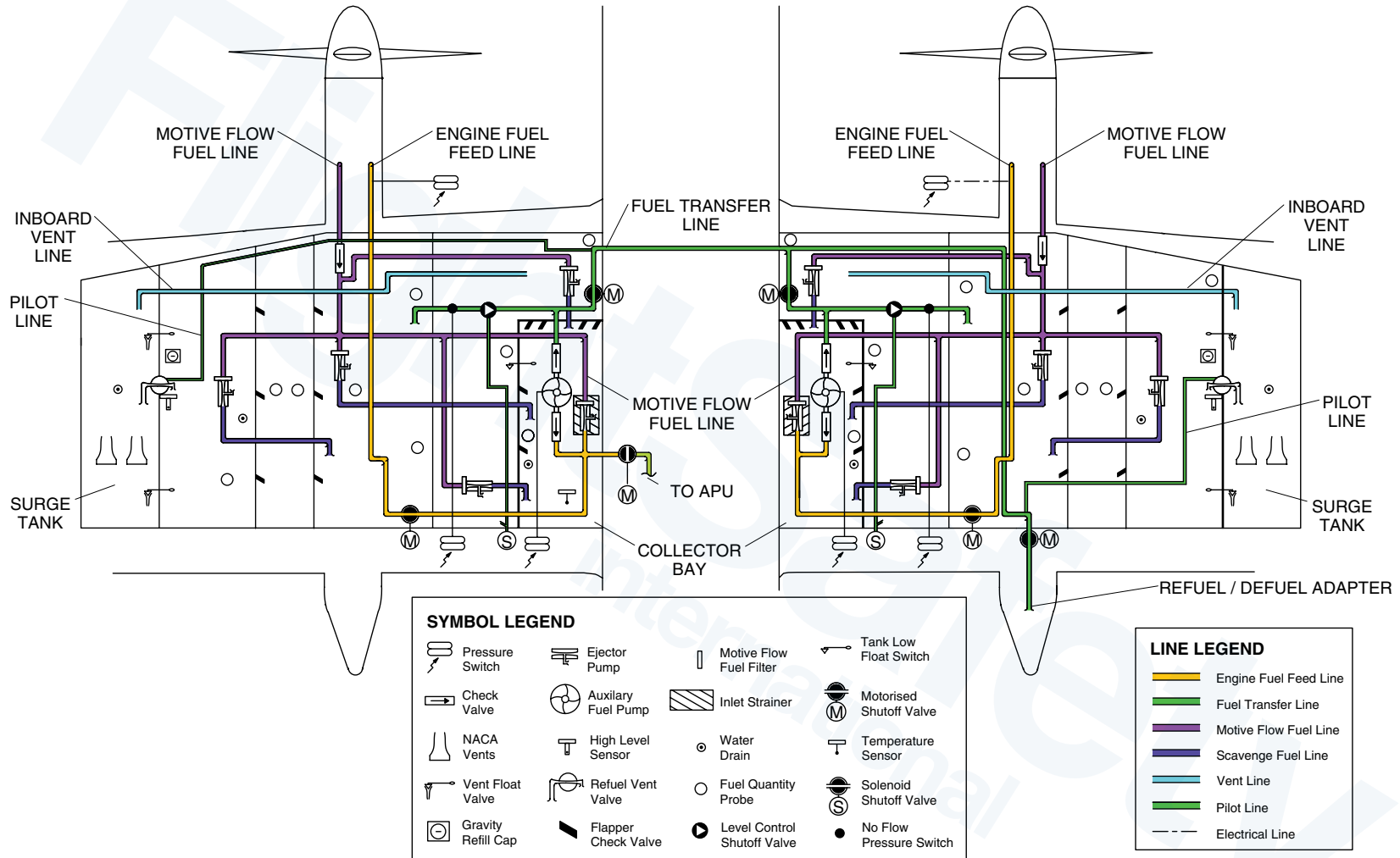
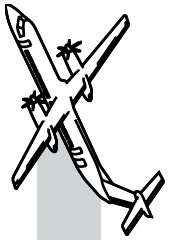
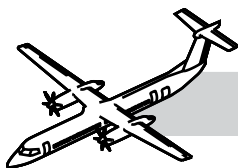


Figure 28-60. Fuel System Synoptic





On the Refuel/Defuel panel the fuel indications are shown on the RDI and the valve position lights.

Refuel/Defuel Panel indications:

- MASTER VALVE CLOSED light (the Refuel/Defuel SOV)
- DUMP VALVE OPEN TANK 1 and 2 lights (the two Vent Valves)
- REFUEL Shut-off VALVE 1 and 2 lights (the no-flow pressure switches).

The Refuel/Defuel panel RDI shows:

- The quantity of each tank,
- The total quantity, and
- The fuel load quantity selected
- The fault codes.

The FQC does not process faulted probe signals (their output data is not valid). The indicated display for data not valid is as follows:

- The ED digital quantity is replaced with white dashes
- The MFD FUEL page QTY needle pointer, scale marks and numbers go out of view
- The MFD FUEL page TOTAL FUEL quantity is replaced with white dashes
- The RDI digital quantity is replaced with yellow dashes.

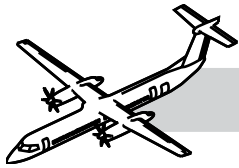
A blank display indicates an FQC or RDI failure.

The Caution and Warning Panel (C&W panel) lights show the fuel system faults that follow:

- FUELING ON comes on when the refuel/defuel panel access door is open and the door switch sends a ground signal to the C&W panel. (Normal operation, no flashing master CAUTION)
- No.1 or No.2 TANK FUEL LOW comes on when the collector bay fuel level is less than

150 kg (305 lb) and the PARK BRAKE is off and appropriate engine is running

- No.1 or No.2 ENG FUEL PRESS comes on when the engine inlet pressure is less than 5.5 psi (38 kPa). The pressure switch sends a ground signal through the FADEC to the ECIU which sends a ground signal to the C&W panel. (Flashing master CAUTION)
- No.1 or No.2 FUEL FLTR BYPASS comes on when the FMU fuel filter is blocked. The impending bypass switch sends a closed signal through the FADEC to the ECIU which sends the signal to the C&W panel. (Flashing master CAUTION).



## 28-00-00 APPENDIX

NOTES

### MAINTENANCE CONSIDERATION

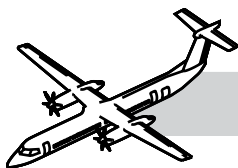
#### CDL

From CDL, ATA 28, page 6-41-5.

#### Unscheduled Inspection

Refer to the Bombardier published AMM Part 2 PSM 1-84-2:

TASK 05-53-00-750-802 Engine Inspection after Hydraulic Fluid in the Fuel System.



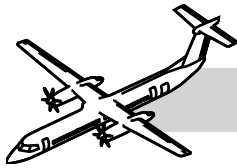
## **28-00-00 SPECIAL TOOLS & TEST EQUIPMENT**

- GSB2800001 Kit, Spark-free Breathing Apparatus
- GSB2810003 Gas Detector (explosimeter) or equivalent.
- Commercially Available Digital Low Resistance ohmmeter (Biddle DLRO #247001 or equivalent)
- Commercially Available Loop Resistance Tester (BAE Systems P/N 906-10247-ALL)
- Commercially Available Accessory Assembly LRT (Battery Charger for Loop Resistance Tester) (BAE Systems P/N 906-10271-3)
- GSB2810001 Fuel Drain Valve Tool
- GSB2000001 Borescope – 110 Volt, 60 Hz
- GSB2000015 Borescope – 220 Volt, 50 Hz
- GSB1216012 Kit, Nitrogen Charging and Gauging, Low Pressure, 0 to 500 psi (0 to 3447 kPa)
- GSB2840006 Clamping Ring Wrench, Aux Pump
- GSB2840006 Clamp Ring Wrench, Aux Pump
- Commercially available Leader Line
- GSB 1216012 Nitrogen, Gauging, Low Pressure (0 to 500 psi)
- GSB2840002 Digital Fuel Quantity Test Set
- GSB2840008A Adapter Cable set for D8-400 Fuel Quantity Test

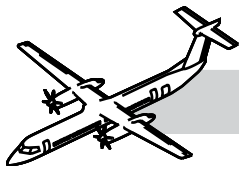
## **28-00-00 MAINTENANCE PRACTICES**

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

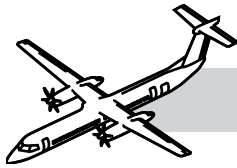
- AMM 28-11-00-680-801: Drain the Water from the Wing Tanks and the Dry Bay.
- AMM 28-11-00-680-802: Drain Water from the Fuel Tanks and Fuel Tank Surge Bays (MRB #281000-201).
- AMM 12-10-28-750-801: Water Contamination Fuel Check.
- AMM 28-10-00-720-801: Functional Check of the Fuel Quantity Metal Overall Shield for Electrical Bonding, LH and RH (FSL#284000-411).
- AMM 28-10-00-720-802: Functional Check of the Fuel Tank Water Drain Valve for Electrical Bonding, LH and RH (FSL#284000-412).
- AMM 28-10-00-720-803: Functional Check of the Magnetic Dipstick for Electrical Bonding, LH and RH (FSL#284000-413).
- AMM 28-10-00-720-804: Functional Check of the Auxiliary Boost Pump for Electrical Bonding, LH and RH (FSL#284000-415).
- AMM 28-10-00-720-805: Functional Check of the High Level Sensor for Electrical Bonding, LH and RH (FSL#284000-416).



- AMM 28-10-00-720-806: Functional Check of the Fuel Tank Components and Plumbing Lines for Electrical Bonding, LH and RH (FSL#284000-417).
- AMM 28-21-31-000-802: Removal of the Auxiliary Pump Canister.
- AMM 28-21-31-400-802: Installation of the Auxiliary Pump Canister.
- AMM 28-21-00-780-801: Operational Check of the Engine Fuel Shut-off Valve (MRB #282000-202).
- AMM 28-21-31-710-801: Operational Test of the Auxiliary Pump.
- AMM 28-22-01-710-802: Operational Check of the APU Fuel Feed Shut-off Valve (MRB#282000-203).
- AMM 28-23-00-710-801: Operational Check of the Fuel Transfer System (CMR# 282000-101).
- AMM 12-10-28-650-801: Pressure Refueling.
- AMM 12-10-28-650-802: Pressure Defueling.
- AMM 28-24-00-710-802: Operational Test of the High Level Control Shut-off.
- AMM 28-24-01-710-802: Operational Test of the Refuel / Defuel Control Panel.
- AMM 28-24-02-742-801: Retrieval of Data from the Refuel/Defuel Indicator.
- AMM 28-24-02-743-801: Erase the Data from the Refuel/Defuel Indicator.
- AMM 28-40-26-610-801: Check of the Fuel Quantity with the Magnastick.



PAGE INTENTIONALLY LEFT BLANK



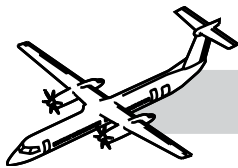
For engines subject to sand erosion and FOD, P&WC recommends the following:

## NOTES

- Restrict CLA movement when the aircraft is stationary
- Taxi using both engines with equal PLA movement
- Use intake blanking covers when the aircraft is parked overnight
- Avoid using reverse on landing (except in emergency)
- Avoid running into sand wakes during taxi.

For engines exposed to severe marine environment and/or volcanic fumes P&WC recommend the following:

- Perform compressor/turbine wash every 50FH. Adjust washing interval based on level of exposure to the salt laden atmosphere
- Perform an initial borescope inspection on the HP, LP and PT sections at 2000FH TSN/TSO for evidence of sulphidation attack on the turbine blades. Adjust inspection interval as necessary
- Perform external engine washes every 1200FH to reduce salt deposits on the engine
- Perform an initial visual inspection of all magnesium parts at 1200 hours TSN/TSO
  - A. Inspect external surfaces and inlet gas path for corrosion or damage
  - B. Repair magnesium surfaces i.a.w. the AMM . After treatment apply 2 or 3 coats of paint
  - C. Pay particular attention to bolted flanges or other areas where the painted surface may be compromised.



## **70-00-00 SPECIAL TOOLS & TEST EQUIPMENT**

- PWC57689 Air Deflector Shield
- PWC58104 Wrench, mini-strap
- Glenair TG70 Wrench, mini-strap
- Glenair TG69 Pliers, soft-jawed
- GSB2000020 or equivalent Variable temperature, electronically controlled heat gun (120 Vac - 60 Hz)
- GSB2000022 or equivalent Variable temperature, electronically
- Controlled heat gun (230 Vac - 50 Hz)
- PNo. 07051 (39 mm) (Steinel) or equivalent Heat gun reflector nozzle
- GSB2400016 Tool kit, electrical and avionics